# OPEN PROM

Model Version 0.1

GIANNOUSAKIS **ANASTASIS** | PLESSIAS **GEORGIOS**

created with goxygen on Sep 13 2023

## Contents

**OPEN PROM** **2**
Authors . . . . . 4
How to cite . . . . . 4
Bibtex format . . . . . 4
Citation File Format . . . . . 5

**declarations (declarations.gms)** **6**
Description . . . . . 6

**Equations (equations.gms)** **29**
Description . . . . . 29

**input (input.gms)** **76**
Description . . . . . 76

**Preloop (preloop.gms)** **106**
Description . . . . . 106

**sets (sets.gms)** **115**
Description . . . . . 115

**solve (solve.gms)** **154**
Description . . . . . 154

**References** **154**

# OPEN PROM

This is the OPEN-PROM model, the open version of the world energy model PROMETHEUS of E3Modelling.

Some preliminary documentation of OPEN-PROM can be found in the Wiki of this repository, extensive documentation of the PROMETHEUS model (on which both MENA-EDS and OPEN-PROM are based) can be found here: https://e3modelling.com/modelling-tools/prometheus/

*** preliminaries

```
option limcol = 16;
```

*** number of rows that are listed for each equation in the equation listing

```
option limrow = 16;
```

*** “dollar” ($) commands section: define GAMS flags & code control & compilation-time options

*** onEolCom: turn on end-of-line comments (starting with !!, i.e. the GAMS default)

```
$onEolCom
```

*** onEnd: turn on alternative flow control syntax (more readable loop, for, if etc.)

```
$onEnd
```

*** onEmpty allows declarations of empty parameters

```
$onEmpty
```

*** TODO: check if the contents of this block are actually used later

*** GAMS “flags” definitions

*** Maximum number of solver attempts

```
$evalGlobal SolverTryMax 1
```

```
$setGlobal fCountries 'RAS,RWO,REU,REP,RAF,RLA,ARG,OCE,AUT,BEL,BRA,BGR,
    CAN,CHA,HRV,CYP,CZE,DNK,EGY,EST,FIN,FRA,DEU,GRC,HUN,NSI,IND,IDN,IRN,
    IRL,ISR,ITA,JPN,KOR,LVA,LTU,LUX,MLT,MEX,MAR,NLD,NGA,POL,PRT,ROU,RUS,
    SAU,SVK,SVN,ZAF,ESP,SWE,TUN,TUR,GBR,USA'
```

```
$setGlobal fSingleCountryRun 'yes'
```

```
$setGlobal fCountryList %countries%
```

```
$setGlobal fScenario 'DECARB_400'
$setGlobal fBaseline 'BASE'
```

```
$evalGlobal fPeriodOfYears 1
```

```
$setGlobal fIncludeNonCO2 no
```

```
$setGlobal fReadCommonDB 'yes'
$setGlobal fReadCountryDB 'yes'
$setGlobal fReadCountryCalib 'yes'
```

```
$evalGlobal fStartHorizon 2010
$evalGlobal fEndHorizon 2100
$evalGlobal fEndY 2020
$evalGlobal fStartY 2018
$evalGlobal fBaseY %fStartY% - %fPeriodOfYears%
```

*** end of dollar commands section, no further flag definitions allowed

*** load input data files

```
$include sets.gms
$include declarations.gms
$include input.gms
$include equations.gms
$include preloop.gms
$include solve.gms
```

## Authors

Giannousakis Anastasis bat@m.an, Plessias Georgios bat@m.an

## How to cite

Anastasis G, Georgios P (2019). “OPEN PROM - Version 0.1.”

### Bibtex format

```
@Misc{,
  title = {OPEN PROM - Version 0.1},
  author = {Giannousakis Anastasis and Plessias Georgios},
  date = {2019-05-13},
  year = {2019},
}
```

### Citation File Format

```yaml
cff-version: 1.0.3
message: If you use this model, please cite it as below.
authors:
- family-names: Anastasis
  given-names: Giannousakis
  affiliation: E3Modelling
  email: bat@m.an
- family-names: Georgios
  given-names: Plessias
  affiliation: E3Modelling
  email: bat@m.an
title: OPEN PROM
version: '0.1'
date-released: '2019-05-13'
repository-code: https://github.com/e3modelling/OPEN-PROM
keywords: testing
license: CC0
```

## declarations (declarations.gms)

### Description

```
Parameters
iCGI(allCy,YTIME)                                "Capital Goods Index (defined as CGI(Scenario)/CGI(Baseline)) (1)"
iNPDL(SBS)                                       "Number of Polynomial Distribution Lags (PDL) (1)"
iFPDL(SBS,KPDL)                                  "Polynomial Distribution Lags (PDL) Coefficients per subsector (1)"
iLifChpPla(allCy,DSBS,CHP)                          "Technical Lifetime for CHP plants (years)"
iPlantEffByType(allCy,PGALL,YTIME)                  "Plant efficiency per plant type (1)"
iCo2EmiFac(allCy,SBS,EF,YTIME)                      "CO2 emission factors per subsector (kgCO2/kgoe fuel burned)"
iNcon(SBS)                                          "Number of consumers (1)"
iDisFunConSize(allCy,DSBS,rCon)                     "Distribution function of consumer size groups (1)"
iAnnCons(allCy,DSBS,conSet)                         "Annual consumption of the smallest,modal,largest consumer, average for all countries (various)"
                                                     !! For passenger cars (Million km/vehicle)
                                                     !! For other passenger tranportation modes (Mpkm/vehicle)
                                                     !! For goods transport, (Mtkm/vehicle)
iCumDistrFuncConsSize(allCy,DSBS)                   "Cummulative distribution function of consumer size groups (1)"
iRateLossesFinCons(allCy,EF,YTIME)                  "Rate of losses over Available for Final Consumption (1)"
iEneProdRDscenarios(allCy,SBS,YTIME)                "Energy productivity index used in R&D scenarios (1)"
iEffDHPlants(allCy,EF,YTIME)                        "Efficiency of District Heating Plants (1)"
iShareFueTransfInput(allCy,EF)                      "Share of fuels in transformation input to Gasworks, Blast Furnances, Briquetting
```

```
    plants in base year (1)"
iTransfInpGasworks(allCy,EF,YTIME)              "Transformation Input
    in Gasworks, Blast Furnances, Briquetting plants (Mtoe)"
iResRefCapacity(allCy,YTIME)                    "Residual in Refineries
     Capacity (1)"
iRefCapacity(allCy,YTIME)                       "Refineries Capacity (
    Million Barrels/day)"
iResTransfOutputRefineries(allCy,EF,YTIME)      "Residual in
    Transformation Output from Refineries (Mtoe)"
iRateEneBranCons(allCy,EF,YTIME)                "Rate of Energy Branch
    Consumption over total transformation output (1)"
iResFeedTransfr(allCy,YTIME)                    "Residual for
    Feedstocks in Transfers (1)"
iFeedTransfr(allCy,EFS,YTIME)                   "Feedstocks in
    Transfers (Mtoe)"
iResHcNgOilPrProd(allCy,EF,YTIME)               "Residuals for Hard
    Coal, Natural Gas and Oil Primary Production (1)"
iNatGasPriProElst(allCy)                        "Natural Gas primary
    production elasticity related to gross inland consumption (1)" /
$ondelim
$include "./iNatGasPriProElst.csv"
$offdelim
/
iFuelPriPro(allCy,EF,YTIME)                     "Fuel Primary
    Production (Mtoe)"
iPolDstrbtnLagCoeffPriOilPr(kpdl)               "Polynomial
    Distribution Lag Coefficients for primary oil production (1)"/
$ondelim
$include "./iPolDstrbtnLagCoeffPriOilPr.csv"
$offdelim
/
iRatePriProTotPriNeeds(allCy,EF,YTIME)          "Rate of Primary
    Production in Total Primary Needs (1)"
iFuelExprts(allCy,EF,YTIME)                     "Fuel Exports (Mtoe)"
iSuppExports(allCy,EF,YTIME)                    "Supplementary
    parameter for  exports (Mtoe)"
iRatioImpFinElecDem(allCy,YTIME)                "Ratio of imports in
    final electricity demand (1)"
iElastCO2Seq(allCy,CO2SEQELAST)                 "Elasticities for CO2
    sequestration cost curve (1)"
iBaseLoadShareDem(allCy,DSBS,YTIME)             "Baseload share of
    demand per sector (1)"
```

```
iLoadFacElecDem(allCy,DSBS,YTIME)                "Load factor of electricity demand per sector (1)"
iMxmLoadFacElecDem(allCy,YTIME)                  "Maximum load factor of electricity demand (1)"
iBslCorrection(allCy,YTIME)                      "Parameter of baseload correction (1)"
iResMargTotAvailCap(allCy,PGRES,YTIME)           "Reserve margins on total available capacity and peak load (1)"
iTechLftPlaType(allCy,PGALL)                              "Technical Lifetime per plant type (year)"
iElecImp(allCy,YTIME)                       "Electricity Imports (1)"
iPlantAvailRate(allCy,PGALL,YTIME)          "Plant availability rate (1)"
iTotAvailCapBsYr(allCy)               "Total installed available capacity in base year (GW)"
iGrossCapCosSubRen(allCy,PGALL,YTIME)       "Gross Capital Cost per Plant Type with subsidy for renewables (kEuro2005/KW)"
iCO2CaptRate(allCy,PGALL,YTIME)             "Plant CO2 capture rate (1)"
iScaleEndogScrap(allCy,PGALL,YTIME) "Scale parameter for endogenous scrapping applied to the sum of full costs (1)"
iPremReplacem(allCy,PGALL)            "Premature replacement (1)"/
$ondelim
$include"./iPremReplacem.csv"
$offdelim
/
iPlantDecomSched(allCy,PGALL,YTIME)       "Decided plant decomissioning schedule (GW)"
iDecInvPlantSched(allCy,PGALL,YTIME)      "Decided plant investment schedule (GW)"
iMinRenPotential(allCy,PGRENEF,YTIME)     "Minimum renewable potential (GW)"
iMaxRenPotential(allCy,PGRENEF,YTIME)     "Maximum enewable potential (GW)"
iMatFacPlaAvailCap(allCy,PGALL,YTIME)     "Maturity factor related to plant available capacity (1)"
iMatureFacPlaDisp(allCy,PGALL,YTIME)      "Maturity factor related to plant dispatching (1)"
iEffValueInEuro(allCy,SBS,YTIME)          "Efficiency value (Euro05/toe)"
iFacElecPriConsu(allCy,ELCPCHAR,YTIME)    "Factors affecting electricity prices to consumers (1)"
iPriceTragets(allCy,SBS,EF,YTIME)         "Price Targets (1)"
iPriceReform(allCy,SBS,EF,YTIME)          "Price reformation (1)"
```

```
iScenarioPri(WEF,NAP,YTIME)                "Scenario prices (KEuro1990/toe)"
iWgtSecAvgPriFueCons(allCy,SBS,EF)         "Weights for sector's average price, based on fuel consumption (1)"
iFuelConsPerFueSub(allCy,SBS,EF,YTIME)     "Fuel consumption per fuel and subsector (Mtoe)"
iTotFinEneDemSubBaseYr(allCy,SBS,YTIME)    "Total Final Energy Demand per subsector in Base year (Mtoe)"
iResNonSubsElecDem(allCy,SBS,YTIME  )      "Residuals in Non Substitutable Electricity Demand (1)"
iResFuelConsPerSubAndFuel(allCy,SBS,EF,YTIME)   "Residuals in fuel consumption per subsector and fuel (1)"
iTransfOutputGasw(allCy,YTIME)                  "Transformation Output from Gasworks, Blast Furnances, Briquetting plants Residual (1)"
iResSpecificFuelConsCost(allCy,TRANSE,TTECH,EF,YTIME)   "Residuals on specific fuel consumption cost (1)"
iShareAnnMilePlugInHybrid(allCy,YTIME)                  "Share of annual mileage of a plug-in hybrid which is covered by electricity (1)"
iAvgVehCapLoadFac(allCy,TRANSE,TRANSUSE,YTIME)          "Average capacity/vehicle and load factor (tn/veh or passenegers/veh)"
iUtilRateChpPlants(allCy,CHP,YTIME)                     "Utilisation rate of CHP Plants (1)"
iHydrogenPri(allCy,SBS,YTIME)                           "Total Hydrogen Cost Per Sector (Euro/toe)"
iTechLft(allCy,SBS,EF,YTIME)                        "Technical Lifetime. For passenger cars it is a variable (1)"
iSpeFuelConsCostBy(allCy,SBS,TTECH,TEA,EF)              "Specific fuel consumption cost in Base year (ktoe/Gpkm or ktoe/Gtkm or ktoe/Gvkm)"
iShrNonSubElecInTotElecDem(allCy,SBS)                   "Share of non substitutable electricity in total electricity demand per subsector (1)"
iElecIndex(allCy,YTIME)                                 "Electricity Index (1)"
iExogDemOfBiomass(allCy,SBS,YTIME)                      "Demand of tranditional biomass defined exogenously ()"
iFinEneConsPrevYear(allCy,EF,YTIME)                     "Final energy consumption used for holding previous year results (Mtoe)"
iCarbValYrExog(ytime)                                   "Carbon value for each year when it is exogenous (Euro05/tn CO2)"
iShrHeatPumpElecCons(allCy,SBS)                         "Share of heat pump electricity consumption in total substitutable electricity (1)"
```

```
iTranfOutGasworks(allCy,EF,YTIME)                          "Transformation Output from Gasworks, Blast Furnances and Briquetting plants (Mtoe)"
iDistrLosses(allCy,EF,YTIME)                               "Distribution Losses (Mtoe)"
iTransfOutputRef(allCy,EF,YTIME)                           "Transformation Output from Refineries (Mtoe)"
iFinEneCons(allCy,EF,YTIME)                                "Final energy consumption (Mtoe)"
iTotEneBranchCons(allCy,EF,YTIME)                          "Total Energy Branch Consumption (Mtoe)"
iFuelImports(allCy,EF,YTIME)                               "Fuel Imports (Mtoe)"
iGrosInlCons(allCy,EF,YTIME)                               "Gross Inland Consumtpion (Mtoe)"
iGrossInConsNoEneBra(allCy,EF,YTIME)                       "Gross Inland Consumption,excluding energy branch (Mtoe)"
iPeakBsLoadBy(allCy,PGLOADTYPE)                            "Peak and Base load for base year (GW)"
iHisChpGrCapData(allCy,CHP,YTIME)                          "Historical CHP gross capacity data (GW)"
iTransfInputRef(allCy,EF,YTIME)                            "Transformation Input in Refineries (Mtoe)"
iSigma(allCy,SG)                                           "S parameters of Gompertz function for passenger cars vehicle km (1)"
iPassCarsMarkSat(allCy)                                    "Passenger cars market saturation (1)"
iGdpPassCarsMarkExt(allCy)                                 "GDP-dependent passenger cars market extension (GDP/capita)"
iPassCarsScrapRate(allCy)                                  "Passenger cars scrapping rate (1)"
iVarCostTech(allCy,SBS,EF,YTIME)                           "Variable Cost of technology ()"
                                                           !! For transport (kEuro05/vehicle)
                                                           !! For Industrial sectors, except Iron and Steel (Euro05/toe-
```

```
                                                                    year)
                                                                !! For Iron and Steel (Euro05/tn-of-steel)
                                                                !! For Domestic sectors (Euro05/toe-year)
iFixOMCostTech(allCy,SBS,EF,YTIME)          "Fixed O&M cost of technology (Euro2005/toe-year)"
                                                 !! Fixed O&M cost of technology for Transport (kEuro2005/vehicle)
                                                 !! Fixed O&M cost of technology for Industrial sectors-except Iron and Steel (Euro2005/toe-year)"
                                                 !! Fixed O&M cost of technology for Iron and Steel (Euro2005/tn-of-steel)"
                                                 !! Fixed O&M cost of technology for Domestic sectors (Euro2005/toe-year)"
iUsfEneConvSubTech(allCy,SBS,EF,YTIME)                "Useful Energy Conversion Factor per subsector and technology (1)"
iInvCostChp(allCy,DSBS,CHP,YTIME)                     "Capital Cost per CHP plant type (kEuro05/KW)"
iFixOMCostPerChp(allCy,DSBS,CHP,YTIME)                "Fixed O&M cost per CHP plant type (Euro2005/KW)"
iVarCostChp(allCy,DSBS,CHP,YTIME)                     "Variable (other than fuel) cost per CHP Type (Gross Euro05/KW)"
iAvailRateChp(allCy,DSBS,CHP)                         "Availability rate of CHP Plants ()"
iBoiEffChp(allCy,CHP,YTIME)                           "Boiler efficiency (typical) used in adjusting CHP efficiency ()"
iCapCostTech(allCy,SBS,EF,YTIME)                       "Capital Cost of technology (various)"
                                                   !! - For transport is expressed in
```

```
                                                                kEuro05 per vehicle
                                                        !! - For Industrial sectors (except Iron and Steel) is expressed in kEuro05/toe-year
                                                        !! - For Iron and Steel is expressed in kEuro05/tn-of-steel
                                                        !! - For Domestic Sectors is expressed in kEuro05/toe-year
iInstCapPast(allCy,PGALL,YTIME)        "Installed capacity past (GW)"
iDiffFuelsInSec(SBS) "auxiliary parameter holding the number of different fuels in a sector"
iResElecIndex(allCy,YTIME)             "Residual for electricity Index (1)"
;
Equations
```

*** Power Generation

```
QChpElecPlants(allCy,CHP,YTIME)            "Compute CHP electric capacity"
QElecDem(allCy,YTIME)                      "Compute total electricity demand"
QElecConsAll(allCy,DSBS,YTIME)             "Compute electricity consumption per final demand sector"
QEstBaseLoad(allCy,YTIME)                  "Compute estimated base load"
QLoadFacDom(allCy,YTIME)                   "Compute load factor of entire domestic system"
QElecPeakLoad(allCy,YTIME)                 "Compute elerctricity peak load"
QBslMaxmLoad(allCy,YTIME)                  "Compute baseload corresponding to maximum load"
QElecBaseLoad(allCy,YTIME)                 "Compute electricity base load"
QShrcap(allCy,PGALL,YTIME)                 "Compute SHRCAP"
QElecGenCap(allCy,PGALL,YTIME)             "Compute electricity generation capacity"
```

```
QLambda(allCy,YTIME)                          "Compute Lambda parameter"
QScalFacPlantDispatch(allCy,HOUR,YTIME)   "Compute the scaling factor for plant dispatching"
QCurrRenPot(allCy,PGRENEF,YTIME)              "Compute current renewable potential"
QScalWeibullSum(allCy,PGALL,YTIME)            "Compute sum (over hours) of temporary variable facilitating the scaling in Weibull equation"
QElecPriIndResNoCliPol(allCy,ESET,YTIME)   "Compute electricity price in Industrial and Residential Consumers excluding climate policies"
QShortPowGenCost(allCy,ESET,YTIME)            "Compute short term power generation cost"
QLonPowGenCostNoClimPol(allCy,ESET,YTIME)  "Compute long term power generation cost excluding climate policies"
QLonAvgPowGenCostNoClimPol(allCy,PGALL,ESET,YTIME)  "Compute long term average power generation cost excluding climate policies"
QLongPowGenCost(allCy,ESET,YTIME)             "Compute long term power generation cost"
QShoPowGenIntPri(allCy,PGALL,ESET,YTIME)   "Compute short term power generation cost of technologies including international Prices of main fuels"
QLongPowGenIntPri(allCy,PGALL,ESET,YTIME)  "Compute long term power generation cost of technologies including international Prices of main fuels"
QLonMnmpowGenCost(allCy,PGALL,YTIME)          "Long-term minimum power generation cost"
QLonPowGenCostTechNoCp(allCy,PGALL,ESET,YTIME)  "Compute long term power generation cost of technologies excluding climate policies"
QShareRenGrossElecProd(allCy,YTIME)           "Compute the share of renewables in gross electricity production for subsdidized renewables"
QSecContrTotChpProd(allCy,SBS,CHP,YTIME)   "Compute sector contribution to total CHP production"
QElecProdPowGenPlants(allCy,PGALL,YTIME)   "Compute electricity production from power generation plants"
QReqElecProd(allCy,YTIME)                     "Compute total required electricity production"
QElecChpPlants(allCy,YTIME)                   "Estimate the electricity of CHP Plants"
QOverallCap(allCy,PGALL,YTIME)                "Compute overall capacity"
QNonChpElecProd(allCy,YTIME)                  "Compute non CHP electricity production"
QAvgCapFacRes(allCy,PGALL,YTIME)              "Compute the average capacity factor of RES"
```

```
QNewCapYearly(allCy,PGALL,YTIME)            "Compute the new capacity added every year"
QElectrPeakLoad(allCy,PGALL,YTIME)          "Compute Power plants sorting according to variable cost to decide the plant dispatching"
QVarCostTechnology(allCy,PGALL,YTIME)       "Compute variable cost of technology"
QElecPeakLoads(allCy,YTIME)                 "Compute Electricity peak loads"
QElecGenCapacity(allCy,PGALL,YTIME)         "Compute electricity generation capacity"
QPowPlaShaNewEquip(allCy,PGALL,YTIME)       "Compute the power plant share in new equipment"
QNewInvDecis(allCy,YTIME)                   "Compute for Power Plant new investment decision"
QMaxmAllowRenPotent(allCy,PGRENEF,YTIME)    "Compute maximum allowed renewable potential"
QMnmAllowRenPot(allCy,PGRENEF,YTIME)        "Compute minimum allowed renewable potential"
QRenTechMatMult(allCy,PGALL,YTIME)          "Compute renewable technologies maturity multiplier"
QElecProdChpPlants(allCy,CHP,YTIME)         "Compute electricity production from CHP plants"
QScalWeibull(allCy,PGALL,HOUR,YTIME)        "Compute temporary variable facilitating the scaling in Weibull equation"
QElecGenNoChp(allCy,YTIME)                  "Compute total electricity generation capacity excluding CHP plants"
QRenPotSupplyCurve(allCy,PGRENEF,YTIME)     "Compute renewable potential supply curve"
QEndogScrapIndex(allCy,PGALL,YTIME)         "Compute endogenous scrapping index"
QProdCostTechPreReplacAvail(allCy,PGALL,PGALL2,YTIME)   "Compute production cost of technology  used in premature replacement including plant availability rate"
QVarCostTechNotPGSCRN(allCy,PGALL,YTIME)    "Compute variable cost of technology excluding PGSCRN"
QVarCostTech(allCy,PGALL,YTIME)             "Compute variable cost of technology"
QTotReqElecProd(allCy,YTIME)                "Compute total required electricity production"
QTotEstElecGenCap(allCy,YTIME)              "Compute Estimated total electricity generation capacity"
QTotElecGenCap(allCy,YTIME)                 "Compute total electricity generation capacity"
```

```
QHourProdCostInv(allCy,PGALL,HOUR,YTIME)   "Compute hourly production cost used in investment decisions"
QHourProdCostInvDec(allCy,PGALL,HOUR,YTIME)"Compute hourly production cost used in investment decisions"
QGammaInCcsDecTree(allCy,YTIME)            "Compute gamma parameter used in CCS/No CCS decision tree"
QHourProdCostInvDecisionsAfterCCS(allCy,PGALL,HOUR,YTIME)   "Compute hourly production cost used in investment decisions taking into account CCS acceptance"
QProdCostInvDecis(allCy,PGALL,YTIME)                "Compute production cost used in investment decisions"
QShrcapNoCcs(allCy,PGALL,YTIME)                     "Compute SHRCAP excluding CCs"
QProdCostTechPreReplac(allCy,PGALL,YTIME)           "Compute production cost of technology  used in premature replacement"
QGapPowerGenCap(allCy,YTIME)                        "Compute the gap in power generation capacity"
```

*** Transport

```
QMExtV(allCy,YTIME)            "Compute passenger cars market extension (GDP dependent)"
QMExtF(allCy,YTIME)            "Compute passenger cars market extension (GDP independent)"
QNumVeh(allCy,YTIME)           "Compute stock of passenger cars (in million vehicles)"
QNewReg(allCy,YTIME)           "Compute new registrations of passenger cars"
QTrnspActiv(allCy,TRANSE,YTIME)"Compute passenger transport acitivity"
QScrap(allCy,YTIME)            "Compute scrapped passenger cars"
QLevl(allCy,YTIME)             "Compute ratio of car ownership over saturation car ownership"
QScrRate(allCy,YTIME)          "Compute passenger cars scrapping rate"
QPassCarsLft(allCy,DSBS,EF,TEA,YTIME)   "Compute the lifetime of passenger cars"
QGoodsTranspActiv(allCy,TRANSE,YTIME)   "Compute goods transport activity"
QGapTranspActiv(allCy,TRANSE,YTIME)     "Compute the gap in transport activity"
QSpecificFuelCons(allCy,TRANSE,TTECH,TEA,EF,YTIME)  "Compute Specific Fuel Consumption"
QTranspCostPerMeanConsSize(allCy,TRANSE,RCon,TTECH,TEA,YTIME)   "Compute transportation cost per mean and consumer size in KEuro per
```

```
vehicle"
QTechSortVarCost(allCy,TRANSE,Rcon,YTIME)                    "Compute technology sorting based on variable cost"
QTechSortVarCostNewEquip(allCy,TRANSE,EF,TEA,YTIME)          "Compute technology sorting based on variable cost and new equipment"
QTranspCostPerVeh(allCy,TRANSE,RCon,TTECH,TEA,YTIME)         "Compute transportation cost per mean and consumer size in KEuro per vehicle"
QTranspCostMatFac(allCy,TRANSE,RCon,TTECH,TEA,YTIME)         "Compute transportation cost including maturity factor"
QConsEachTechTransp(allCy,TRANSE,TTECH,EF,TEA,YTIME)         "Compute consumption of each technology in transport sectors"
QFinEneDemTranspPerFuel(allCy,TRANSE,EF,YTIME)               "Compute final energy demand in transport per fuel"
QFinEneDemTransp(allCy,TRANSE,YTIME)                         "Compute final energy demand in transport"
```

*** INDUSTRY - DOMESTIC - NON ENERGY USES - BUNKERS EQUATIONS

```
QDemSub(allCy,DSBS,YTIME)                    "Compute total final demand (of substitutable fuels) per subsector"
QElecConsInd(allCy,YTIME)                    "Compute Consumption of electricity in industrial sectors"
QDemInd(allCy,YTIME)                         "Copmpute total final demand (of substitutable fuels) in industrial sectors"
QElecIndPrices(allCy,YTIME)                  "Compute electricity industry prices"
QElecConsHeatPla(allCy,DSBS,YTIME)           "Compute electricity consumed in heatpump plants"
QFuelCons(allCy,DSBS,EF,YTIME)               "Compute fuel consumption"
QElecIndPricesEst(allCy,YTIME)               "Compute Electricity index - a function of industry price - Estimate"
QFuePriSubChp(allCy,DSBS,EF,TEA,YTIME)       "Compute fuel prices per subsector and fuel especially for chp plants"
QElecProdCosChp(allCy,DSBS,CHP,YTIME)        "Compute electricity production cost per CHP plant and demand sector"
QTechCost(allCy,DSBS,rCon,EF,TEA,YTIME)      "Compute technology cost"
QTechCostIntrm(allCy,DSBS,rCon,EF,TEA,YTIME) "Compute intermediate technology cost"
QTechCostMatr(allCy,DSBS,rCon,EF,TEA,YTIME)  "Compute the technology cost including the maturity factor per technology and subsector"
QTechSort(allCy,DSBS,rCon,YTIME)             "Compute Technology
```

```
sorting based on variable cost"
QGapFinalDem(allCy,DSBS,YTIME)                "Compute the gap in final demand of industry, tertiary, non-energy uses and bunkers"
QTechShareNewEquip(allCy,DSBS,EF,TEA,YTIME)   "Compute technology share in new equipment"
QFuelConsInclHP(allCy,DSBS,EF,YTIME)          "Equation for fuel consumption in Mtoe (including heat from heatpumps)"
QVarProCostPerCHPDem(allCy,DSBS,CHP,YTIME)    "Compute  variable including fuel electricity production cost per CHP plant and demand sector "
QAvgElcProCostCHP(allCy,CHP,YTIME)            "Compute Average Electricity production cost per CHP plant"
QAvgVarElecProd(allCy,CHP,YTIME)              "Compute Average variable including fuel electricity production cost per CHP plant"
QElecConsNonSub(allCy,INDDOM,YTIME)           "Compute non-substituable electricity consumption in Industry and Tertiary"
QConsOfRemSubEquip(allCy,DSBS,EF,YTIME)       "Equation for consumption of remaining substitutble equipment"
```

*** REST OF ENERGY BALANCE SECTORS EQUATIONS

```
QTotFinEneCons(allCy,EFS,YTIME)               "Compute total final energy consumption"
QTransfOutputPatFuel(allCy,EFS,YTIME)         "Compute the transfomration output from patent fuel and briquetting plants,coke-oven plants,blast furnace plants and gas works"
QTotFinEneConsAll(YTIME)                      "Compute total final energy consumption in ALL countries"
QFinNonEneCons(allCy,EFS,YTIME)               "Compute final non-energy consumption"
QDistrLosses(allCy,EFS,YTIME)                 "Compute distribution losses"
QTranfOutputDHPlants(allCy,EFS,YTIME)         "Compute the transformation output from district heating plants"
QTransfInputDHPlants(allCy,EFS,YTIME)         "Compute the transformation input to distrcit heating plants"
QTransfInputPatFuel(allCy,EFS,YTIME)          "Compute the transfomration input to patent fuel and briquetting plants,coke-oven plants,blast furnace plants and gas works"
QRefCapacity(allCy,YTIME)                     "Compute refineries capacity"
QTranfOutputRefineries(allCy,EFS,YTIME)       "Compute the transformation output from refineries"
```

```
QTransfInputRefineries(allCy,EFS,YTIME)      "Compute the transformation input to refineries"
QTransfOutputNuclear(allCy,EFS,YTIME)        "Compute transformation output from nuclear plants"
QTransfInNuclear(allCy,EFS,YTIME)            "Compute transformation input to nuclear plants"
QTransfInPowerPls(allCy,EFS,YTIME)           "Compute transformation input to power plants"
QTransfOutThermPP(allCy,EFS,YTIME)           "Compute transformation output from thermal power plants"
QTotTransfInput(allCy,EFS,YTIME)             "Compute total transformation input"
QTotTransfOutput(allCy,EFS,YTIME)            "Compute total transformation output"
QTransfers(allCy,EFS,YTIME)                  "Compute transfers"
QGrsInlConsNotEneBranch(allCy,EFS,YTIME)     "Compute gross inland consumption not including consumption of energy branch"
QGrssInCons(allCy,EFS,YTIME)                 "Compute gross inland consumption"
QPrimProd(allCy,EFS,YTIME)                   "Compute primary production"
QFakeExp(allCy,EFS,YTIME)                    "Compute fake exports"
QFakeImprts(allCy,EFS,YTIME)                 "Compute fake imports"
QNetImports(allCy,EFS,YTIME)                 "Compute net imports"
QEneBrnchEneCons(allCy,EFS,YTIME)            "Compute energy branch final consumption"
```

*** CO2 SEQUESTRATION COST CURVES EQUATIONS

```
QCO2ElcHrg(allCy,YTIME)                      "Compute CO2 captured by electricity and hydrogen production plants (Mtn CO2)"
QCumCO2Capt(allCy,YTIME)                     "Compute cumulative CO2 captured (Mtn of CO2)"
QWghtTrnstLinToExpo(allCy,YTIME)             "Transtition weight for shifting from linear to exponential CO2 sequestration cost curve"
QCstCO2SeqCsts(allCy,YTIME)                  "Compute cost curve for CO2 sequestration costs"
```

*** Emissions Constraints Equations

```
QTotCo2AllCoun(YTIME)                       "Compute total CO2eq GHG emissions in all countries"
QTotGhgEmisAllCountrNap(NAP,YTIME)          "Compute total CO2eq GHG
```

```
    emissions in all countries per NAP sector"
QHouseExpEne(allCy,YTIME)                    "Compute households
    expenditures on energy"
QFuelPriSubSepCarbVal(allCy,SBS,EF,YTIME)  "Compute fuel prices per
    subsector and fuel, separate carbon value in each sector"
```

*** Prices

```
QFuelPriSepCarbon(allCy,SBS,EF,YTIME)   "Compute fuel prices per
    subsector and fuel, separate carbon value in each sector"
QAvgFuelPriSub(allCy,DSBS,YTIME)        "Compute average fuel price per
     subsector"
QElecPriIndResCons(allCy,ESET,YTIME)    "Compute electricity price in
    Industrial and Residential Consumers"
```

*** Miscellaneous

```
qDummyObj                                   "Define dummy objective
    function"
;
Variables
```

*** Power Generation Variables

```
VElecCapChpPla(allCy,CHP,YTIME)           "Capacity of CHP Plants (GW)"
VLoadCurveConstr(allCy,YTIME)             "Parameter for load curve
    construction (1)"
VEstBaseLoad(allCy,YTIME)                 "Estimated base load (GW)"
VElecDem(allCy,YTIME)                     "Total electricity demand (
    TWh)"
VCapChpPlants(allCy,YTIME)                "Electricity load factor for
    entire domestic system"
VElecPeakLoad(allCy,YTIME)                "Electricity peak load (GW)"
VBslMaxmLoad(allCy,YTIME)                 "Baseload corresponding to
    Maximum Load Factor (1)"
VHourProdCostTechAfterCCS(allCy,PGALL,HOUR,YTIME)"Hourly production
    cost of technology accounting for CCS acceptance (Euro/KWh)"
VResShareGrossElecProd(allCy,YTIME)        "Renewables share in gross
    electricity production for subsdidized renewables"
VLongPowGenCost(allCy,PGALL,ESET,YTIME)    "Long-term average power
    generation cost (Euro2005/kWh)"
VCurrRenPot(allCy,PGRENEF,YTIME)           "Current renewable potential
     (GW)"
```

```
VReqElecProd(allCy,YTIME)                    "Total required electricity production (TWh)"
VOverallCap(allCy,PGALL,YTIME)               "Overall Capacity (MW)"
VVarCostTechNotPGSCRN(allCy,PGALL,YTIME)  "Variable cost of technology excluding PGSCRN (Euro/KWh)"
VAvgPowerGenCostShoTrm(allCy,ESET,YTIME)  "Short-term average power generation cost (Euro2005/kWh)"
VLongAvgPowGenCost(allCy,ESET,YTIME)      "Long-term average power generation cost (Euro2005/kWh)"
VLongPowGenIntPri(allCy,PGALL,ESET,YTIME) "Long term power generation cost of technologies including international prices of main fuels (kEuro05/toe)"
VRenShareElecProdSub(allCy,YTIME)         "Renewables share in gross electricity production for subsdidized renewables (1)"
VElecPriIndResNoCliPol(allCy,ESET,YTIME)  "Electricity price to Industrial and Residential Consumers (Euro2005/KWh)"
VLonPowGenCostNoClimPol(allCy,ESET,YTIME) "Long-term average power generation cost  excluding climate policies (Euro2005/kWh)"
VLonAvgPowGenCostNoClimPol(allCy,PGALL,ESET,YTIME)  "Long-term average power generation cost excluding climate policies(Euro2005/kWh)"
VShoPowGenIntPri(allCy,PGALL,ESET,YTIME)  "Short term power generation cost of technologies including international Prices of main fuels (kEuro05/toe)"
VLonMnmpowGenCost(allCy,PGALL,YTIME)      "Long-term minimum power generation cost (Euro2005/kWh)"
VChpElecProd(allCy,CHP,YTIME)             "CHP electricity production (TWh)"
VScalFacPlaDisp(allCy,HOUR,YTIME)         "Scaling factor for plant dispatching (1)"
VSecContrTotChpProd(allCy,SBS,CHP,YTIME)  "Contribution of each sector in total CHP production (1)"
VPowPlantSorting(allCy,PGALL,YTIME)       "Power plants sorting according to variable cost to decide the plant dispatching (1)"
VElecProdPowGenPlants(allCy,PGALL,YTIME)  "Electricity production (TWh)"
VAvgCapFacRes(allCy,PGALL,YTIME)          "The average capacity factor of RES (1)"
VNewCapYearly(allCy,PGALL,YTIME)          "The new capacity added every year (MW)"
VNonChpElecProd(allCy,YTIME)              "Non CHP total electricity production (TWh)"
VHourProdCostTech(allCy,PGALL,HOUR,YTIME)     "Hourly production cost of technology (Euro/KWh)"
```

```
VElecChpPlants(allCy,YTIME)                "Estimate the electricity of CHP Plants (1)"
VProdCostTechnology(allCy,PGALL,YTIME)     "Production cost of technology (Euro/KWh)"
VElecCapChpPla(allCy,CHP,YTIME)            "Capacity of CHP Plants (GW)"
VScalWeibullSum(allCy,PGALL,YTIME)         "Sum (over hours) of temporary variable facilitating the scaling in Weibull equation (1)"
VElecPeakLoads(allCy,YTIME)                "Electricity peak loads (GW)"
VVarCostTechnology(allCy,PGALL,YTIME)      "Variable cost of technology (Euro/KWh)"
VNewInvDecis(allCy,YTIME)                  "Power plant sorting for new investment decision according to total cost (1)"
VPowPlaShaNewEquip(allCy,PGALL,YTIME)      "Power plant share in new equipment (1)"
VRenTechMatMult(allCy,PGALL,YTIME)         "Renewable technologies maturity multiplier (1)"
VRenPotSupplyCurve(allCy,PGRENEF,YTIME)    "Renewable potential supply curve (1)"
VMaxmAllowRenPotent(allCy,PGRENEF,YTIME)   "Maximum allowed renewable potential (GW)"
VMnmAllowRenPot(allCy,PGRENEF,YTIME)       "Minimum allowed renewable potential (GW)"
VScalWeibull(allCy,PGALL,HOUR,YTIME)       "Temporary variable facilitating the scaling in Weibull equation"
VElecGenPlantsCapac(allCy,PGALL,YTIME)     "Electricity generation plants capacity (GW)"
VGapPowerGenCap(allCy,YTIME)               "Gap in total generation capacity to be filled by new equipment (GW)"
VProdCostTechPreReplac(allCy,PGALL,YTIME)  "Production cost of technology used in premature replacement (Euro/KWh)"
VEndogScrapIndex(allCy,PGALL,YTIME)        "Index used for endogenous power plants scrapping (1)"
VSensCcs(allCy,YTIME)                      "Variable that controlls the sensitivity of CCS acceptance (1)"
VCorrBaseLoad(allCy,YTIME)                 "Corrected base load (GW)"
VElecGenPlanCap(allCy,PGALL,YTIME)         "Electricity generation plants capacity (GW)"
VElecGenNoChp(allCy,YTIME)                 "Total electricity generation capacity excluding CHP (GW)"
VVarCostTech(allCy,PGALL,YTIME)            "Variable cost of technology (Euro/KWh)"
VPowerPlantNewEq(allCy,PGALL,YTIME)        "Power plant share in new equipment (1)"
```

```
VPowerPlaShrNewEq(allCy,PGALL,YTIME)      "Power plant share in new equipment (1)"
VHourProdCostTechNoCCS(allCy,PGALL,HOUR,YTIME) "Hourly production cost of technology (Euro/KWh)"
VProdCostTechPreReplacAvail(allCy,PGALL,PGALL2,YTIME)   "Production cost of technology used in premature replacement including plant availability rate (Euro/KWh)"
VTotReqElecProd(allCy,YTIME)          "Total required electricity production (TWh)"
VTotElecGenCapEst(allCy,YTIME)        "Estimated Total electricity generation capacity (GW)"
VTotElecGenCap(allCy,YTIME)           "Total electricity generation capacity (GW)"
VFeCons(allCy,EF,YTIME)               "Total final energy consumnption (Mtoe)"
VFNonEnCons(allCy,EFS,YTIME)          "Final non energy consumption (Mtoe)"
VLosses(allCy,EFS,YTIME)              "Distribution losses (Mtoe)"
VEnCons(allCy,EFS,YTIME)              "Final consumption in energy sector (Mtoe)"
VNetImp(allCy,EFS,YTIME)              "Net imports (Mtoe)"
VMExtV(allCy,YTIME)                   "Passenger cars market extension (GDP dependent)"
VMExtF(allCy,YTIME)                   "Passenger cars market extension (GDP independent)"
VLamda(allCy,YTIME)                   "Ratio of car ownership over saturation car ownership (1)"
VNumVeh(allCy,YTIME)                  "Stock of passenger cars (million vehicles)"
VNewReg(allCy,YTIME)                  "Passenger cars new registrations (million vehicles)"
VScrap(allCy,YTIME)                   "Scrapped passenger cars (million vehicles)"
VTrnspActiv(allCy,TRANSE,YTIME)       "Passenger transport acitivity (1)"
                                         !! - Activity for passenger cars is measured in (000)km
                                         !! - Activity for all other passenger transportation modes is measured in Gpkm
VFuelPriceAvg(allCy,DSBS,YTIME)           "Average fuel prices per subsector (k$2015/toe)"
VScrRate(allCy,YTIME)                  "Scrapping rate of passenger
```

```
cars (1)"
VElecConsAll(allCy,DSBS,YTIME)         "Electricity demand per final sector (Mtoe)"
VConsFuel(allCy,DSBS,EF,YTIME)         "Consumption of fuels in each demand subsector, excluding heat from heatpumps (Mtoe)"
VDemTr(allCy,TRANSE,EF,YTIME)          "Final energy demand in transport subsectors per fuel (Mtoe)"
VLifeTimeTech(allCy,DSBS,EF,TEA,YTIME) "Lifetime of technologies (years)"
```

*** Transport Variables

```
VGoodsTranspActiv(allCy,TRANSE,YTIME)       "Goods transport acitivity (Gtkm)"
VGapTranspFillNewTech(allCy,TRANSE,YTIME)   "Gap in transport activity to be filled by new technologies ()"
                                            !! Gap for passenger cars (million vehicles)
                                            !! Gap for all other passenger transportation modes (Gpkm)
                                            !! Gap for all goods transport is measured (Gtkm)
VSpecificFuelCons(allCy,TRANSE,TTECH,TEA,EF,YTIME)  "Specific Fuel Consumption ()"
                                                    !! SFC for passenger cars (ktoe/Gkm)
                                                    !! SFC for other passsenger transportation modes (ktoe/Gpkm)
                                                    !! SFC for trucks is measured (ktoe/Gtkm)
VTranspCostPermeanConsSize(allCy,TRANSE,RCon,TTECH,TEA,YTIME)   "Transportation cost per mean and consumer size (Keuro/vehicle)"
VTranspCostPerVeh(allCy,TRANSE,RCon,TTECH,TEA,YTIME)    "Transportation cost per mean and consumer size (Keuro/vehicle)"
VTranspCostMatFac(allCy,TRANSE,RCon,TTECH,TEA,YTIME)    "Transportation cost including maturity factor (Keuro/vehicle)"
```

```
VTechSortVarCost(allCy,TRANSE,Rcon,YTIME)                "Technology sorting based on variable cost (1)"
VTechSortVarCostNewEquip(allCy,TRANSE,EF,TEA,YTIME)      "Technology share in new equipment (1)"
VConsEachTechTransp(allCy,TRANSE,TTECH,EF,TEA,YTIME)     "Consumption of each technology and subsector (Mtoe)"
VFinEneDemTranspSub(allCy,TRANSE,YTIME)                  "Final energy demand in transport subsectors (Mtoe)"
```

*** INDUSTRY - DOMESTIC - NON ENERGY USES - BUNKERS VARIABLES

```
VElecNonSub(allCy,DSBS,YTIME)           "Consumption of non-substituable electricity in Industry and Tertiary (Mtoe)"
VElecConsInd(allCy,YTIME)               "Total Consumption of Electricity in industrial sectors (Mtoe)"
VDemInd(allCy,YTIME)                    "Total final demand (of substitutable fuels) in industrial sectors (Mtoe)"
VDemSub(allCy,DSBS,YTIME)               "Total final demand (of substitutable fuels) per subsector (Mtoe)"
VElecIndPrices(allCy,YTIME)             "Electricity index - a function of industry price (1)"
VElecConsHeatPla(allCy,DSBS,YTIME)      "Electricity consumed in heatpump plants (Mtoe)"
VConsFuelSub(allCy,DSBS,EF,YTIME)       "Consumption of fuels in each demand subsector (including heat from heatpumps) (Mtoe)"
VElecIndPricesEst(allCy,YTIME)          "Electricity index - a function of industry price - Estimate (1)"
VFuelPriceSub(allCy,SBS,EF,YTIME)       "Fuel prices per subsector and fuel (k$2015/toe)"
VFuePriSubChp(allCy,DSBS,EF,TEA,YTIME) "Fuel prices per subsector and fuel for CHP plants (kUS$2005/toe)"
VTechCostVar(allCy,SBS,EF,TEA,YTIME)    "Variable Cost of technology (various)"
                                         !! - For transport (kEuro05/vehicle)
                                         !! - For Industrial sectors except Iron and Steel (Euro05/toe-year)
                                         !! - For Iron and Steel (Euro05/tn-of-steel)
                                         !! - For Domestic sectors (Euro05/toe-year)
VElecProdCostChp(allCy,DSBS,CHP,YTIME)                   "Electricity
```

```
production cost per CHP plant and demand sector (Euro/KWh)"
VTechCost(allCy,DSBS,rCon,EF,TEA,YTIME)                    "Technology cost (Keuro/toe)"
VTechCostIntrm(allCy,DSBS,rcon,EF,TEA,YTIME)               "Intermediate technology cost (Keuro/toe)"
VTechCostMatr(allCy,DSBS,rCon,EF,TEA,YTIME)                "Technology cost including maturity factor (Keuro/toe)"
VTechSort(allCy,DSBS,rCon,YTIME)                           "Technology sorting based on variable cost (1)"
VConsRemSubEquip(allCy,DSBS,EF,YTIME)                      "Consumption of remaining substitutable equipment (Mtoe)"
VGapFinalDem(allCy,DSBS,YTIME)                             "Final demand gap to be filed by new technologies (Mtoe)"
VTechShareNewEquip(allCy,DSBS,EF,TEA,YTIME)                "Technology share in new equipment (1)"
VFuelConsInclHP(allCy,DSBS,EF,YTIME)                       "Consumption of fuels in each demand subsector including heat from heatpumps (Mtoe)"
VProCostCHPDem(allCy,DSBS,CHP,YTIME)                       "Variable including fuel electricity production cost per CHP plant and demand sector (Euro/KWh)"
VAvgElcProCHP(allCy,CHP,YTIME)                             "Average Electricity production cost per CHP plant (Euro/KWh)"
VAvgVarProdCostCHP(allCy,CHP,YTIME)                        "Average variable including fuel electricity production cost per CHP plant (Euro/KWh)"
```

*** REST OF ENERGY BALANCE SECTORS VARIABLES

```
VPlantEffPlantType(allCy,PGALL,YTIME)                      "Plant efficiency per plant type (1)"
VTotFinEneConsAll(YTIME)                                   "Total final energy Consumption in ALL COUNTRIES (Mtoe)"
VDistrLosses(allCy,EFS,YTIME)                              "Distribution losses (Mtoe)"
VTransfOutputDHPlants(allCy,EFS,YTIME)                     "Transformation output from District Heating Plants (Mtoe)"
VTransfInputDHPlants(allCy,EFS,YTIME)                      "Transformation input to District Heating Plants (Mtoe)"
VTransfInputPatFuel(allCy,EFS,YTIME)                       "Transformation input to patent fuel and briquetting plants,coke-oven plants,blast furnace plants and gas works (1)"
VTransfOutputPatFuel(allCy,EFS,YTIME)                      "Transformation
```

```
 input to patent fuel and briquetting plants,coke-oven plants,blast
    furnace plants and gas works (1)"
VRefCapacity(allCy,YTIME)                                "Refineries
    capacity (Million barrels/day)"
VTransfOutputRefineries(allCy,EFS,YTIME)                 "Transformation
     output from refineries (Mtoe)"
VTransfInputRefineries(allCy,EFS,YTIME)                  "Transformation
     input to refineries (Mtoe)"
VTransfOutputNuclear(allCy,EFS,YTIME)                    "Transformation
     output from nuclear plants (Mtoe)"
VElecProd(allCy,PGALL,YTIME)                             "Electricity
    production (TWh)"
VTransfInNuclear(allCy,EFS,YTIME)                        "Transformation
     input to nuclear plants (Mtoe)"
VTransfInThermPowPls(allCy,EFS,YTIME)                    "Transformation
     input to thermal power plants (Mtoe)"
VChpElecProd(allCy,CHP,YTIME)                            "CHP
    electricity production (TWh)"
VTransfOutThermPowSta(allCy,EFS,YTIME)                   "Transformation
     output from thermal power stations (Mtoe)"
VTotTransfInput(allCy,EFS,YTIME)                         "Total
    transformation input (Mtoe)"
VTotTransfOutput(allCy,EFS,YTIME)                        "Total
    transformation output (Mtoe)"
VTransfers(allCy,EFS,YTIME)                              "Transfers (
    Mtoe)"
VGrsInlConsNotEneBranch(allCy,EFS,YTIME)                 "Gross Inland
    Consumption not including consumption of energy branch (Mtoe)"
VGrssInCons(allCy,EFS,YTIME)                             "Gross Inland
    Consumption (Mtoe)"
VPrimProd(allCy,EFS,YTIME)                               "Primary
    Production (Mtoe)"
VExportsFake(allCy,EFS,YTIME)                            "Exports fake (
    Mtoe)"
VFkImpAllFuelsNotNatGas(allCy,EFS,YTIME)                 "Fake Imports
    for all fuels except natural gas (Mtoe)"
VNetImports(allCy,EFS,YTIME)                             "Net Imports (
    Mtoe)"
```

*** Prices Variables

```
VElecPriInduResConsu(allCy,ESET,YTIME)                   "Electricity
    price to Industrial and Residential Consumers (Euro2005/KWh)"
```

```
VFuelPriMultWgt(allCy,SBS,EF,YTIME)                    "Fuel prices per subsector and fuel  mutliplied by weights (kdollars2005/toe)"
VAvgPowerGenLongTrm(allCy,ESET,YTIME)                  "Long-term average power generation cost (Euro2005/kWh)"
VFuelPriSubNoCarb(allCy,SBS,EF,YTIME)                  "Fuel prices per subsector and fuel  without carbon value (kdollars2005/toe)"
```

*** CO2 SEQUESTRATION COST CURVES VARIABLES

```
VCO2ElcHrgProd(allCy,YTIME)                            "CO2 captured by electricity and hydrogen production plants (Mtn CO2)"
VCumCO2Capt(allCy,YTIME)                               "Cumulative CO2 captured (Mtn CO2)"
VWghtTrnstLnrToExpo(allCy,YTIME)                       "Weight for transtition from linear CO2 sequestration cost curve to exponential (1)"
VCO2CO2SeqCsts(allCy,YTIME)                            "Cost curve for CO2 sequestration costs (Euro/tn of CO2 sequestrated)"
```

*** Emissions Constraints Variables

```
VTotCo2AllCoun(YTIME)                                  "Total CO2eq GHG emissions in all countries (1)"
VTotGhgEmisAllCountrNap(NAP,YTIME)                     "Total CO2eq GHG emissions in all countries per NAP sector (1)"
VHouseExpEne(allCy,YTIME)                              "Households expenditures on energy (billions)"
```

*** Miscellaneous

```
vDummyObj                                              "Dummy maximisation variable (1)"
;
Positive Variables
VCarVal(allCy,NAP,YTIME)                               "Carbon value for all countries (Euro2005/tn CO2)"
VRenValue(YTIME)                                       "Renewable value (Euro2005/KWh)"
;
Scalars
sTWhToMtoe          "TWh to Mtoe conversion factor" /0.086/
sElecToSteRatioChp "Technical maximum of electricity to steam ratio in CHP plants" /1.15/
```

```
sGwToTwhPerYear     "convert GW mean power into TWh/y" /8.76/
s                   "time step iterator" /0/
sSolverTryMax       "maximum attempts to solve each time step" /%
    SolverTryMax%/
sModelStat          "helper parameter for solver status"
sExogCarbValue      "switch for exogenous carbon value (0=exogenous 1=
    endogenous)" /1/
;
```

## Equations (equations.gms)

### Description

Declaration of variables and equation.

- Power Generation
- Compute current renewable potential

```
QCurrRenPot(runCy,PGRENEF,YTIME)$TIME(YTIME)..
         VCurrRenPot(runCy,PGRENEF,YTIME)
         =E=
         ( VMaxmAllowRenPotent(runCy,PGRENEF,YTIME) + iMinRenPotential(runCy,PGRENEF,YTIME))/2;
```

- Compute CHP electric capacity

```
QChpElecPlants(runCy,CHP,YTIME)$TIME(YTIME)..
         VElecCapChpPla(runCy,CHP,YTIME)
         =E=
         1/sTWhToMtoe * sum(INDDOM,VConsFuel(runCy,INDDOM,CHP,YTIME)) * VElecIndPrices(runCy,YTIME)/
         sum(PGALL$CHPtoEON(CHP,PGALL),iAvailRate(PGALL,YTIME)) /
         iUtilRateChpPlants(runCy,CHP,YTIME) /sGwToTwhPerYear;
```

- Compute Lambda parameter

```
QLambda(runCy,YTIME)$TIME(YTIME)..
         (1 - exp( -VLoadCurveConstr(runCy,YTIME)*sGwToTwhPerYear))  / VLoadCurveConstr(runCy,YTIME)
             =E=
         (VElecDem(runCy,YTIME) - sGwToTwhPerYear*VCorrBaseLoad(runCy,YTIME))
         / (VElecPeakLoad(runCy,YTIME) - VCorrBaseLoad(runCy,YTIME));
```

- Compute electricity final demand

```
QElecDem(runCy,YTIME)$TIME(YTIME)..
         VElecDem(runCy,YTIME)
             =E=
         1/sTWhToMtoe *
         ( VFeCons(runCy,"ELC",YTIME) + VFNonEnCons(runCy,"ELC",YTIME) + VLosses(runCy,"ELC",YTIME)
           + VEnCons(runCy,"ELC",YTIME) - VNetImp(runCy,"ELC",YTIME)
```

```
        );
```

- Compute estimated base load

```
QEstBaseLoad(runCy,YTIME)$TIME(YTIME)..
        VEstBaseLoad(runCy,YTIME)
            =E=
        (
            sum(DSBS, iBaseLoadShareDem(runCy,DSBS,YTIME)*VElecConsAll(runCy,DSBS,YTIME))*(1+iRateLossesFinCons(runCy,"ELC",YTIME))*
            (1 - VNetImports(runCy,"ELC",YTIME)/(sum(DSBS, VElecConsAll(runCy,DSBS,YTIME))+VLosses(runCy,"ELC",YTIME)))
            + 0.5*VEnCons(runCy,"ELC",YTIME)
        ) / sTWhToMtoe / sGwToTwhPerYear;
```

- Compute load factor of entire domestic system

```
QLoadFacDom(runCy,YTIME)$TIME(YTIME)..
        VCapChpPlants(runCy,YTIME)
            =E=
        (sum(INDDOM,VConsFuel(runCy,INDDOM,"ELC",YTIME)) + sum(TRANSE, VDemTr(runCy,TRANSE,"ELC",YTIME)))/
        (sum(INDDOM,VConsFuel(runCy,INDDOM,"ELC",YTIME)/iLoadFacElecDem(runCy,INDDOM,YTIME)) +
        sum(TRANSE, VDemTr(runCy,TRANSE,"ELC",YTIME)/iLoadFacElecDem(runCy,TRANSE,YTIME)));
```

- Compute elerctricity peak load

```
QElecPeakLoad(runCy,YTIME)$TIME(YTIME)..
        VElecPeakLoad(runCy,YTIME)
            =E=
        VElecDem(runCy,YTIME)/(VCapChpPlants(runCy,YTIME)*sGwToTwhPerYear);
```

- Compute baseload corresponding to maximum load

```
QBslMaxmLoad(runCy,YTIME)$TIME(YTIME)..
        (VElecDem(runCy,YTIME)-VBslMaxmLoad(runCy,YTIME)*sGwToTwhPerYear)
            =E=
```

```
        iMxmLoadFacElecDem(runCy,YTIME)*(VElecPeakLoad(runCy,YTIME)-VBslMaxmLoad(runCy,YTIME))*sGwToTwhPerYear;
```

- Compute electricity base load

```
QElecBaseLoad(runCy,YTIME)$TIME(YTIME)..
        VCorrBaseLoad(runCy,YTIME)
            =E=
        (1/(1+Exp(iBslCorrection(runCy,YTIME)*(VEstBaseLoad(runCy,YTIME)-VBslMaxmLoad(runCy,YTIME)))))*VEstBaseLoad(runCy,YTIME)
       +(1-1/(1+Exp(iBslCorrection(runCy,YTIME)*(VEstBaseLoad(runCy,YTIME)-VBslMaxmLoad(runCy,YTIME)))))*VBslMaxmLoad(runCy,YTIME);
```

- Compute total required electricity production

```
QTotReqElecProd(runCy,YTIME)$TIME(YTIME)..
        VTotReqElecProd(runCy,YTIME)
            =E=
        sum(HOUR, (VElecPeakLoad(runCy,YTIME)-VCorrBaseLoad(runCy,YTIME))
                  * exp(-VLoadCurveConstr(runCy,YTIME)*(0.25+(ord(HOUR)-1)))
            ) + 9*VCorrBaseLoad(runCy,YTIME);
```

- Compute Estimated total electricity generation capacity

```
QTotEstElecGenCap(runCy,YTIME)$TIME(YTIME)..
       VTotElecGenCapEst(runCy,YTIME)
            =E=
       iResMargTotAvailCap(runCy,"TOT_CAP_RES",YTIME) * VTotElecGenCap(runCy,YTIME-1)
       * VElecPeakLoad(runCy,YTIME)/VElecPeakLoad(runCy,YTIME-1);
```

- Compute total electricity generation capacity

```
QTotElecGenCap(runCy,YTIME)$TIME(YTIME)..
       VTotElecGenCap(runCy,YTIME)
       =E=
    VTotElecGenCapEst(runCy,YTIME);
```

- Compute hourly production cost used in investment decisions

```
QHourProdCostInv(runCy,PGALL,HOUR,YTIME)$(TIME(YTIME)) ..
         VHourProdCostTech(runCy,PGALL,HOUR,YTIME)
                 =E=

                   ( ( iDisc(runCy,"PG",YTIME) * exp(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL))
                       / (exp(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL)) -1))
                     * iGrossCapCosSubRen(runCy,PGALL,YTIME)* 1E3 * iCGI(runCy,YTIME)  + iFixOandMCost(PGALL,YTIME)
                   )/iAvailRate(PGALL,YTIME) / (1000*(ord(HOUR)-1+0.25))
                   + iVarCost(PGALL,YTIME)/1E3 + (VRenValue(YTIME)*8.6e-5)$( not ( PGREN(PGALL)
                   $(not sameas("PGASHYD",PGALL)) $(not sameas("PGSHYD",PGALL)) $(not sameas("PGLHYD",PGALL)) ))
                   + sum(PGEF$PGALLtoEF(PGALL,PGEF), (VFuelPriceSub(runCy,"PG",PGEF,YTIME)+
                       iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,YTIME)*1e-3*
                   iCo2EmiFac(runCy,"PG",PGEF,YTIME)
                        +(1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)*
                        (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))))
                        *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME))$(not PGREN(PGALL));
```

- Compute hourly production cost used in investment decisions excluding CCS

```
QHourProdCostInvDec(runCy,PGALL,HOUR,YTIME)$(TIME(YTIME) $NOCCS(PGALL)) ..
         VHourProdCostTechNoCCS(runCy,PGALL,HOUR,YTIME) =E=
         VPowerPlantNewEq(runCy,PGALL,YTIME)*VHourProdCostTech(runCy,PGALL,HOUR,YTIME)+
         sum(CCS$CCS_NOCCS(CCS,PGALL), VPowerPlaShrNewEq(runCy,CCS,YTIME)*VHourProdCostTech(runCy,CCS,HOUR,YTIME));
```

- Compute gamma parameter used in CCS/No CCS decision tree

```
QGammaInCcsDecTree(runCy,YTIME)$TIME(YTIME)..
```

```
        VSensCcs(runCy,YTIME) =E= 20+25*EXP(-0.06*((sum(
           NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME-1)))));
```

- Compute hourly production cost used in investment decisions taking into account CCS acceptance

```
QHourProdCostInvDecisionsAfterCCS(runCy,PGALL,HOUR,YTIME)$(TIME(YTIME)
   $(CCS(PGALL) or NOCCS(PGALL))) ..
        VHourProdCostTechAfterCCS(runCy,PGALL,HOUR,YTIME)
        =E=
         VHourProdCostTech(runCy,PGALL,HOUR,YTIME)**(-VSensCcs(runCy,
            YTIME));
```

- Compute production cost used in investment decisions

```
QProdCostInvDecis(runCy,PGALL,YTIME)$(TIME(YTIME) $(CCS(PGALL) or NOCCS
   (PGALL)) ) ..
        VProdCostTechnology(runCy,PGALL,YTIME)
        =E=
        sum(HOUR,VHourProdCostTech(runCy,PGALL,HOUR,YTIME)**(-VSensCcs
           (runCy,YTIME))) ;
```

- Compute SHRCAP

```
QShrcap(runCy,PGALL,YTIME)$(TIME(YTIME) $CCS(PGALL))..
        VPowerPlaShrNewEq(runCy,PGALL,YTIME) =E=
        1.1 *VProdCostTechnology(runCy,PGALL,YTIME)
        /(1.1*VProdCostTechnology(runCy,PGALL,YTIME)
          + sum(PGALL2$CCS_NOCCS(PGALL,PGALL2),VProdCostTechnology(
             runCy,PGALL2,YTIME))
          );
```

- Compute SHRCAP excluding CCs

```
QShrcapNoCcs(runCy,PGALL,YTIME)$(TIME(YTIME) $NOCCS(PGALL))..
        VPowerPlantNewEq(runCy,PGALL,YTIME)
        =E=
        1 - sum(CCS$CCS_NOCCS(CCS,PGALL), VPowerPlaShrNewEq(runCy,CCS,
           YTIME));
```

- Compute variable cost of technology

```
QVarCostTech(runCy,PGALL,YTIME)$(time(YTIME))..
```

```
        VVarCostTech(runCy,PGALL,YTIME)
            =E=
        (iVarCost(PGALL,YTIME)/1E3 + sum(PGEF$PGALLtoEF(PGALL,PGEF), (VFuelPriceSub(runCy,"PG",PGEF,YTIME)/1.2441+
        iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)
        + (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)
         *(sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))))
         *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME))$(not PGREN(PGALL)));
```

- Compute variable cost of technology excluding PGSCRN

```
QVarCostTechNotPGSCRN(runCy,PGALL,YTIME)$(time(YTIME) $(not PGSCRN(PGALL))).
        VVarCostTechNotPGSCRN(runCy,PGALL,YTIME)
            =E=
         VVarCostTech(runCy,PGALL,YTIME)**(-5);
```

- Compute production cost of technology used in premature replacement

```
QProdCostTechPreReplac(runCy,PGALL,YTIME)$TIME(YTIME)..
        VProdCostTechPreReplac(runCy,PGALL,YTIME) =e=
                        (
                          ((iDisc(runCy,"PG",YTIME) * exp(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL))/
                          (exp(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL)) -1))
                            * iInvCost(PGALL,YTIME)* 1E3 * iCGI(runCy,YTIME)  +
                            iFixOandMCost(PGALL,YTIME))/(8760*iAvailRate(PGALL,YTIME))
                           + (iVarCost(PGALL,YTIME)/1E3 + sum(PGEF$PGALLtoEF(PGALL,PGEF),
                           (VFuelPriceSub(runCy,"PG",PGEF,YTIME)+
                            iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME) +
                             (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)*
                          (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))))
```

```
                                    *sTWhToMtoe/iPlantEffByType(runCy,
                                       PGALL,YTIME))$(not PGREN(PGALL)))
                          );
```

- Compute production cost of technology used in premature replacement including plant availability rate

```
QProdCostTechPreReplacAvail(runCy,PGALL,PGALL2,YTIME)$TIME(YTIME)..
         VProdCostTechPreReplacAvail(runCy,PGALL,PGALL2,YTIME) =E=
         iAvailRate(PGALL,YTIME)/iAvailRate(PGALL2,YTIME)*
            VProdCostTechPreReplac(runCy,PGALL,YTIME)+
         VVarCostTech(runCy,PGALL,YTIME)*(1-iAvailRate(PGALL,YTIME)/
            iAvailRate(PGALL2,YTIME));
```

- Compute endogenous scrapping index

```
QEndogScrapIndex(runCy,PGALL,YTIME)$(TIME(YTIME) $(not PGSCRN(PGALL)))
   ..
         VEndogScrapIndex(runCy,PGALL,YTIME)
                 =E=
         VVarCostTechNotPGSCRN(runCy,PGALL,YTIME)/
         (VVarCostTechNotPGSCRN(runCy,PGALL,YTIME)+(iScaleEndogScrap(
            runCy,PGALL,YTIME)*
         sum(PGALL2,VProdCostTechPreReplacAvail(runCy,PGALL,PGALL2,
            YTIME)))**(-5));
```

- Compute total electricity generation capacity excluding CHP plants

```
QElecGenNoChp(runCy,YTIME)$TIME(YTIME)..
         VElecGenNoChp(runCy,YTIME)
          =E=
VTotElecGenCap(runCy,YTIME) - SUM(CHP,VElecCapChpPla(runCy,CHP,YTIME)
   *0.85);
```

- Compute the gap in power generation capacity

```
QGapPowerGenCap(runCy,YTIME)$TIME(YTIME)..
         VGapPowerGenCap(runCy,YTIME)
             =E=
 (        (  VElecGenNoChp(runCy,YTIME) - VElecGenNoChp(runCy,YTIME-1)
    + sum(PGALL,VElecGenPlanCap(runCy,PGALL,YTIME-1) *
 (1 - VEndogScrapIndex(runCy,PGALL,YTIME))) +
```

```
          sum(PGALL, (iPlantDecomSched(runCy,PGALL,YTIME)-iDecInvPlantSched(runCy,PGALL,YTIME))*iAvailRate(PGALL,YTIME))
          + Sum(PGALL$PGSCRN(PGALL), (VElecGenPlantsCapac(runCy,PGALL,YTIME-1)-iPlantDecomSched(runCy,PGALL,YTIME))/
          iTechLftPlaType(runCy,PGALL))
        )
   + 0 + SQRT( SQR(       (  VElecGenNoChp(runCy,YTIME) - VElecGenNoChp(runCy,YTIME-1) +
        sum(PGALL,VElecGenPlanCap(runCy,PGALL,YTIME-1) * (1 - VEndogScrapIndex(runCy,PGALL,YTIME))) +
          sum(PGALL, (iPlantDecomSched(runCy,PGALL,YTIME)-iDecInvPlantSched(runCy,PGALL,YTIME))*iAvailRate(PGALL,YTIME))
          + Sum(PGALL$PGSCRN(PGALL), (VElecGenPlantsCapac(runCy,PGALL,YTIME-1)-iPlantDecomSched(runCy,PGALL,YTIME))/
          iTechLftPlaType(runCy,PGALL))
        ) -0) + SQR(1e-10) ) )/2;
```

- Compute temporary variable facilitating the scaling in Weibull equation

```
QScalWeibull(runCy,PGALL,HOUR,YTIME)$((not CCS(PGALL))$TIME(YTIME))..
          VScalWeibull(runCy,PGALL,HOUR,YTIME)
         =E=
         (VHourProdCostTech(runCy,PGALL,HOUR,YTIME)$(not NOCCS(PGALL))
         +
          VHourProdCostTechNoCCS(runCy,PGALL,HOUR,YTIME)$NOCCS(PGALL))**(-6);
```

- Compute renewable potential supply curve

```
QRenPotSupplyCurve(runCy,PGRENEF,YTIME)$TIME(YTIME)..
         VRenPotSupplyCurve(runCy,PGRENEF,YTIME) =E=
         iMinRenPotential(runCy,PGRENEF,YTIME) +(VCarVal(runCy,"Trade",YTIME))/(70)*
         (iMaxRenPotential(runCy,PGRENEF,YTIME)-iMinRenPotential(runCy,PGRENEF,YTIME));
```

- Compute maximum allowed renewable potential

```
QMaxmAllowRenPotent(runCy,PGRENEF,YTIME)$TIME(YTIME)..
         VMaxmAllowRenPotent(runCy,PGRENEF,YTIME) =E=
```

```
        ( VRenPotSupplyCurve(runCy,PGRENEF,YTIME)+ iMaxRenPotential(runCy,PGRENEF,YTIME))/2;
```

- Compute minimum allowed renewable potential

```
QMnmAllowRenPot(runCy,PGRENEF,YTIME)$TIME(YTIME)..
        VMnmAllowRenPot(runCy,PGRENEF,YTIME) =E=
        ( VRenPotSupplyCurve(runCy,PGRENEF,YTIME) + VMnmAllowRenPot(runCy,PGRENEF,YTIME))/2;
```

- Compute renewable technologies maturity multiplier

```
QRenTechMatMult(runCy,PGALL,YTIME)$TIME(YTIME)..
        VRenTechMatMult(runCy,PGALL,YTIME)
         =E=
        1$(NOT PGREN(PGALL))
        +
        (
          1/(1+EXP(9*(
                sum(PGRENEF$PGALLtoPGRENEF(PGALL,PGRENEF),
                sum(PGALL2$(PGALLtoPGRENEF(PGALL2,PGRENEF) $PGREN(PGALL2)),
                VElecGenPlanCap(runCy,PGALL2,YTIME-1))/VRenPotSupplyCurve(runCy,PGRENEF,YTIME))-0.6)))
          )$PGREN(PGALL);
```

- Compute temporary variable facilitating the scaling in Weibull equation

```
QScalWeibullSum(runCy,PGALL,YTIME)$((not CCS(PGALL)) $TIME(YTIME))..
        VScalWeibullSum(runCy,PGALL,YTIME)
        =E=
             iMatFacPlaAvailCap(runCy,PGALL,YTIME) * VRenTechMatMult(runCy,PGALL,YTIME)*
             sum(HOUR,
                (VHourProdCostTech(runCy,PGALL,HOUR,YTIME)$(not NOCCS(PGALL))
                +
                VHourProdCostTechNoCCS(runCy,PGALL,HOUR,YTIME)$NOCCS(PGALL)
                )**(-6)
             );
```

- Compute for Power Plant new investment decision

```
QNewInvDecis(runCy,YTIME)$TIME(YTIME)..
         VNewInvDecis(runCy,YTIME)
             =E=
         sum(PGALL$(not CCS(PGALL)),VScalWeibullSum(runCy,PGALL,YTIME))
             ;
```

- Compute the power plant share in new equipment

```
QPowPlaShaNewEquip(runCy,PGALL,YTIME)$(TIME(YTIME)) ..
        VPowPlaShaNewEquip(runCy,PGALL,YTIME)
             =E=
         ( VScalWeibullSum(runCy,PGALL,YTIME)/ VNewInvDecis(runCy,YTIME
            ))$(not CCS(PGALL))
          +
          sum(NOCCS$CCS_NOCCS(PGALL,NOCCS),VPowPlaShaNewEquip(runCy,
             NOCCS,YTIME))$CCS(PGALL);
```

- Compute electricity generation capacity

```
QElecGenCapacity(runCy,PGALL,YTIME)$TIME(YTIME)..
         VElecGenPlantsCapac(runCy,PGALL,YTIME)
             =E=
         (VElecGenPlanCap(runCy,PGALL,YTIME-1)*VEndogScrapIndex(runCy,
            PGALL,YTIME-1)
          +(VPowPlaShaNewEquip(runCy,PGALL,YTIME) * VGapPowerGenCap(
             runCy,YTIME))$( (not CCS(PGALL)) AND (not NOCCS(PGALL)))
          +(VPowPlaShaNewEquip(runCy,PGALL,YTIME) * VPowerPlantNewEq(
             runCy,PGALL,YTIME) * VGapPowerGenCap(runCy,YTIME))$NOCCS(
             PGALL)
          +(VPowPlaShaNewEquip(runCy,PGALL,YTIME) * VPowerPlantNewEq(
             runCy,PGALL,YTIME) * VGapPowerGenCap(runCy,YTIME))$CCS(
             PGALL)
          + iDecInvPlantSched(runCy,PGALL,YTIME) * iAvailRate(PGALL,
             YTIME)
          - iPlantDecomSched(runCy,PGALL,YTIME) * iAvailRate(PGALL,
             YTIME)
         )
         - ((VElecGenPlantsCapac(runCy,PGALL,YTIME-1)-iPlantDecomSched(
            runCy,PGALL,YTIME-1))*
         iAvailRate(PGALL,YTIME)*(1/iTechLftPlaType(runCy,PGALL)))
            $PGSCRN(PGALL);
```

- Compute electricity generation capacity

```
QElecGenCap(runCy,PGALL,YTIME)$TIME(YTIME)..
         VElecGenPlanCap(runCy,PGALL,YTIME)
             =E=
         ( VElecGenPlantsCapac(runCy,PGALL,YTIME) + 1e-6 + SQRT( SQR(
            VElecGenPlantsCapac(runCy,PGALL,YTIME)-1e-6) + SQR(1e-4) )
            )/2;
```

- Compute variable cost of technology

```
QVarCostTechnology(runCy,PGALL,YTIME)$TIME(YTIME)..
         VVarCostTechnology(runCy,PGALL,YTIME)
         =E=
          iMatureFacPlaDisp(runCy,PGALL,YTIME)*VVarCostTechnology(runCy
             ,PGALL,YTIME)**(-2);
```

- Compute Electricity peak loads

```
QElecPeakLoads(runCy,YTIME)$TIME(YTIME)..
         VElecPeakLoads(runCy,YTIME)
         =E=
         sum(PGALL, VVarCostTechnology(runCy,PGALL,YTIME));
```

- Compute Power plants sorting according to variable cost to decide the plant dispatching

```
QElectrPeakLoad(runCy,PGALL,YTIME)$TIME(YTIME)..
         VPowPlantSorting(runCy,PGALL,YTIME)
                 =E=
         VVarCostTechnology(runCy,PGALL,YTIME)
         /
         VElecPeakLoads(runCy,YTIME);
```

- Compute the new capacity added every year

```
QNewCapYearly(runCy,PGALL,YTIME)$(PGREN(PGALL)$TIME(YTIME))..
         VNewCapYearly(runCy,PGALL,YTIME) =e=
VElecGenPlanCap(runCy,PGALL,YTIME)- VElecGenPlanCap(runCy,PGALL,YTIME
    -1);
```

- Compute the average capacity factor of RES

```
QAvgCapFacRes(runCy,PGALL,YTIME)$(PGREN(PGALL)$TIME(YTIME))..
   VAvgCapFacRes(runCy,PGALL,YTIME)
```

```
      =E=
    (iAvailRate(PGALL,YTIME)*VNewCapYearly(runCy,PGALL,YTIME)+
     iAvailRate(PGALL,YTIME-1)*VNewCapYearly(runCy,PGALL,YTIME-1)+
     iAvailRate(PGALL,YTIME-2)*VNewCapYearly(runCy,PGALL,YTIME-2)+
     iAvailRate(PGALL,YTIME-3)*VNewCapYearly(runCy,PGALL,YTIME-3)+
     iAvailRate(PGALL,YTIME-4)*VNewCapYearly(runCy,PGALL,YTIME-4)+
     iAvailRate(PGALL,YTIME-5)*VNewCapYearly(runCy,PGALL,YTIME-5)+
     iAvailRate(PGALL,YTIME-6)*VNewCapYearly(runCy,PGALL,YTIME-6)+
     iAvailRate(PGALL,YTIME-7)*VNewCapYearly(runCy,PGALL,YTIME-7))/
(VNewCapYearly(runCy,PGALL,YTIME)+VNewCapYearly(runCy,PGALL,YTIME-1)+
    VNewCapYearly(runCy,PGALL,YTIME-2)+
VNewCapYearly(runCy,PGALL,YTIME-3)+VNewCapYearly(runCy,PGALL,YTIME-4)+
    VNewCapYearly(runCy,PGALL,YTIME-5)+
VNewCapYearly(runCy,PGALL,YTIME-6)+VNewCapYearly(runCy,PGALL,YTIME-7));
```

- Compute overall capacity

```
QOverallCap(runCy,PGALL,YTIME)$TIME(YTIME)..
     VOverallCap(runCy,PGALL,YTIME)
     =E=
VElecGenPlanCap(runCy,pgall,ytime)$ (not PGREN(PGALL))
+VAvgCapFacRes(runCy,PGALL,YTIME-1)*(VNewCapYearly(runCy,PGALL,YTIME)/
    iAvailRate(PGALL,YTIME)+
VOverallCap(runCy,PGALL,YTIME-1)
/VAvgCapFacRes(runCy,PGALL,YTIME-1))$PGREN(PGALL);
```

- Compute the scaling factor for plant dispatching

```
QScalFacPlantDispatch(runCy,HOUR,YTIME)$TIME(YTIME)..
         sum(PGALL,
                 (VOverallCap(runCy,PGALL,YTIME)+
                 sum(CHP$CHPtoEON(CHP,PGALL),VElecCapChpPla(runCy,CHP,
                    YTIME)))*
                 exp(-VScalFacPlaDisp(runCy,HOUR,YTIME)/
                    VPowPlantSorting(runCy,PGALL,YTIME))
                 )
         =E=
         (VElecPeakLoad(runCy,YTIME) - VCorrBaseLoad(runCy,YTIME))
         * exp(-VLoadCurveConstr(runCy,YTIME)*(0.25 + ord(HOUR)-1))
         + VCorrBaseLoad(runCy,YTIME);
```

- Compute estimated electricity of CHP Plants

```
QElecChpPlants(runCy,YTIME)$TIME(YTIME)..
         VElecChpPlants(runCy,YTIME)
         =E=
         ( (1/0.086 * sum((INDDOM,CHP),VConsFuel(runCy,INDDOM,CHP,YTIME)) * VElecIndPrices(runCy,YTIME)) +
         iMxmShareChpElec(runCy,YTIME)*VElecDem(runCy,YTIME) - SQRT(SQR((1/0.086 * sum((INDDOM,CHP),VConsFuel(runCy,INDDOM,CHP,YTIME)) *
         VElecIndPrices(runCy,YTIME)) -
         iMxmShareChpElec(runCy,YTIME)*VElecDem(runCy,YTIME)) + SQR(1E-4) ) )/2;
```

- Compute non CHP electricity production

```
QNonChpElecProd(runCy,YTIME)$TIME(YTIME)..
         VNonChpElecProd(runCy,YTIME)
         =E=
  (VElecDem(runCy,YTIME) - VElecChpPlants(runCy,YTIME));
```

- Compute total required electricity production

```
QReqElecProd(runCy,YTIME)$TIME(YTIME)..
VReqElecProd(runCy,YTIME)
                   =E=
         sum(hour, sum(CHP,VElecCapChpPla(runCy,CHP,YTIME)*exp(-VScalFacPlaDisp(runCy,HOUR,YTIME)/
         sum(pgall$chptoeon(chp,pgall),VPowPlantSorting(runCy,PGALL,YTIME)))));
```

- Compute electricity production from power generation plants

```
QElecProdPowGenPlants(runCy,PGALL,YTIME)$TIME(YTIME)..
         VElecProdPowGenPlants(runCy,PGALL,YTIME)
                 =E=
         VNonChpElecProd(runCy,YTIME) /
         (VTotReqElecProd(runCy,YTIME)- VReqElecProd(runCy,YTIME))
         * VElecGenPlanCap(runCy,PGALL,YTIME)* sum(HOUR, exp(-VScalFacPlaDisp(runCy,HOUR,YTIME)/VPowPlantSorting(runCy,PGALL,YTIME)));
```

- Compute sector contribution to total CHP production

```
QSecContrTotChpProd(runCy,INDDOM,CHP,YTIME)$(TIME(YTIME) $SECTTECH(INDDOM,CHP))..
```

```
        VSecContrTotChpProd(runCy,INDDOM,CHP,YTIME)
         =E=
        VConsFuel(runCy,INDDOM,CHP,YTIME)/(1e-6+SUM(INDDOM2,VConsFuel(
           runCy,INDDOM2,CHP,YTIME)));
```

- Compute electricity production from CHP plants

```
QElecProdChpPlants(runCy,CHP,YTIME)$TIME(YTIME)..
        VChpElecProd(runCy,CHP,YTIME)
                =E=
       sum(INDDOM,VConsFuel(runCy,INDDOM,CHP,YTIME)) / SUM(chp2,sum(
          INDDOM,VConsFuel(runCy,INDDOM,CHP2,YTIME)))*
       (VElecDem(runCy,YTIME) - SUM(PGALL,VElecProd(runCy,PGALL,YTIME)
          ));
```

- Compute the share of renewables in gross electricity production for subsdidized renewables

```
QShareRenGrossElecProd(runCy,YTIME)$TIME(YTIME)..
                VResShareGrossElecProd(runCy,YTIME)
                =E=
                (SUM(PGNREN$((not sameas("PGASHYD",PGNREN)) $(not
                   sameas("PGSHYD",PGNREN)) $(not sameas("PGLHYD",
                   PGNREN)) ),
                       VElecProdPowGenPlants(runCy,PGNREN,YTIME)))/
                (SUM(PGALL,VElecProdPowGenPlants(runCy,PGALL,YTIME))+
                1e-3*sum(DSBS,sum(CHP$SECTTECH(DSBS,CHP),VConsFuel(
                   runCy,DSBS,CHP,YTIME)))/8.6e-5*VElecIndPrices(runCy
                   ,YTIME) +
                1/0.086 *VNetImports(runCy,"ELC",YTIME));
```

- Compute long term power generation cost of technologies excluding climate policies

```
QLonPowGenCostTechNoCp(runCy,PGALL,ESET,YTIME)$TIME(YTIME)..
        VLongPowGenCost(runCy,PGALL,ESET,YTIME)
                =E=
            (iDisc(runCy,"PG",YTIME)*EXP(iDisc(runCy,"pg",YTIME)*
               iTechLftPlaType(runCy,PGALL)) /
            (EXP(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL))
               -1)*iGrossCapCosSubRen(runCy,PGALL,YTIME)*1000*iCGI(
               runCy,YTIME) +
            iFixOandMCost(PGALL,YTIME))/iAvailRate(PGALL,YTIME)
             / (1000*(6$ISET(ESET)+4$RSET(ESET))) +
            sum(PGEF$PGALLTOEF(PGALL,PGEF),
```

```
                    (iVarCost(PGALL,YTIME)/1000+(VFuelPriceSub(runCy,"PG",PGEF,YTIME)/1.2441+
                    iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME) +
                    (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)*
                    (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))))
                    *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME)));
```

- Long-term minimum power generation cost

```
QLonMnmpowGenCost(runCy,PGALL,YTIME)$TIME(YTIME)..
         VLonMnmpowGenCost(runCy,PGALL,YTIME)
                 =E=
             (iDisc(runCy,"PG",YTIME)*EXP(iDisc(runCy,"pg",YTIME)*iTechLftPlaType(runCy,PGALL)) /
             (EXP(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL))-1)*iGrossCapCosSubRen(runCy,PGALL,YTIME)*1000*iCGI(runCy,YTIME) +
             iFixOandMCost(PGALL,YTIME))/iAvailRate(PGALL,YTIME)
             / (1000*sGwToTwhPerYear) +
             sum(PGEF$PGALLTOEF(PGALL,PGEF),
                 (iVarCost(PGALL,YTIME)/1000+(VFuelPriceSub(runCy,"PG",PGEF,YTIME)/1.2441+
                 iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME) +
                 (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)*
                 (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))))
                 *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME)));
```

- Compute long term power generation cost of technologies including international Prices of main fuels

```
QLongPowGenIntPri(runCy,PGALL,ESET,YTIME)$TIME(YTIME)..
         VLongPowGenIntPri(runCy,PGALL,ESET,YTIME)
                 =E=
             (iDisc(runCy,"PG",YTIME)*EXP(iDisc(runCy,"pg",YTIME)*iTechLftPlaType(runCy,PGALL)) /
             (EXP(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL))-1)*iGrossCapCosSubRen(runCy,PGALL,YTIME)/1.5*1000*iCGI
```

```
                    (runCy,YTIME) +
                iFixOandMCost(PGALL,YTIME))/iAvailRate(PGALL,YTIME)
                / (1000*(7.25$ISET(ESET)+2.25$RSET(ESET))) +
                sum(PGEF$PGALLTOEF(PGALL,PGEF),
                    (iVarCost(PGALL,YTIME)/1000+((
  SUM(EF,sum(WEF$EFtoWEF("PG",EF,WEF), iPriceFuelsInt(WEF,YTIME))*
      sTWhToMtoe/1000*1.5))$(not PGREN(PGALL))    +
                    iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,
                        YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME) +
                    (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(
                        runCy,"PG",PGEF,YTIME)*
                    (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME
                        ))))
                    *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME)));
```

- Compute short term power generation cost of technologies including international Prices of main fuels

```
QShoPowGenIntPri(runCy,PGALL,ESET,YTIME)$TIME(YTIME)..
        VShoPowGenIntPri(runCy,PGALL,ESET,YTIME)
                =E=
                sum(PGEF$PGALLTOEF(PGALL,PGEF),
                    (iVarCost(PGALL,YTIME)/1000+((
  SUM(EF,sum(WEF$EFtoWEF("PG",EF,WEF), iPriceFuelsInt(WEF,YTIME))*
      sTWhToMtoe/1000*1.5))$(not PGREN(PGALL))    +
                    iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,
                        YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME) +
                    (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(
                        runCy,"PG",PGEF,YTIME)*
                    (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME
                        ))))
                    *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME)));
```

- Compute long term power generation cost

```
QLongPowGenCost(runCy,ESET,YTIME)$TIME(YTIME)..
        VLongAvgPowGenCost(runCy,ESET,YTIME)
                =E=
        (
        SUM(PGALL, VElecProdPowGenPlants(runCy,PGALL,YTIME)*
            VLongPowGenCost(runCy,PGALL,ESET,YTIME))
       +
        sum(CHP, VAvgElcProCHP(runCy,CHP,YTIME)*VChpElecProd(runCy,CHP
```

```
                ,YTIME))
          )
/VElecDem(runCy,YTIME);
```

- Compute long term average power generation cost excluding climate policies

```
QLonAvgPowGenCostNoClimPol(runCy,PGALL,ESET,YTIME)$TIME(YTIME)..
         VLonAvgPowGenCostNoClimPol(runCy,PGALL,ESET,YTIME)
                 =E=
             (iDisc(runCy,"PG",YTIME)*EXP(iDisc(runCy,"pg",YTIME)*iTechLftPlaType(runCy,PGALL)) /
             (EXP(iDisc(runCy,"PG",YTIME)*iTechLftPlaType(runCy,PGALL))-1)*iGrossCapCosSubRen(runCy,PGALL,YTIME)*1000*iCGI(runCy,YTIME) +
             iFixOandMCost(PGALL,YTIME))/iAvailRate(PGALL,YTIME)
             / (1000*(7.25$ISET(ESET)+2.25$RSET(ESET))) +
             sum(PGEF$PGALLTOEF(PGALL,PGEF),
                 (iVarCost(PGALL,YTIME)/1000+((VFuelPriceSub(runCy,"PG",PGEF,YTIME)-iEffValueInEuro(runCy,"pg",ytime)/1000-iCo2EmiFac(runCy,"PG",PGEF,YTIME)*
                 sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))/1000 )/1.2441+
                 iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME) +
                 (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)*
                 (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))))
                 *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME)));
```

- Compute long term power generation cost excluding climate policies

```
QLonPowGenCostNoClimPol(runCy,ESET,YTIME)$TIME(YTIME)..
         VLonPowGenCostNoClimPol(runCy,ESET,YTIME)
                 =E=
         (
         SUM(PGALL, (VElecProdPowGenPlants(runCy,PGALL,YTIME))*VLonAvgPowGenCostNoClimPol(runCy,PGALL,ESET,YTIME))
        +
         sum(CHP, VAvgElcProCHP(runCy,CHP,YTIME)*VChpElecProd(runCy,CHP,YTIME))
         )
/(VElecDem(runCy,YTIME));
```

- Compute electricity price in Industrial and Residential Consumers excluding climate policies

```
QElecPriIndResNoCliPol(runCy,ESET,YTIME)$TIME(YTIME)..   !! The electricity price is based on previous year's production costs
        VElecPriIndResNoCliPol(runCy,ESET,YTIME)
                 =E=
        (1 + iFacElecPriConsu(runCy,"VAT",YTIME)) *
        (
           (
             (VFuelPriceSub(runCy,"OI","ELC",YTIME-1)*sTWhToMtoe) $TFIRST(YTIME-1) +
             (  iFacElecPriConsu(runCy,"IND_RES",YTIME-1) + VRenShareElecProdSub(runCy,YTIME-1)*(VRenValue(YTIME)*8.6e-5)+
                iFacElecPriConsu(runCy,"W_INDU",YTIME-1)* VLonPowGenCostNoClimPol(runCy,"i",YTIME-1) +
               (1-iFacElecPriConsu(runCy,"W_INDU",YTIME-1))* VAvgPowerGenCostShoTrm(runCy,"i",YTIME-1)
              )$(not TFIRST(YTIME-1))
           )$ISET(ESET)
        +
           (
             (VFuelPriceSub(runCy,"HOU","ELC",YTIME-1)*sTWhToMtoe) $TFIRST(YTIME-1) +
             (  iFacElecPriConsu(runCy,"TERT_RES",YTIME-1) + VRenShareElecProdSub(runCy,YTIME-1)*(VRenValue(YTIME)*8.6e-5)+
                iFacElecPriConsu(runCy,"W_TERT",YTIME-1)* VLonPowGenCostNoClimPol(runCy,"r",YTIME-1) +
                (1-iFacElecPriConsu(runCy,"W_TERT",YTIME-1))* VAvgPowerGenCostShoTrm(runCy,"r",YTIME-1)
             )$(not TFIRST(YTIME-1))
           )$RSET(ESET)
        );
```

- Compute short term power generation cost

```
QShortPowGenCost(runCy,ESET,YTIME)$TIME(YTIME)..
        VAvgPowerGenCostShoTrm(runCy,ESET,YTIME)
                 =E=
        (
        sum(PGALL,
        VElecProdPowGenPlants(runCy,PGALL,YTIME)*
```

```
        (
        sum(PGEF$PGALLtoEF(PGALL,PGEF),
        (iVarCost(PGALL,YTIME)/1000+(VFuelPriceSub(runCy,"PG",PGEF,YTIME)/1.2441+
         iCO2CaptRate(runCy,PGALL,YTIME)*VCO2CO2SeqCsts(runCy,YTIME)*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME) +
         (1-iCO2CaptRate(runCy,PGALL,YTIME))*1e-3*iCo2EmiFac(runCy,"PG",PGEF,YTIME)*
         (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))))
                *sTWhToMtoe/iPlantEffByType(runCy,PGALL,YTIME)))
        ))
        +
         sum(CHP, VAvgVarProdCostCHP(runCy,CHP,YTIME)*VChpElecProd(runCy,CHP,YTIME))
         )
         /VElecDem(runCy,YTIME);
```

- Transport
- Compute the lifetime of passenger cars

```
QPassCarsLft(runCy,DSBS,EF,TEA,YTIME)$(TIME(YTIME) $sameas(DSBS,"PC") $SECTTECH(DSBS,EF))..
         VLifeTimeTech(runCy,DSBS,EF,TEA,YTIME)
                =E=
         1/VScrRate(runCy,YTIME);
```

- Compute goods transport activity

```
QGoodsTranspActiv(runCy,TRANSE,YTIME)$(TIME(YTIME) $TRANG(TRANSE))..
         VGoodsTranspActiv(runCy,TRANSE,YTIME)
                =E=
         (
           iResActiv(runCy,TRANSE,YTIME) * VGoodsTranspActiv(runCy,TRANSE,YTIME-1)
           * [(iGDP(YTIME,runCy)/iPop(YTIME,runCy))/(iGDP(YTIME-1,runCy)/iPop(YTIME-1,runCy))]**iElastA(runCy,TRANSE,"a",YTIME)
           * (iPop(YTIME,runCy)/iPop(YTIME-1,runCy))
           * (VFuelPriceAvg(runCy,TRANSE,YTIME)/VFuelPriceAvg(runCy,TRANSE,YTIME-1))**iElastA(runCy,TRANSE,"c1",YTIME)
           * (VFuelPriceAvg(runCy,TRANSE,YTIME-1)/VFuelPriceAvg(runCy,TRANSE,YTIME-2))**iElastA(runCy,TRANSE,"c2",YTIME)
           * prod(kpdl,
                 [(VFuelPriceAvg(runCy,TRANSE,YTIME-ord(kpdl))/
```

```
                    VFuelPriceAvg(runCy,TRANSE,YTIME-(ord(kpdl)+1)))/
                    (iCGI(runCy,YTIME)**(1/6))]**(iElastA(runCy,TRANSE,"c3",YTIME)*iFPDL(TRANSE,KPDL))
                )
        )$sameas(TRANSE,"GU")        !!trucks
        +
        (
          iResActiv(runCy,TRANSE,YTIME) * VGoodsTranspActiv(runCy,TRANSE,YTIME-1)
          * [(iGDP(YTIME,runCy)/iPop(YTIME,runCy))/(iGDP(YTIME-1,runCy)/iPop(YTIME-1,runCy))]**iElastA(runCy,TRANSE,"a",YTIME)
          * (VFuelPriceAvg(runCy,TRANSE,YTIME)/VFuelPriceAvg(runCy,TRANSE,YTIME-1))**iElastA(runCy,TRANSE,"c1",YTIME)
          * (VFuelPriceAvg(runCy,TRANSE,YTIME-1)/VFuelPriceAvg(runCy,TRANSE,YTIME-2))**iElastA(runCy,TRANSE,"c2",YTIME)
          * prod(kpdl,
                [(VFuelPriceAvg(runCy,TRANSE,YTIME-ord(kpdl))/
                  VFuelPriceAvg(runCy,TRANSE,YTIME-(ord(kpdl)+1)))/
                  (iCGI(runCy,YTIME)**(1/6))]**(iElastA(runCy,TRANSE,"c3",YTIME)*iFPDL(TRANSE,KPDL))
                )
          * (VGoodsTranspActiv(runCy,"GU",YTIME)/VGoodsTranspActiv(runCy,"GU",YTIME-1))**iElastA(runCy,TRANSE,"c4",YTIME)
        )$(not sameas(TRANSE,"GU"));                        !!other freight transport
```

- Compute the gap in transport activity

```
QGapTranspActiv(runCy,TRANSE,YTIME)$TIME(YTIME)..
        VGapTranspFillNewTech(runCy,TRANSE,YTIME)
                =E=
        VNewReg(runCy,YTIME)$sameas(TRANSE,"PC")
        +
        (
        ( [VTrnspActiv(runCy,TRANSE,YTIME) - VTrnspActiv(runCy,TRANSE,YTIME-1) + VTrnspActiv(runCy,TRANSE,YTIME-1)/
        (sum((TTECH,TEA)$SECTTECH(TRANSE,TTECH),VLifeTimeTech(runCy,TRANSE,TTECH,TEA,YTIME-1))/TECHS(TRANSE))] +
         SQRT( SQR([VTrnspActiv(runCy,TRANSE,YTIME) - VTrnspActiv(runCy,TRANSE,YTIME-1) + VTrnspActiv(runCy,TRANSE,YTIME-1)/
         (sum((TTECH,TEA)$SECTTECH(TRANSE,TTECH),VLifeTimeTech(runCy,TRANSE,TTECH,TEA,YTIME-1))/TECHS(TRANSE))]) + SQR(1e-4) ))/2
```

```
          )$(TRANP(TRANSE) $(not sameas(TRANSE,"PC")))
          +
          (
          ( [VGoodsTranspActiv(runCy,TRANSE,YTIME) - VGoodsTranspActiv(runCy,TRANSE,YTIME-1) + VGoodsTranspActiv(runCy,TRANSE,YTIME-1)/
          (sum((EF,TEA)$SECTTECH(TRANSE,EF),VLifeTimeTech(runCy,TRANSE,EF,TEA,YTIME-1))/TECHS(TRANSE))] + SQRT( SQR([VGoodsTranspActiv(runCy,TRANSE,YTIME) -
           VGoodsTranspActiv(runCy,TRANSE,YTIME-1) + VGoodsTranspActiv(runCy,TRANSE,YTIME-1)/
           (sum((EF,TEA)$SECTTECH(TRANSE,EF),VLifeTimeTech(runCy,TRANSE,EF,TEA,YTIME-1))/TECHS(TRANSE))]) + SQR(1e-4) ) )/2
          )$TRANG(TRANSE);
```

- Compute Specific Fuel Consumption

```
QSpecificFuelCons(runCy,TRANSE,TTECH,TEA,EF,YTIME)$(TIME(YTIME) $SECTTECH(TRANSE,EF) $TTECHtoEF(TTECH,EF) )..
          VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,EF,YTIME)
                  =E=
          iResSpecificFuelConsCost(runCy,TRANSE,TTECH,EF,YTIME)*VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,EF,YTIME-1)
           * prod(KPDL,
                      (
                         VFuelPriceSub(runCy,TRANSE,EF,YTIME-ord(KPDL))/VFuelPriceSub(runCy,TRANSE,EF,YTIME-(ord(KPDL)+1))
                       )**(iElastA(runCy,TRANSE,"c5",YTIME)*iFPDL(TRANSE,KPDL))
           );
```

- Compute transportation cost per mean and consumer size in KEuro per vehicle

```
QTranspCostPerMeanConsSize(runCy,TRANSE,RCon,TTECH,TEA,YTIME)$(TIME(YTIME) $SECTTECH(TRANSE,TTECH) $(ord(Rcon) le iNcon(TRANSE)+1))..
          VTranspCostPermeanConsSize(runCy,TRANSE,RCon,TTECH,TEA,YTIME)
          =E=
                        (
                          (
                            (iDisc(runCy,TRANSE,YTIME)*exp(iDisc(runCy,TRANSE,YTIME)*VLifeTimeTech(runCy,TRANSE,TTECH,TEA,YTIME)))
```

```
                                /
                                (exp(iDisc(runCy,TRANSE,YTIME)*VLifeTimeTech(runCy,TRANSE,TTECH,TEA,YTIME)) - 1)
                            ) * iCapCostTech(runCy,TRANSE,TTECH,YTIME)  * iCGI(runCy,YTIME)
                            + iFixOMCostTech(runCy,TRANSE,TTECH,YTIME)  +
                            (
                                (sum(EF$TTECHtoEF(TTECH,EF), VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,EF,YTIME)*VFuelPriceSub(runCy,TRANSE,EF,YTIME)) )$(not PLUGIN(TTECH))
                                +
                                (sum(EF$(TTECHtoEF(TTECH,EF) $(not sameas("ELC",EF))),
                                    (1-iShareAnnMilePlugInHybrid(runCy,YTIME))*VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,EF,YTIME)*VFuelPriceSub(runCy,TRANSE,EF,YTIME))
                                  + iShareAnnMilePlugInHybrid(runCy,YTIME)*VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,"ELC",YTIME)*VFuelPriceSub(runCy,TRANSE,"ELC",YTIME)
                                )$PLUGIN(TTECH)
                                + iVarCostTech(runCy,TRANSE,TTECH,YTIME)
                                + (VRenValue(YTIME)/1000)$( not RENEF(TTECH))
                            )
                            *  iAnnCons(runCy,TRANSE,"smallest") * (iAnnCons(runCy,TRANSE,"largest")/iAnnCons(runCy,TRANSE,"smallest"))**((ord(Rcon)-1)/iNcon(TRANSE))
                        );
```

- Compute transportation cost per mean and consumer size

```
QTranspCostPerVeh(runCy,TRANSE,rCon,TTECH,TEA,YTIME)$(TIME(YTIME) $SECTTECH(TRANSE,TTECH) $(ord(rCon) le iNcon(TRANSE)+1))..
        VTranspCostPerVeh(runCy,TRANSE,rCon,TTECH,TEA,YTIME)
        =E=
        VTranspCostPermeanConsSize(runCy,TRANSE,rCon,TTECH,TEA,YTIME)**(-4);
```

- Compute transportation cost including maturity factor

```
QTranspCostMatFac(runCy,TRANSE,RCon,TTECH,TEA,YTIME)$(TIME(YTIME)
    $SECTTECH(TRANSE,TTECH) $(ord(rCon) le iNcon(TRANSE)+1))..
        VTranspCostMatFac(runCy,TRANSE,RCon,TTECH,TEA,YTIME)
        =E=
        iMatrFactor(runCy,TRANSE,TTECH,YTIME) * VTranspCostPerVeh(
            runCy,TRANSE,rCon,TTECH,TEA,YTIME);
```

- Compute technology sorting based on variable cost

```
QTechSortVarCost(runCy,TRANSE,rCon,YTIME)$(TIME(YTIME) $(ord(rCon) le
    iNcon(TRANSE)+1))..
        VTechSortVarCost(runCy,TRANSE,rCon,YTIME)
                =E=
        sum((TTECH,TEA)$SECTTECH(TRANSE,TTECH), VTranspCostMatFac(
            runCy,TRANSE,rCon,TTECH,TEA,YTIME));
```

- Compute the share of each technology in total sectoral use

```
QTechSortVarCostNewEquip(runCy,TRANSE,TTECH,TEA,YTIME)$(TIME(YTIME)
    $SECTTECH(TRANSE,TTECH) )..
        VTechSortVarCostNewEquip(runCy,TRANSE,TTECH,TEA,YTIME)
        =E=
        iMatrFactor(runCy,TRANSE,TTECH,YTIME) / iCumDistrFuncConsSize(
            runCy,TRANSE)
        * sum( Rcon$(ord(Rcon) le iNcon(TRANSE)+1),
               VTranspCostPerVeh(runCy,TRANSE,RCon,TTECH,TEA,YTIME)
               * iDisFunConSize(runCy,TRANSE,RCon) / VTechSortVarCost(
                   runCy,TRANSE,RCon,YTIME)
             );
```

- Compute consumption of each technology in transport sectors

```
QConsEachTechTransp(runCy,TRANSE,TTECH,EF,TEA,YTIME)$(TIME(YTIME)
    $SECTTECH(TRANSE,TTECH) $TTECHtoEF(TTECH,EF) )..
        VConsEachTechTransp(runCy,TRANSE,TTECH,EF,TEA,YTIME)
                =E=
        VConsEachTechTransp(runCy,TRANSE,TTECH,EF,TEA,YTIME-1) *
        (
                (
                    ((VLifeTimeTech(runCy,TRANSE,TTECH,TEA,YTIME-1)-1)
                        /VLifeTimeTech(runCy,TRANSE,TTECH,TEA,YTIME-1))
                     *(iAvgVehCapLoadFac(runCy,TRANSE,"CAP",YTIME-1)*
                         iAvgVehCapLoadFac(runCy,TRANSE,"LF",YTIME-1))
```

```
                        /(iAvgVehCapLoadFac(runCy,TRANSE,"CAP",YTIME)*
                           iAvgVehCapLoadFac(runCy,TRANSE,"LF",YTIME))
                  )$(not sameas(TRANSE,"PC"))
                  +
                  (1 - VScrRate(runCy,YTIME))$sameas(TRANSE,"PC")
          )
          +
          VTechSortVarCostNewEquip(runCy,TRANSE,TTECH,TEA,YTIME) *
          (
                  VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,EF,YTIME)$(
                     not PLUGIN(TTECH))
                  +
                  ( ((1-iShareAnnMilePlugInHybrid(runCy,YTIME))*
                     VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,EF,YTIME))
                     $(not sameas("ELC",EF))
                    + iShareAnnMilePlugInHybrid(runCy,YTIME)*
                       VSpecificFuelCons(runCy,TRANSE,TTECH,TEA,"ELC",
                       YTIME))$PLUGIN(TTECH)
          )/1000
          * VGapTranspFillNewTech(runCy,TRANSE,YTIME) *
          (
                  (
                   (iAvgVehCapLoadFac(runCy,TRANSE,"CAP",YTIME-1)*
                      iAvgVehCapLoadFac(runCy,TRANSE,"LF",YTIME-1))
                   / (iAvgVehCapLoadFac(runCy,TRANSE,"CAP",YTIME)*
                      iAvgVehCapLoadFac(runCy,TRANSE,"LF",YTIME))
                  )$(not sameas(TRANSE,"PC"))
                  +
                  (VTrnspActiv(runCy,TRANSE,YTIME))$sameas(TRANSE,"PC")
          );
```

- Compute final energy demand in transport per fuel

```
QFinEneDemTranspPerFuel(runCy,TRANSE,EF,YTIME)$(TIME(YTIME) $SECTTECH(
   TRANSE,EF))..
          VDemTr(runCy,TRANSE,EF,YTIME)
                  =E=
          sum((TTECH,TEA)$(SECTTECH(TRANSE,TTECH) $TTECHtoEF(TTECH,EF) )
             , VConsEachTechTransp(runCy,TRANSE,TTECH,EF,TEA,YTIME));
```

- Compute final energy demand in transport

```
QFinEneDemTransp(runCy,TRANSE,YTIME)$(TIME(YTIME)) ..
```

```
        VFinEneDemTranspSub(runCy,TRANSE,YTIME)
                =E=
        sum(EF,VDemTr(runCy,TRANSE,EF,YTIME));
```

- Compute passenger cars market extension (GDP dependent)

```
QMExtV(runCy,YTIME)$TIME(YTIME)..
        VMExtV(runCy,YTIME)
                =E=
        iTransChar(runCy,"RES_MEXTV",YTIME) * VMExtV(runCy,YTIME-1) *
        [(iGDP(YTIME,runCy)/iPop(YTIME,runCy)) / (iGDP(YTIME-1,runCy)/
           iPop(YTIME-1,runCy))] ** iElastA(runCy,"PC","a",YTIME);
```

- Compute passenger cars market extension (GDP independent)

```
QMExtF(runCy,YTIME)$TIME(YTIME)..
        VMExtF(runCy,YTIME)
                =E=
        iTransChar(runCy,"RES_MEXTF",YTIME) * iSigma(runCy,"S1") * EXP
           (iSigma(runCy,"S2") * EXP(iSigma(runCy,"S3") * VLamda(runCy
           ,YTIME))) *
        VNumVeh(runCy,YTIME-1) / (iPop(YTIME-1,runCy) * 1000);
```

- Compute stock of passenger cars (in million vehicles)

```
QNumVeh(runCy,YTIME)$TIME(YTIME)..
        VNumVeh(runCy,YTIME)
                =E=
        (VNumVeh(runCy,YTIME-1)/(iPop(YTIME-1,runCy)*1000) + VMExtF(
           runCy,YTIME) + VMExtV(runCy,YTIME)) *
        iPop(YTIME,runCy) * 1000;
```

- Compute new registrations of passenger cars

```
QNewReg(runCy,YTIME)$TIME(YTIME)..
        VNewReg(runCy,YTIME)
                =E=
        (VMExtF(runCy,YTIME) + VMExtV(runCy,YTIME)) * (iPop(YTIME,
           runCy)*1000)  !! new cars due to GDP
        - VNumVeh(runCy,YTIME-1)*(1 - iPop(YTIME,runCy)/iPop(YTIME-1,
           runCy))    !! new cars due to population
        + VScrap(runCy,YTIME);
                                                          !! new
           cars due to scrapping
```

- Compute passenger transport activity

```
QTrnspActiv(runCy,TRANSE,YTIME)$(TIME(YTIME) $TRANP(TRANSE))..
         VTrnspActiv(runCy,TRANSE,YTIME)
                 =E=
         (  !! passenger cars
           iResActiv(runCy,TRANSE,YTIME) * VTrnspActiv(runCy,TRANSE,YTIME-1) *
           (VFuelPriceAvg(runCy,TRANSE,YTIME)/VFuelPriceAvg(runCy,TRANSE,YTIME-1))**iElastA(runCy,TRANSE,"b1",YTIME) *
           (VFuelPriceAvg(runCy,TRANSE,YTIME-1)/VFuelPriceAvg(runCy,TRANSE,YTIME-2))**iElastA(runCy,TRANSE,"b2",YTIME) *
           [(VNumVeh(runCy,YTIME-1)/(iPop(YTIME-1,runCy)*1000))/(VNumVeh(runCy,YTIME)/(iPop(YTIME,runCy)*1000))]**iElastA(runCy,TRANSE,"b3",YTIME) *
           [(iGDP(YTIME,runCy)/iPop(YTIME,runCy))/(iGDP(YTIME-1,runCy)/iPop(YTIME-1,runCy))]**iElastA(runCy,TRANSE,"b4",YTIME)
         )$sameas(TRANSE,"PC") +
         (  !! passenger aviation
           iResActiv(runCy,TRANSE,YTIME) * VTrnspActiv(runCy,TRANSE,YTIME-1) *
           [(iGDP(YTIME,runCy)/iPop(YTIME,runCy))/(iGDP(YTIME-1,runCy)/iPop(YTIME-1,runCy))]**iElastA(runCy,TRANSE,"a",YTIME) *
           (VFuelPriceAvg(runCy,TRANSE,YTIME)/VFuelPriceAvg(runCy,TRANSE,YTIME-1))**iElastA(runCy,TRANSE,"c1",YTIME) *
           (VFuelPriceAvg(runCy,TRANSE,YTIME-1)/VFuelPriceAvg(runCy,TRANSE,YTIME-2))**iElastA(runCy,TRANSE,"c2",YTIME)
         )$sameas(TRANSE,"PA") +
         (   !! other passenger transportation modes
          iResActiv(runCy,TRANSE,YTIME) * VTrnspActiv(runCy,TRANSE,YTIME-1) *
           [(iGDP(YTIME,runCy)/iPop(YTIME,runCy))/(iGDP(YTIME-1,runCy)/iPop(YTIME-1,runCy))]**iElastA(runCy,TRANSE,"a",YTIME) *
           (VFuelPriceAvg(runCy,TRANSE,YTIME)/VFuelPriceAvg(runCy,TRANSE,YTIME-1))**iElastA(runCy,TRANSE,"c1",YTIME) *
           (VFuelPriceAvg(runCy,TRANSE,YTIME-1)/VFuelPriceAvg(runCy,TRANSE,YTIME-2))**iElastA(runCy,TRANSE,"c2",YTIME) *
           [(VNumVeh(runCy,YTIME)*VTrnspActiv(runCy,"PC",YTIME))/(VNumVeh(runCy,YTIME-1)*VTrnspActiv(runCy,"PC",YTIME-1))]**iElastA(runCy,TRANSE,"c4",YTIME) *
           prod(kpdl,
```

```
                    [(VFuelPriceAvg(runCy,TRANSE,YTIME-ord(kpdl))/
                      VFuelPriceAvg(runCy,TRANSE,YTIME-(ord(kpdl)+1)))/
                      (iCGI(runCy,YTIME)**(1/6))]**(iElastA(runCy,TRANSE,
                         "c3",YTIME)*iFPDL(TRANSE,KPDL))
                  )
          )$(NOT (sameas(TRANSE,"PC") or sameas(TRANSE,"PA")));
```

- Compute scrapped passenger cars

```
QScrap(runCy,YTIME)$TIME(YTIME)..
         VScrap(runCy,YTIME)
                 =E=
         VScrRate(runCy,YTIME) * VNumVeh(runCy,YTIME-1);
```

- Compute ratio of car ownership over saturation car ownership

```
QLevl(runCy,YTIME)$TIME(YTIME)..
         VLamda(runCy,YTIME) !! level of saturation of gompertz
            function
                 =E=
         ( (VNumVeh(runCy,YTIME-1)/(iPop(YTIME-1,runCy)*1000)) /
            iPassCarsMarkSat(runCy) + 1 - SQRT( SQR((VNumVeh(runCy,
            YTIME-1)/(iPop(YTIME-1,runCy)*1000)) /  iPassCarsMarkSat(
            runCy) - 1)  ) )/2;
```

- Compute passenger cars scrapping rate

```
QScrRate(runCy,YTIME)$TIME(YTIME)..
         VScrRate(runCy,YTIME)
                  =E=
         [(iGDP(YTIME,runCy)/iPop(YTIME,runCy)) / (iGDP(YTIME-1,runCy)/
            iPop(YTIME-1,runCy))]**0.5
         * VScrRate(runCy,YTIME-1);
```

- Compute electricity consumption per final demand sector

```
QElecConsAll(runCy,DSBS,YTIME)$TIME(YTIME)..
         VElecConsAll(runCy,DSBS,YTIME)
             =E=
         sum(INDDOM $SAMEAS(INDDOM,DSBS), VConsFuel(runCy,INDDOM,"ELC",
            YTIME)) + sum(TRANSE $SAMEAS(TRANSE,DSBS), VDemTr(runCy,
            TRANSE,"ELC",YTIME));
```

- INDUSTRY - DOMESTIC - NON ENERGY USES - BUNKERS VARIABLES
- Compute non-substituable electricity consumption in Industry and Tertiary

```
QElecConsNonSub(runCy,INDDOM,YTIME)$TIME(YTIME)..
         VElecNonSub(runCy,INDDOM,YTIME)
                 =E=
         [
         iResNonSubsElecDem(runCy,INDDOM,YTIME) * VElecNonSub(runCy,INDDOM,YTIME-1) * ( iActv(YTIME,runCy,INDDOM)/iActv(YTIME-1,runCy,INDDOM) )**
         iElastNonSubElec(runCy,INDDOM,"a",YTIME)
         * ( VFuelPriceSub(runCy,INDDOM,"ELC",YTIME)/VFuelPriceSub(runCy,INDDOM,"ELC",YTIME-1) )**iElastNonSubElec(runCy,INDDOM,"b1",YTIME)
         * ( VFuelPriceSub(runCy,INDDOM,"ELC",YTIME-1)/VFuelPriceSub(runCy,INDDOM,"ELC",YTIME-2) )**iElastNonSubElec(runCy,INDDOM,"b2",YTIME)
         * prod(KPDL,
                  ( VFuelPriceSub(runCy,INDDOM,"ELC",YTIME-ord(KPDL))/VFuelPriceSub(runCy,INDDOM,"ELC",YTIME-(ord(KPDL)+1))
                  )**( iElastNonSubElec(runCy,INDDOM,"c",YTIME)*iFPDL(INDDOM,KPDL))
                )      ]$iActv(YTIME-1,runCy,INDDOM)+0;
```

- Compute the consumption of the remaining substitutable equipment

```
QConsOfRemSubEquip(runCy,DSBS,EF,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS)) $SECTTECH(DSBS,EF))..
         VConsRemSubEquip(runCy,DSBS,EF,YTIME)
                 =E=
         [
         iResFuelConsPerSubAndFuel(runCy,DSBS,EF,YTIME) * (sum(TEA,VLifeTimeTech(runCy,DSBS,EF,TEA,YTIME)-1)/sum(TEA,VLifeTimeTech(runCy,DSBS,EF,TEA,YTIME)))
         * (VFuelConsInclHP(runCy,DSBS,EF,YTIME-1) - (VElecNonSub(runCy,DSBS,YTIME-1)$(ELCEF(EF) $INDDOM(DSBS)) + 0$(not (ELCEF(EF) $INDDOM(DSBS)) )))
         * (iActv(YTIME,runCy,DSBS)/iActv(YTIME-1,runCy,DSBS))**iElastA(runCy,DSBS,"a",YTIME)
         * (VFuelPriceSub(runCy,DSBS,EF,YTIME)/VFuelPriceSub(runCy,DSBS,EF,YTIME-1))**iElastA(runCy,DSBS,"b1",YTIME)
         * (VFuelPriceSub(runCy,DSBS,EF,YTIME-1)/VFuelPriceSub(runCy,
```

```
                DSBS,EF,YTIME-2))**iElastA(runCy,DSBS,"b2",YTIME)
          * prod(KPDL,
                  (VFuelPriceSub(runCy,DSBS,EF,YTIME-ord(KPDL))/
                     VFuelPriceSub(runCy,DSBS,EF,YTIME-(ord(KPDL)+1)))
                     **(iElastA(runCy,DSBS,"c",YTIME)*iFPDL(DSBS,KPDL))
                )  ]$iActv(YTIME-1,runCy,DSBS)+0;
```

- Compute total final demand (of substitutable fuels) per subsector

```
QDemSub(runCy,DSBS,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS)))..
         VDemSub(runCy,DSBS,YTIME)
                 =E=
         [
         VDemSub(runCy,DSBS,YTIME-1)
         * ( iActv(YTIME,runCy,DSBS)/iActv(YTIME-1,runCy,DSBS) )**
            iElastA(runCy,DSBS,"a",YTIME)
         * ( VFuelPriceAvg(runCy,DSBS,YTIME)/VFuelPriceAvg(runCy,DSBS,
            YTIME-1) )**iElastA(runCy,DSBS,"b1",YTIME)
         * ( VFuelPriceAvg(runCy,DSBS,YTIME-1)/VFuelPriceAvg(runCy,DSBS
            ,YTIME-2) )**iElastA(runCy,DSBS,"b2",YTIME)
         * prod(KPDL,
                  ( (VFuelPriceAvg(runCy,DSBS,YTIME-ord(KPDL))/
                     VFuelPriceAvg(runCy,DSBS,YTIME-(ord(KPDL)+1)))/(
                     iCGI(runCy,YTIME)**(1/6))
                  )**( iElastA(runCy,DSBS,"c",YTIME)*iFPDL(DSBS,KPDL))
                )  ]$iActv(YTIME-1,runCy,DSBS)
;
```

- Compute Consumption of electricity in industrial sectors

```
QElecConsInd(runCy,YTIME)$TIME(YTIME)..
         VElecConsInd(runCy,YTIME)
         =E=
         SUM(INDSE,VElecNonSub(runCy,INDSE,YTIME));
```

- Compute total final demand (of substitutable fuels) in industrial sectors (Mtoe)

```
QDemInd(runCy,YTIME)$TIME(YTIME)..
        VDemInd(runCy,YTIME)=E= SUM(INDSE,VDemSub(runCy,INDSE,YTIME));
```

- Compute electricity industry prices

```
QElecIndPrices(runCy,YTIME)$TIME(YTIME)..
```

```
         VElecIndPrices(runCy,YTIME) =E=
        ( VElecIndPricesEst(runCy,YTIME) + sElecToSteRatioChp - SQRT(
           SQR(VElecIndPricesEst(runCy,YTIME)-sElecToSteRatioChp) + SQR
           (1E-4) ) )/2;
```

- Compute electricity consumed in heatpump plants

```
QElecConsHeatPla(runCy,INDDOM,YTIME)$time(ytime) ..
         VElecConsHeatPla(runCy,INDDOM,YTIME)
         =E=   1E-7;
```

- Compute fuel consumption (Mtoe)

```
QFuelCons(runCy,DSBS,EF,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS))
   $SECTTECH(DSBS,EF) $(not HEATPUMP(EF)) )..
         VConsFuel(runCy,DSBS,EF,YTIME)
                 =E=
         VConsFuelSub(runCy,DSBS,EF,YTIME)$(not ELCEF(EF)) +
         (VConsFuelSub(runCy,DSBS,EF,YTIME) + VElecConsHeatPla(runCy,
            DSBS,YTIME))$ELCEF(EF);
```

- Compute the estimated electricity index of industry price

```
QElecIndPricesEst(runCy,YTIME)$TIME(YTIME)..
         VElecIndPricesEst(runCy,YTIME)
                 =E=
         iResElecIndex(runCy,YTIME) * VElecIndPrices(runCy,YTIME-1) *
        ((VFuelPriceSub(runCy,"OI","ELC",YTIME-1)/VFuelPriceAvg(runCy,"
           OI",YTIME-1))/
        (VFuelPriceSub(runCy,"OI","ELC",YTIME-2)/VFuelPriceAvg(runCy,"
           OI",YTIME-2)))**(0.3) *
        ((VFuelPriceSub(runCy,"OI","ELC",YTIME-2)/VFuelPriceAvg(runCy,"
           OI",YTIME-2))/
        (VFuelPriceSub(runCy,"OI","ELC",YTIME-3)/VFuelPriceAvg(runCy,"
           OI",YTIME-3)))**(0.3);
```

- Compute fuel prices per subsector and fuel especially for chp plants (take into account the profit of electricity sales)

```
QFuePriSubChp(runCy,DSBS,EF,TEA,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS))
      $SECTTECH(DSBS,EF) )..
VFuePriSubChp(runCy,DSBS,EF,TEA,YTIME)
                 =E=
```

```
              (((VFuelPriceSub(runCy,DSBS,EF,YTIME) + (VRenValue(YTIME)
                 /1000)$(not RENEF(EF))+VTechCostVar(runCy,DSBS,EF,TEA,
                 YTIME)/1000)/iUsfEneConvSubTech(runCy,DSBS,EF,YTIME)-
                (0$(not CHP(EF)) + (VFuelPriceSub(runCy,"OI","ELC",YTIME
                  )*iFracElecPriChp(runCy,YTIME)*VElecIndPrices(runCy,
                  YTIME))$CHP(EF)))  + SQRT( SQR(((VFuelPriceSub(runCy,
                  DSBS,EF,YTIME)+VTechCostVar(runCy,DSBS,EF,TEA,YTIME)
                  /1000)/VTechCostVar(runCy,DSBS,EF,TEA,YTIME)-
               (0$(not CHP(EF)) + (VFuelPriceSub(runCy,"OI","ELC",YTIME)
                 *iFracElecPriChp(runCy,YTIME)*VElecIndPrices(runCy,
                 YTIME))$CHP(EF))))  ) )/2;
```

- Compute electricity production cost per CHP plant and demand sector

```
QElecProdCosChp(runCy,DSBS,CHP,YTIME)$(TIME(YTIME) $INDDOM(DSBS))..
         VElecProdCostChp(runCy,DSBS,CHP,YTIME)
                 =E=
                    ( ( iDisc(runCy,"PG",YTIME) * exp(iDisc(runCy,"PG",
                       YTIME)*iLifChpPla(runCy,DSBS,CHP))
                        / (exp(iDisc(runCy,"PG",YTIME)*iLifChpPla(runCy
                           ,DSBS,CHP)) -1))
                      * iInvCostChp(runCy,DSBS,CHP,YTIME)* 1000 * iCGI(
                         runCy,YTIME)  + iFixOMCostPerChp(runCy,DSBS,
                         CHP,YTIME)
                    )/(iAvailRateChp(runCy,DSBS,CHP)*(1000*sTWhToMtoe))
                       /1000
                    + iVarCostChp(runCy,DSBS,CHP,YTIME)/1000
                    + sum(PGEF$CHPtoEF(CHP,PGEF), (VFuelPriceSub(runCy,
                       "PG",PGEF,YTIME)+0.001*iCo2EmiFac(runCy,"PG",
                       PGEF,YTIME)*
                         (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,
                            NAP,YTIME))))
                         * sTWhToMtoe /  (iBoiEffChp(runCy,CHP,YTIME) *
                             VElecIndPrices(runCy,YTIME)) );
```

- Compute technology cost

```
QTechCost(runCy,DSBS,rCon,EF,TEA,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS)
    ) $(ord(rCon) le iNcon(DSBS)+1) $SECTTECH(DSBS,EF) )..
VTechCost(runCy,DSBS,rCon,EF,TEA,YTIME)
                 =E=
                 VTechCostIntrm(runCy,DSBS,rCon,EF,TEA,YTIME)**(-
                    iElaSub(runCy,DSBS)) ;
```

- Compute technology cost including lifetime

```
QTechCostIntrm(runCy,DSBS,rCon,EF,TEA,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS)) $(ord(rCon) le iNcon(DSBS)+1) $SECTTECH(DSBS,EF))..
        VTechCostIntrm(runCy,DSBS,rCon,EF,TEA,YTIME) =E=
                 ( (( (iDisc(runCy,DSBS,YTIME)$(not CHP(EF)) + iDisc(runCy,"PG",YTIME)$CHP(EF)) !! in case of chp plants we use the discount rate of power generation sector
                     * exp((iDisc(runCy,DSBS,YTIME)$(not CHP(EF)) + iDisc(runCy,"PG",YTIME)$CHP(EF))* VLifeTimeTech(runCy,DSBS,EF,TEA,YTIME))
                    )
                     / (exp((iDisc(runCy,DSBS,YTIME)$(not CHP(EF)) + iDisc(runCy,"PG",YTIME)$CHP(EF))*VLifeTimeTech(runCy,DSBS,EF,TEA,YTIME))- 1)
                   ) * iCapCostTech(runCy,DSBS,EF,YTIME) * iCGI(runCy,YTIME)
                   +
                   iFixOMCostTech(runCy,DSBS,EF,YTIME)/1000
                   +
                   VFuePriSubChp(runCy,DSBS,EF,TEA,YTIME)
                   * iAnnCons(runCy,DSBS,"smallest") * (iAnnCons(runCy,DSBS,"largest")/iAnnCons(runCy,DSBS,"smallest"))**((ord(rCon)-1)/iNcon(DSBS))
                 )$INDDOM(DSBS)
                +
                 ( (( iDisc(runCy,DSBS,YTIME)
                     * exp(iDisc(runCy,DSBS,YTIME)*VLifeTimeTech(runCy,DSBS,EF,TEA,YTIME))
                    )
                     / (exp(iDisc(runCy,DSBS,YTIME)*VLifeTimeTech(runCy,DSBS,EF,TEA,YTIME))- 1)
                   ) * iCapCostTech(runCy,DSBS,EF,YTIME) * iCGI(runCy,YTIME)
                   +
                   iFixOMCostTech(runCy,DSBS,EF,YTIME)/1000
                   +
                   (
                     (VFuelPriceSub(runCy,DSBS,EF,YTIME)+VTechCostVar(runCy,DSBS,EF,TEA,YTIME)/1000)/iUsfEneConvSubTech(runCy,DSBS,EF,YTIME)
                   )
```

```
                    * iAnnCons(runCy,DSBS,"smallest") * (iAnnCons(runCy,DSBS,"largest")/iAnnCons(runCy,DSBS,"smallest"))**((ord(rCon)-1)/iNcon(DSBS))
                  )$NENSE(DSBS);
```

- Compute technology cost including Maturity factor per technology and subsector

```
QTechCostMatr(runCy,DSBS,rCon,EF,TEA,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS)) $(ord(rCon) le iNcon(DSBS)+1) $SECTTECH(DSBS,EF) )..
VTechCostMatr(runCy,DSBS,rCon,EF,TEA,YTIME)
                                                    =E=
iMatrFactor(runCy,DSBS,EF,YTIME) * VTechCost(runCy,DSBS,rCon,EF,TEA,YTIME) ;
```

- Compute Technology sorting based on variable cost

```
QTechSort(runCy,DSBS,rCon,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS)) $(ord(rCon) le iNcon(DSBS)+1) )..
VTechSort(runCy,DSBS,rCon,YTIME)
                        =E=
sum((EF,TEA)$(SECTTECH(DSBS,EF) ),VTechCostMatr(runCy,DSBS,rCon,EF,TEA,YTIME));
```

- Compute the gap in final demand on industry, tertiary, non-energy uses and bunkers

```
QGapFinalDem(runCy,DSBS,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS)))..
         VGapFinalDem(runCy,DSBS,YTIME)
                 =E=
         VDemSub(runCy,DSBS,YTIME) - sum(EF$SECTTECH(DSBS,EF), VConsRemSubEquip(runCy,DSBS,EF,YTIME))
         + SQRT( SQR(VDemSub(runCy,DSBS,YTIME) - sum(EF$SECTTECH(DSBS,EF), VConsRemSubEquip(runCy,DSBS,EF,YTIME)))) /2;
```

- Compute technology share in new equipment

```
QTechShareNewEquip(runCy,DSBS,EF,TEA,YTIME)$(TIME(YTIME) $SECTTECH(DSBS,EF) $(not TRANSE(DSBS)) )..
         VTechShareNewEquip(runCy,DSBS,EF,TEA,YTIME) =E=
         iMatrFactor(runCy,DSBS,EF,YTIME) / iCumDistrFuncConsSize(runCy,DSBS) *
         sum(rCon$(ord(rCon) le iNcon(DSBS)+1),
                  VTechCost(runCy,DSBS,rCon,EF,TEA,YTIME)
                  * iDisFunConSize(runCy,DSBS,rCon)/VTechSort(runCy,DSBS,rCon,YTIME));
```

*Compute fuel consumption including heat from heatpumps

```
QFuelConsInclHP(runCy,DSBS,EF,YTIME)$(TIME(YTIME) $(not TRANSE(DSBS))
    $SECTTECH(DSBS,EF) )..
         VFuelConsInclHP(runCy,DSBS,EF,YTIME)
                 =E=
         VConsRemSubEquip(runCy,DSBS,EF,YTIME)+
         sum(TEA,VTechShareNewEquip(runCy,DSBS,EF,TEA,YTIME)*
             VGapFinalDem(runCy,DSBS,YTIME))
         + (VElecNonSub(runCy,DSBS,YTIME))$(INDDOM(DSBS) $(ELCEF(EF)));
```

- Compute variable including fuel electricity production cost per CHP plant and demand sector

```
QVarProCostPerCHPDem(runCy,DSBS,CHP,YTIME)$(TIME(YTIME) $INDDOM(DSBS))
    ..
         VProCostCHPDem(runCy,DSBS,CHP,YTIME)
                 =E=
         iVarCostChp(runCy,DSBS,CHP,YTIME)/1E3
                    + sum(PGEF$CHPtoEF(CHP,PGEF), (VFuelPriceSub(runCy,
                        "PG",PGEF,YTIME)+1e-3*iCo2EmiFac(runCy,"PG",PGEF
                        ,YTIME)*
                          (sum(NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,
                              NAP,YTIME))))
                          *sTWhToMtoe/(   iBoiEffChp(runCy,CHP,YTIME)*
                              VElecIndPrices(runCy,YTIME)    ));
```

- Compute Average Electricity production cost per CHP plant

```
QAvgElcProCostCHP(runCy,CHP,YTIME)$TIME(YTIME)..
         VAvgElcProCHP(runCy,CHP,YTIME)
         =E=
         (sum(INDDOM, VConsFuel(runCy,INDDOM,CHP,YTIME-1)/SUM(INDDOM2,
             VConsFuel(runCy,INDDOM2,CHP,YTIME-1))*VElecProdCostChp(
             runCy,INDDOM,CHP,YTIME)))
         $SUM(INDDOM2,VConsFuel.L(runCy,INDDOM2,CHP,YTIME-1))+0$(NOT
             SUM(INDDOM2,VConsFuel.L(runCy,INDDOM2,CHP,YTIME-1)));
```

- Compute Average variable including fuel electricity production cost per CHP plant

```
QAvgVarElecProd(runCy,CHP,YTIME)$(TIME(YTIME) ) ..
         VAvgVarProdCostCHP(runCy,CHP,YTIME)
         =E=
```

```
        (sum(INDDOM, VConsFuel(runCy,INDDOM,CHP,YTIME-1)/SUM(INDDOM2,
            VConsFuel(runCy,INDDOM2,CHP,YTIME-1))
        *VProCostCHPDem(runCy,INDDOM,CHP,YTIME)))
        $SUM(INDDOM2,VConsFuel.L(runCy,INDDOM2,CHP,YTIME-1))+0$(NOT
            SUM(INDDOM2,VConsFuel.L(runCy,INDDOM2,CHP,YTIME-1)));
```

- REST OF ENERGY BALANCE SECTORS
- Compute the transfomration output from patent fuel and briquetting plants,coke-oven plants,blast furnace plants and gas works

```
QTransfOutputPatFuel(runCy,EFS,YTIME)$TIME(YTIME)..
        VTransfOutputPatFuel(runCy,EFS,YTIME)
            =E=
        [
        iTransfOutputGasw(runCy,YTIME) * VTransfOutputPatFuel(runCy,
            EFS,YTIME-1) * (iActv(YTIME,runCy,"IS")/iActv(YTIME-1,runCy
            ,"IS"))**iElastA(runCy,"IS","a",YTIME)
        ]$iActv(YTIME-1,runCy,"IS")+0;
```

- Compute total final energy consumption

```
QTotFinEneCons(runCy,EFS,YTIME)$TIME(YTIME)..
        VFeCons(runCy,EFS,YTIME)
            =E=
        sum(INDDOM,
            sum(EF$(EFtoEFS(EF,EFS) $SECTTECH(INDDOM,EF) ), VConsFuel(
                runCy,INDDOM,EF,YTIME)))
        +
        sum(TRANSE,
            sum(EF$(EFtoEFS(EF,EFS) $SECTTECH(TRANSE,EF)), VDemTr(
                runCy,TRANSE,EF,YTIME)));
```

- Compute total final energy consumption in ALL countries

```
QTotFinEneConsAll(YTIME)$TIME(YTIME)..
        VTotFinEneConsAll(YTIME) =E= sum((runCy,EFS), VFeCons(runCy,
            EFS,YTIME) );
```

- Compute final non-energy consumption

```
QFinNonEneCons(runCy,EFS,YTIME)$TIME(YTIME)..
        VFNonEnCons(runCy,EFS,YTIME)
            =E=
```

```
        sum(NENSE$(not sameas("BU",NENSE)),
            sum(EF$(EFtoEFS(EF,EFS) $SECTTECH(NENSE,EF) ), VConsFuel(
                runCy,NENSE,EF,YTIME)));
```

- Compute distribution losses

```
QDistrLosses(runCy,EFS,YTIME)$TIME(YTIME)..
        VDistrLosses(runCy,EFS,YTIME)
            =E=
        (iRateLossesFinCons(runCy,EFS,YTIME) * (VFeCons(runCy,EFS,
            YTIME) + VFNonEnCons(runCy,EFS,YTIME)))$(not H2EF(EFS));
```

- Compute the transformation output from district heating plants

```
QTranfOutputDHPlants(runCy,STEAM,YTIME)$TIME(YTIME)..
        VTransfOutputDHPlants(runCy,STEAM,YTIME)
            =E=
        sum(DOMSE,
            sum(DH$(EFtoEFS(DH,STEAM) $SECTTECH(DOMSE,DH)), VConsFuel(
                runCy,DOMSE,DH,YTIME)));
```

- Compute the transformation input to distrcit heating plants

```
QTransfInputDHPlants(runCy,EFS,YTIME)$TIME(YTIME)..
        VTransfInputDHPlants(runCy,EFS,YTIME)
            =E=
        sum(DH$DHtoEF(DH,EFS),
            sum(DOMSE$SECTTECH(DOMSE,DH),VConsFuel(runCy,DOMSE,DH,
                YTIME)) / iEffDHPlants(runCy,EFS,YTIME));
```

- Compute the transfomration input to patent fuel and briquetting plants,coke-oven plants,blast furnace plants and gas works

```
QTransfInputPatFuel(runCy,EFS,YTIME)$TIME(YTIME)..
        VTransfInputPatFuel(runCy,EFS,YTIME)
            =E=
        sum(EF$(EFS(EF) $iAvgEffGas(runCy,EF,YTIME)) ,
            VTransfOutputPatFuel(runCy,EFS,YTIME)/iAvgEffGas(runCy,EF,
            YTIME)) * iShareFueTransfInput(runCy,EFS);
```

- Compute refineries capacity

```
QRefCapacity(runCy,YTIME)$TIME(YTIME)..
```

```
        VRefCapacity(runCy,YTIME)
            =E=
        [
        iResRefCapacity(runCy,YTIME) * VRefCapacity(runCy,YTIME-1)
        *
        (1$(ord(YTIME) le 16) +
        (prod(rc,
        (sum(EFS$EFtoEFA(EFS,"LQD"),VFeCons(runCy,EFS,YTIME-(ord(rc)
            +1)))/sum(EFS$EFtoEFA(EFS,"LQD"),VFeCons(runCy,EFS,YTIME-(
            ord(rc)+2))))**(0.5/(ord(rc)+1)))
        )
        $(ord(YTIME) gt 16)
        )     ] $iRefCapacity(runCy,"%fStartHorizon%");
```

- Compute the transformation output from refineries

```
QTranfOutputRefineries(runCy,EFS,YTIME)$(TIME(YTIME) $EFtoEFA(EFS,"LQD"
    )).
        VTransfOutputRefineries(runCy,EFS,YTIME)
            =E=
        [
        iResTransfOutputRefineries(runCy,EFS,YTIME) *
            VTransfOutputRefineries(runCy,EFS,YTIME-1)
        * (VRefCapacity(runCy,YTIME)/VRefCapacity(runCy,YTIME-1))**0.3
        * (
            1$(TFIRST(YTIME-1) or TFIRST(YTIME-2))
            +
            (
                sum(EF$EFtoEFA(EF,"LQD"),VFeCons(runCy,EF,YTIME-1))/sum
                    (EF$EFtoEFA(EF,"LQD"),VFeCons(runCy,EF,YTIME-2))
            )$(not (TFIRST(YTIME-1) or TFIRST(YTIME-2)))
          )**(0.7)  ]$iRefCapacity(runCy,"%fStartHorizon%");
```

- Compute the transformation input to refineries

```
QTransfInputRefineries(runCy,"CRO",YTIME)$(TIME(YTIME) )..
        VTransfInputRefineries(runCy,"CRO",YTIME)
            =E=
        [
        VTransfInputRefineries(runCy,"CRO",YTIME-1) *
        sum(EFS$EFtoEFA(EFS,"LQD"), VTransfOutputRefineries(runCy,EFS,
            YTIME)) /
        sum(EFS$EFtoEFA(EFS,"LQD"), VTransfOutputRefineries(runCy,EFS,
```

```
                YTIME-1))  ]$iRefCapacity(runCy,"%fStartHorizon%");
```

- Compute transformation output from nuclear plants

```
QTransfOutputNuclear(runCy,"ELC",YTIME)$TIME(YTIME) ..
         VTransfOutputNuclear(runCy,"ELC",YTIME) =E=SUM(PGNUCL,
            VElecProd(runCy,PGNUCL,YTIME))*sTWhToMtoe;
```

- Compute transformation input to nuclear plants

```
QTransfInNuclear(runCy,"NUC",YTIME)$TIME(YTIME)..
        VTransfInNuclear(runCy,"NUC",YTIME) =E=SUM(PGNUCL,VElecProd(
           runCy,PGNUCL,YTIME)/iPlantEffByType(runCy,PGNUCL,YTIME))*
           sTWhToMtoe;
```

- Compute transformation input to power plants

```
QTransfInPowerPls(runCy,PGEF,YTIME)$TIME(YTIME)..
         VTransfInThermPowPls(runCy,PGEF,YTIME)
             =E=
        sum(PGALL$(PGALLtoEF(PGALL,PGEF)$((not PGGEO(PGALL)) $(not
           PGNUCL(PGALL)))),
             VElecProd(runCy,PGALL,YTIME) * sTWhToMtoe /
                iPlantEffByType(runCy,PGALL,YTIME))
        +
        sum(PGALL$(PGALLtoEF(PGALL,PGEF)$PGGEO(PGALL)),
             VElecProd(runCy,PGALL,YTIME) * sTWhToMtoe)
        +
        sum(CHP$CHPtoEF(CHP,PGEF),  sum(INDDOM,VConsFuel(runCy,INDDOM,
           CHP,YTIME))+sTWhToMtoe*VChpElecProd(runCy,CHP,YTIME))
           /(0.8+0.1*(ord(YTIME)-16)/32);
```

- Compute transformation output from thermal power plants

```
QTransfOutThermPP(runCy,TOCTEF,YTIME)$TIME(YTIME)..
         VTransfOutThermPowSta(runCy,TOCTEF,YTIME)
             =E=
        (
             sum(PGALL$(not PGNUCL(PGALL)),VElecProd(runCy,PGALL,YTIME)
                ) * sTWhToMtoe
             +
             sum(CHP,VChpElecProd(runCy,CHP,YTIME)*sTWhToMtoe)
         )$ELCEF(TOCTEF)
```

```
        +
        (
          sum(INDDOM,
          sum(CHP$SECTTECH(INDDOM,CHP), VConsFuel(runCy,INDDOM,CHP,
              YTIME)))+
          iRateEneBranCons(runCy,TOCTEF,YTIME)*(VFeCons(runCy,TOCTEF,
              YTIME) + VFNonEnCons(runCy,TOCTEF,YTIME) + VLosses(runCy,
              TOCTEF,YTIME)) +
          VLosses(runCy,TOCTEF,YTIME)
         )$STEAM(TOCTEF);
```

- Computer total transformation input

```
QTotTransfInput(runCy,EFS,YTIME)$TIME(YTIME)..
         VTotTransfInput(runCy,EFS,YTIME)
                 =E=
        (
            VTransfInThermPowPls(runCy,EFS,YTIME) +
               VTransfInputDHPlants(runCy,EFS,YTIME) + VTransfInNuclear
               (runCy,EFS,YTIME) +
             VTransfInputPatFuel(runCy,EFS,YTIME) +
                VTransfInputRefineries(runCy,EFS,YTIME)     !!$H2PRODEF
                (EFS)
        )$(not sameas(EFS,"OGS"))
        +
        (
          VTotTransfOutput(runCy,EFS,YTIME) - VFeCons(runCy,EFS,YTIME)
              - VFNonEnCons(runCy,EFS,YTIME) - iRateEneBranCons(runCy,
              EFS,YTIME)*
          VTotTransfOutput(runCy,EFS,YTIME) - VLosses(runCy,EFS,YTIME)
        )$sameas(EFS,"OGS");
```

- Compute total transformation output

```
QTotTransfOutput(runCy,EFS,YTIME)$TIME(YTIME)..
         VTotTransfOutput(runCy,EFS,YTIME)
                 =E=
         VTransfOutThermPowSta(runCy,EFS,YTIME) + VTransfOutputDHPlants
             (runCy,EFS,YTIME) + VTransfOutputNuclear(runCy,EFS,YTIME) +
              VTransfOutputPatFuel(runCy,EFS,YTIME) +
         VTransfOutputRefineries(runCy,EFS,YTIME);        !!+ TONEW(
             runCy,EFS,YTIME)
```

- Compute transfers

```
QTransfers(runCy,EFS,YTIME)$TIME(YTIME)..
         VTransfers(runCy,EFS,YTIME) =E=
         (( (VTransfers(runCy,EFS,YTIME-1)*iResFeedTransfr(runCy,YTIME)
            *VFeCons(runCy,EFS,YTIME)/VFeCons(runCy,EFS,YTIME-1))
            $EFTOEFA(EFS,"LQD")+
          (
                VTransfers(runCy,"CRO",YTIME-1)*iResFeedTransfr(runCy,
                   YTIME)*SUM(EFS2$EFTOEFA(EFS2,"LQD"),VTransfers(
                   runCy,EFS2,YTIME))/
                SUM(EFS2$EFTOEFA(EFS2,"LQD"),VTransfers(runCy,EFS2,
                   YTIME-1)))$sameas(EFS,"CRO")    )$(iFeedTransfr(
                   runCy,EFS,"%fStartHorizon%"))$(NOT sameas("OLQ",EFS
                   ))
);
```

- Compute gross inland consumption not including consumption of energy branch

```
 QGrsInlConsNotEneBranch(runCy,EFS,YTIME)$TIME(YTIME)..
         VGrsInlConsNotEneBranch(runCy,EFS,YTIME)
                 =E=
         VFeCons(runCy,EFS,YTIME) + VFNonEnCons(runCy,EFS,YTIME) +
            VTotTransfInput(runCy,EFS,YTIME) - VTotTransfOutput(runCy,
            EFS,YTIME) + VLosses(runCy,EFS,YTIME) -
         VTransfers(runCy,EFS,YTIME);
```

- Compute gross inland consumption

```
QGrssInCons(runCy,EFS,YTIME)$TIME(YTIME)..
         VGrssInCons(runCy,EFS,YTIME)
                 =E=
         VFeCons(runCy,EFS,YTIME) + VEnCons(runCy,EFS,YTIME) +
            VFNonEnCons(runCy,EFS,YTIME) + VTotTransfInput(runCy,EFS,
            YTIME) - VTotTransfOutput(runCy,EFS,YTIME) +
          VLosses(runCy,EFS,YTIME) - VTransfers(runCy,EFS,YTIME);
```

- Compute primary production

```
QPrimProd(runCy,PPRODEF,YTIME)$TIME(YTIME)..
         VPrimProd(runCy,PPRODEF,YTIME)
                 =E=  [
         (
```

```
            iRatePriProTotPriNeeds(runCy,PPRODEF,YTIME) * (
               VGrsInlConsNotEneBranch(runCy,PPRODEF,YTIME) +  VEnCons
               (runCy,PPRODEF,YTIME))
         )$(not (sameas(PPRODEF,"CRO")or sameas(PPRODEF,"NGS")))
         +
         (
            iResHcNgOilPrProd(runCy,PPRODEF,YTIME) * VPrimProd(runCy,
               PPRODEF,YTIME-1) *
            (VGrsInlConsNotEneBranch(runCy,PPRODEF,YTIME)/
               VGrsInlConsNotEneBranch(runCy,PPRODEF,YTIME-1))**
               iNatGasPriProElst(runCy)
         )$(sameas(PPRODEF,"NGS") )
        +
         (
           iRatePriProTotPriNeeds(runCy,PPRODEF,YTIME) * VPrimProd(
              runCy,PPRODEF,YTIME-1) *
           ((VGrsInlConsNotEneBranch(runCy,PPRODEF,YTIME) +
              VExportsFake(runCy,PPRODEF,YTIME))/
            (VGrsInlConsNotEneBranch(runCy,PPRODEF,YTIME-1) +
               VExportsFake(runCy,PPRODEF,YTIME-1)))
         )$(sameas(PPRODEF,"NGS") )
         +(
           iResHcNgOilPrProd(runCy,PPRODEF,YTIME) *  iFuelPriPro(runCy,
              PPRODEF,YTIME) *
           prod(kpdl$(ord(kpdl) lt 5),
                        (iPriceFuelsInt("WCRO",YTIME-(ord(kpdl)+1))/
                           iPriceFuelsIntBase("WCRO",YTIME-(ord(kpdl)
                           +1)))
                        **(0.2*iPolDstrbtnLagCoeffPriOilPr(kpdl)))
         )$sameas(PPRODEF,"CRO")   ]$iRatePriProTotPriNeeds(runCy,
            PPRODEF,YTIME);
```

- Compute fake exports

```
QFakeExp(runCy,EFS,YTIME)$(TIME(YTIME) $IMPEF(EFS))..
         VExportsFake(runCy,EFS,YTIME)
                 =E=
         (
                 iFuelExprts(runCy,EFS,YTIME)
         )
+  iFuelExprts(runCy,EFS,YTIME);
```

- Compute fake imports

```
QFakeImprts(runCy,EFS,YTIME)$(TIME(YTIME) $IMPEF(EFS))..
         VFkImpAllFuelsNotNatGas(runCy,EFS,YTIME)
                 =E=
         (
            iRatioImpFinElecDem(runCy,YTIME) * (VFeCons(runCy,EFS,YTIME) + VFNonEnCons(runCy,EFS,YTIME)) + VExportsFake(runCy,EFS,YTIME)
         + iElecImp(runCy,YTIME)
         )$ELCEF(EFS)
         +
         (
            VGrssInCons(runCy,EFS,YTIME)+ VExportsFake(runCy,EFS,YTIME) + VConsFuel(runCy,"BU",EFS,YTIME)$SECTTECH("BU",EFS)
            - VPrimProd(runCy,EFS,YTIME)
         )$(sameas(EFS,"CRO"))
         +
         (
            VGrssInCons(runCy,EFS,YTIME)+ VExportsFake(runCy,EFS,YTIME) + VConsFuel(runCy,"BU",EFS,YTIME)$SECTTECH("BU",EFS)
            - VPrimProd(runCy,EFS,YTIME)
         )$(sameas(EFS,"NGS"))
         +iImpExp(runCy,"NGS",YTIME)$(sameas(EFS,"NGS"))
         +
         (
            (1-iRatePriProTotPriNeeds(runCy,EFS,YTIME)) *
            (VGrssInCons(runCy,EFS,YTIME) + VExportsFake(runCy,EFS,YTIME) + VConsFuel(runCy,"BU",EFS,YTIME)$SECTTECH("BU",EFS) )
         )$(not (ELCEF(EFS) or sameas(EFS,"NGS") or sameas(EFS,"CRO")))
            ;
```

- Compute net imports

```
QNetImports(runCy,EFS,YTIME)$TIME(YTIME)..
         VNetImports(runCy,EFS,YTIME)
                 =E=
         VFkImpAllFuelsNotNatGas(runCy,EFS,YTIME) - VExportsFake(runCy,EFS,YTIME);
```

- Compute energy branch final consumption

```
QEneBrnchEneCons(runCy,EFS,YTIME)$TIME(YTIME)..
```

```
        VEnCons(runCy,EFS,YTIME)
                =E=
        iRateEneBranCons(runCy,EFS,YTIME) *
        (
          (
            VTotTransfOutput(runCy,EFS,YTIME) +
            VPrimProd(runCy,EFS,YTIME)$(sameas(EFS,"CRO") or sameas(EFS,"NGS"))
           )$(not TOCTEF(EFS))
           +
           (
            VFeCons(runCy,EFS,YTIME) + VFNonEnCons(runCy,EFS,YTIME) + VLosses(runCy,EFS,YTIME)
           )$TOCTEF(EFS)
        );
```

- CO2 SEQUESTRATION COST CURVES
- Compute CO2 captured by electricity and hydrogen production plants

```
QCO2ElcHrg(runCy,YTIME)$TIME(YTIME)..
        VCO2ElcHrgProd(runCy,YTIME)
        =E=
        sum(PGEF,sum(CCS$PGALLtoEF(CCS,PGEF),
                VElecProd(runCy,CCS,YTIME)*sTWhToMtoe/iPlantEffByType(runCy,CCS,YTIME)*
                iCo2EmiFac(runCy,"PG",PGEF,YTIME)*iCO2CaptRate(runCy,CCS,YTIME)));
```

- Compute cumulative CO2 captured

```
QCumCO2Capt(runCy,YTIME)$TIME(YTIME)..
        VCumCO2Capt(runCy,YTIME) =E= VCumCO2Capt(runCy,YTIME-1)+ VCO2ElcHrgProd(runCy,YTIME-1);
```

- Compute the transition weight from linear CO2 sequestration cost curve to exponential

```
QWghtTrnstLinToExpo(runCy,YTIME)$TIME(YTIME)..
        VWghtTrnstLnrToExpo(runCy,YTIME)
        =E=
        1/(1+exp(-iElastCO2Seq(runCy,"mc_s")*( VCumCO2Capt(runCy,YTIME)/iElastCO2Seq(runCy,"pot")-iElastCO2Seq(runCy,"mc_m"))));
```

- Compute cost curve for CO2 sequestration costs in Euro per tn of CO2 sequestrated

```
QCstCO2SeqCsts(runCy,YTIME)$TIME(YTIME)..
         VCO2CO2SeqCsts(runCy,YTIME) =E=
       (1-VWghtTrnstLnrToExpo(runCy,YTIME))*(iElastCO2Seq(runCy,"mc_a")
          *VCumCO2Capt(runCy,YTIME)+iElastCO2Seq(runCy,"mc_b"))+
       VWghtTrnstLnrToExpo(runCy,YTIME)*(iElastCO2Seq(runCy,"mc_c")*exp
          (iElastCO2Seq(runCy,"mc_d")*VCumCO2Capt(runCy,YTIME)));
```

- EMISSIONS CONSTRAINTS
- Compute total CO2eq GHG emissions in all countries per NAP sector

```
QTotGhgEmisAllCountrNap(NAP,YTIME)$TIME(YTIME)..
         VTotGhgEmisAllCountrNap(NAP,YTIME)
          =E=
        (
        sum(runCy,
                 sum((EFS,INDSE)$(SECTTECH(INDSE,EFS)  $NAPtoALLSBS(NAP
                    ,INDSE)),
                      VConsFuel(runCy,INDSE,EFS,YTIME) * iCo2EmiFac(
                         runCy,INDSE,EFS,YTIME)) !! final consumption
                +
                 sum(PGEF, VTransfInThermPowPls(runCy,PGEF,YTIME)*
                    iCo2EmiFac(runCy,"PG",PGEF,YTIME)$(not h2f1(pgef)))
                     !! input to power generation sector
                 +
                 sum(EFS, VTransfInputDHPlants(runCy,EFS,YTIME)*
                    iCo2EmiFac(runCy,"PG",EFS,YTIME)) !! input to
                    district heating plants
                 +
                 sum(EFS, VEnCons(runCy,EFS,YTIME)*iCo2EmiFac(runCy,"
                    PCH",EFS,YTIME)) !! consumption of energy branch
                 -
                 sum(PGEF,sum(CCS$PGALLtoEF(CCS,PGEF),
                         VElecProd(runCy,CCS,YTIME)*sTWhToMtoe/
                            iPlantEffByType(runCy,CCS,YTIME)*
                         iCo2EmiFac(runCy,"PG",PGEF,YTIME)*iCO2CaptRate
                            (runCy,CCS,YTIME)))));   !! CO2 captured by
                             CCS plants in power generation
```

- Compute total CO2eq GHG emissions in all countries

```
QTotCo2AllCoun(YTIME)$TIME(YTIME)..
         VTotCo2AllCoun(YTIME)
```

```
        =E=
        sum(NAP, VTotGhgEmisAllCountrNap(NAP,YTIME));
```

- Compute households expenditures on energy

```
QHouseExpEne(runCy,YTIME)$TIME(YTIME)..
                VHouseExpEne(runCy,YTIME)
                =E=
                SUM(DSBS$HOU(DSBS),SUM(EF$SECTTECH(dSBS,EF),
                    VConsRemSubEquip(runCy,DSBS,EF,YTIME)*(
                    VFuelPriceSub(runCy,DSBS,EF,YTIME)-iEffValueInEuro(
                    runCy,DSBS,YTIME)/
                1000-iCo2EmiFac(runCy,"PG",EF,YTIME)*sum(
                    NAP$NAPtoALLSBS(NAP,"PG"),VCarVal(runCy,NAP,YTIME))
                    /1000)))
                                        +VElecPriIndResNoCliPol(runCy
                                            ,"R",YTIME)*VElecNonSub(
                                            runCy,"HOU",YTIME)/
                                            sTWhToMtoe;
```

- Prices
- Compute fuel prices per subsector and fuel, separate carbon value in each sector

```
QFuelPriSubSepCarbVal(runCy,SBS,EF,YTIME)$(SECTTECH(SBS,EF) $TIME(YTIME
    ) $(not sameas("NUC",EF)))..
        VFuelPriceSub(runCy,SBS,EF,YTIME)
                =E=
      (
          (1+sum(WEF$EFtoWEF(SBS,EF,WEF),sum(NAP$NAPtoALLSBS(NAP,SBS),
              iScenarioPri(WEF,NAP,YTIME))))*     !!ONLY FOR THE ETS
              SECTORS
        (
          (1-iPriceReform(runCy,SBS,EF,YTIME))*
          ( VFuelPriceSub(runCy,SBS,EF,YTIME-1))
          + iPriceReform(runCy,SBS,EF,YTIME)*iPriceTragets(runCy,SBS,
              EF,YTIME)
)
         + iCo2EmiFac(runCy,SBS,EF,YTIME) *sum(NAP$NAPtoALLSBS(NAP,SBS
             ),(VCarVal(runCy,NAP,YTIME)))/1000
         + ((iEffValueInEuro(runCy,SBS,YTIME)-iEffValueInEuro(runCy,
             SBS,YTIME-1))/1000)$DSBS(SBS)
        )$( not (ELCEF(EF) or HEATPUMP(EF) or ALTEF(EF)))
        +
```

```
        (
iFuelPrice(runCy,SBS,EF,"2018") $(DSBS(SBS))$ALTEF(EF)
        )
        +
        (
          ( VElecPriInduResConsu(runCy,"i",YTIME)$INDTRANS(SBS)+
             VElecPriInduResConsu(runCy,"r",YTIME)$RESIDENT(SBS))/
             sTWhToMtoe
           +
           ((iEffValueInEuro(runCy,SBS,YTIME))/1000)$DSBS(SBS)
        )$(ELCEF(EF) or HEATPUMP(EF))
        +
        (
                iHydrogenPri(runCy,SBS,YTIME-1)$DSBS(SBS)
        )$(H2EF(EF) or sameas("STE1AH2F",EF));
```

- Compute fuel prices per subsector and fuel multiplied by weights separate carbon value in each sector

```
QFuelPriSepCarbon(runCy,DSBS,EF,YTIME)$(SECTTECH(DSBS,EF) $TIME(YTIME))
    ..
        VFuelPriMultWgt(runCy,DSBS,EF,YTIME)
          =E=
        iWgtSecAvgPriFueCons(runCy,DSBS,EF) * VFuelPriceSub(runCy,DSBS,
           EF,YTIME);
```

- Compute average fuel price per subsector

```
QAvgFuelPriSub(runCy,DSBS,YTIME)$TIME(YTIME)..
        VFuelPriceAvg(runCy,DSBS,YTIME)
                =E=
         sum(EF$SECTTECH(DSBS,EF), VFuelPriMultWgt(runCy,DSBS,EF,YTIME)
            );
```

- Compute electricity price in Industrial and Residential Consumers

```
QElecPriIndResCons(runCy,ESET,YTIME)$TIME(YTIME)..  !! The electricity
    price is based on previous year's production costs
        VElecPriInduResConsu(runCy,ESET,YTIME)
                =E=
        (1 + iFacElecPriConsu(runCy,"VAT",YTIME)) *
        (
```

```
            (
              (VFuelPriceSub(runCy,"OI","ELC",YTIME-1)*sTWhToMtoe)
                 $TFIRST(YTIME-1) +
              (  iFacElecPriConsu(runCy,"IND_RES",YTIME-1) +
                 VRenShareElecProdSub(runCy,YTIME-1)*(VRenValue(YTIME)
                 *8.6e-5)+
                 iFacElecPriConsu(runCy,"W_INDU",YTIME-1)*
                    VAvgPowerGenLongTrm(runCy,"i",YTIME-1) +
                (1-iFacElecPriConsu(runCy,"W_INDU",YTIME-1))*
                   VAvgPowerGenCostShoTrm(runCy,"i",YTIME-1)
               )$(not TFIRST(YTIME-1))
            )$ISET(ESET)
         +
            (
              (VFuelPriceSub(runCy,"HOU","ELC",YTIME-1)*sTWhToMtoe)
                 $TFIRST(YTIME-1) +
              (  iFacElecPriConsu(runCy,"TERT_RES",YTIME-1) +
                 VRenShareElecProdSub(runCy,YTIME-1)*(VRenValue(YTIME)
                 *8.6e-5)+
                 iFacElecPriConsu(runCy,"W_TERT",YTIME-1)*
                    VAvgPowerGenLongTrm(runCy,"r",YTIME-1) +
                 (1-iFacElecPriConsu(runCy,"W_TERT",YTIME-1))*
                    VAvgPowerGenCostShoTrm(runCy,"r",YTIME-1)
              )$(not TFIRST(YTIME-1))
            )$RSET(ESET)
         );
```

- Define dummy objective function

```
qDummyObj.. vDummyObj =e= 1;
```

## input (input.gms)

### Description

```
table iGDP(YTIME,allCy) "GDP (billion US$2015)"
$ondelim
$include "./iGDP.csvr"
$offdelim
;
table iPop(YTIME,allCy) "Population (billion)"
$ondelim
$include "./iPop.csvr"
$offdelim
;
table iActv(YTIME,allCy,SBS) "Sector activity (various)"
                              !! main sectors (Billion Euro05)
                              !! bunkers and households (1)
                              !! transport (Gpkm, or Gvehkm or Gtkm)
$ondelim
$include "./iActv.csvr"
$offdelim
;
table iTransChar(allCy,TRANSPCHAR,YTIME) "km per car, passengers per car and residuals for passenger cars market extension ()"
$ondelim
$include "./iTransChar.csv"
$offdelim
;
table iElastA(allCy,SBS,ETYPES,YTIME) "Activity Elasticities per subsector (1)"
$ondelim
$include "./iElastA.csv"
$offdelim
;
table iResActiv(allCy,TRANSE,YTIME) "Residuals on transport activity ()"
$ondelim
$include "./iResActiv.csv"
$offdelim
;
table iElastNonSubElec(allCy,SBS,ETYPES,YTIME) "Elasticities of Non Substitutable Electricity (1)"
$ondelim
```

```
$include "./iElastNonSubElec.csv"
$offdelim
;
table iFracElecPriChp(allCy, YTIME) "Fraction of Electricity Price at which a CHP sells electricity to network (1)"
$ondelim
$include "./iFracElecPriChp.csv"
$offdelim
;
table iDisc(allCy,SBS,YTIME) "Discount rates per subsector ()"
$ondelim
$include "./iDisc.csv"
$offdelim
;
table iCo2EmiFacAllSbs(allCy,EF) "CO2 emission factors (kgCO2/kgoe fuel burned)"
$ondelim
$include "./iCo2EmiFacAllSbs.csv"
$offdelim
;
iCo2EmiFac(allCy,SBS,EF,YTIME) = iCo2EmiFacAllSbs(allCy,EF);
iCo2EmiFac(allCy,"IS","HCL",YTIME)$(not An(YTIME))   = iCo2EmiFacAllSbs(allCy,"SLD"); !! This is the assignment for coke
table iDataPassCars(allCy,GompSet1,Gompset2)        "Initial Data for Passenger Cars ()"
$ondelim
$include "./iDataPassCars.csv"
$offdelim
;
iSigma(allCy,"S1") = iDataPassCars(allCy,"PC","S1");
iSigma(allCy,"S2") = iDataPassCars(allCy,"PC","S2");
iSigma(allCy,"S3") = iDataPassCars(allCy,"PC","S3");
iPassCarsMarkSat(allCy) = iDataPassCars(allCy,"PC","SAT");
iGdpPassCarsMarkExt(allCy) = iDataPassCars(allCy,"PC","MEXTV");
iPassCarsScrapRate(allCy)  = iDataPassCars(allCy,"PC", "SCR");
table iInitSpecFuelCons(allCy,TRANSE,TTECH,EF,YTIME)        "Initial Specific fuel consumption ()"
$ondelim
$include "./iInitSpecFuelCons.csv"
$offdelim
;
iSpeFuelConsCostBy(allCy,TRANSE,TTECH,TEA,EF) = iInitSpecFuelCons(allCy,TRANSE,TTECH,EF,"2017");
```

```
table iElaSub(allCy,DSBS)                                "Elasticities by subsectors (1)"
$ondelim
$include "./iElaSub.csv"
$offdelim
;
table iConsSizeDistHeat(allCy,conSet)                    "Consumer sizes for district heating (1)"
$ondelim
$include "./iConsSizeDistHeat.csv"
$offdelim
;
table iCapCostTechTr(allCy,TRANSE,EF,YTIME)      "Capital Cost of technology for transport (kEuro2005/vehicle)"
$ondelim$include "./iCapCostTechTr.csv"
$offdelim
;
table iRateLossesFinConsSup(allCy,EF, YTIME)                "Supplementary parameter for Rate of losses over Available for Final Consumption (1)"
$ondelim
$include "./iRateLossesFinConsSup.csv"
$offdelim
;
table iEneProdRDscenariosSupplement(allCy,SBS,YTIME)       "Supplementary Parameter for Energy productivity indices and R&D indices (1)"
$ondelim
$include "./iEneProdRDscenariosSupplement.csv"
$offdelim
;
iEneProdRDscenarios(runCy,SBS,YTIME)=iEneProdRDscenariosSupplement(runCy,SBS,YTIME);
table iParDHEfficiency(PGEFS,YTIME)                      "Parameter of district heating Efficiency (1)"
$ondelim
$include "./iParDHEfficiency.csv"
$offdelim
;
table iAvgEffGas(allCy,EF,YTIME)                         "Average Efficiency of Gasworks, Blast Furnances, Briquetting plants (1)"
$ondelim
$include "./iAvgEffGas.csv"
```

```
$offdelim
;
table iSuppTransfInputPatFuel(EF,YTIME)            "Supplementary Parameter for the transformation input to patent fuel and briquetting plants,coke-oven plants,blast furnace plants and gas works (1)"
$ondelim
$include "./iSuppTransfInputPatFuel.csv"
$offdelim
;
table iSupResRefCapacity(allCy,SUPOTH,YTIME)               "Supplementary Parameter for the residual in refineries Capacity (1)"
$ondelim
$include "./iSupResRefCapacity.csv"
$offdelim
;
table iSuppRefCapacity(allCy,SUPOTH,YTIME)            "Supplementary Parameter for the residual in refineries Capacity (1)"
$ondelim
$include "./iSuppRefCapacity.csv"
$offdelim
;
table iSupTrnasfOutputRefineries(allCy,EF,YTIME)         "Supplementary parameter for the transformation output from refineries (Mtoe)"
$ondelim
$include"./iSupTrnasfOutputRefineries.csv"
$offdelim
;
table iSupRateEneBranCons(allCy,EF,YTIME)             "Rate of Energy Branch Consumption over total transformation output of iRateEneBranCons (1)"
$ondelim
$include"./iSupRateEneBranCons.csv"
$offdelim
;
table iSuppTransfers(allCy,EFS,YTIME)                   "Supplementary Parameter for Transfers (Mtoe)"
$ondelim
$include"./iSuppTransfers.csv"
$offdelim
;
table iSuppPrimProd(allCy,PPRODEF,YTIME)              "Supplementary Parameter for Primary Production (Mtoe)"
```

```
$ondelim
$include"./iSuppPrimProd.csv"
$offdelim
;
iFuelPriPro(allCy,PPRODEF,YTIME) = iSuppPrimProd(allCy,PPRODEF,YTIME);
table iPriceFuelsInt(WEF,YTIME)                "International Fuel
    Prices ($2015/toe)"
$ondelim
$include"./iPriceFuelsInt.csv"
$offdelim
;
table iPriceFuelsIntBase(WEF,YTIME)            "International Fuel
    Prices USED IN BASELINE SCENARIO ($2015/toe)"
$ondelim
$include"./iPriceFuelsIntBase.csv"
$offdelim
;
table iSuppRatePrimProd(allCy,EF,YTIME)               "Supplementary
    Parameter for iRatePrimProd (1)"
$ondelim
$include"./iSuppRatePrimProd.csv"
$offdelim
;
table iSuppExports(allCy,EF,YTIME)                   "Supplementary
    parameter for  exports (Mtoe)"
$ondelim
$include"./iSuppExports.csv"
$offdelim
;
table iImpExp(allCy,EFS,YTIME)                   "Imports of exporting
    countries usually zero (1)"
$ondelim
$include"./iImpExp.csv"
$offdelim
;
table iLoadFactorAdj(allCy,DSBS,YTIME)  "Parameters for load factor
    adjustment iBaseLoadShareDem (1)"
$ondelim
$include"./iLoadFactorAdj.csv"
$offdelim
;
iBaseLoadShareDem(allCy,DSBS,YTIME)$an(YTIME)  = iLoadFactorAdj(allCy,
    DSBS,YTIME);
```

```
table iCO2SeqData(allCy,CO2SEQELAST,YTIME)         "Data for CO2 sequestration (1)"
$ondelim
$include"./iCO2SeqData.csv"
$offdelim
;
iElastCO2Seq(allCy,CO2SEQELAST) = sum(tfirst,iCO2SeqData(allCy,CO2SEQELAST,TFIRST));
table iDataTransTech (TRANSE,EF,ECONCHAR)                  "Technoeconomic characteristics of transport (various)"
$ondelim
$include"./iDataTransTech.csv"
$offdelim
;
table iDataIndTechnology(INDSE,EF,ECONCHAR)                  "Technoeconomic characteristics of industry (various)"
$ondelim
$include"./iDataIndTechnology.csv"
$offdelim
;
table iDataDomTech(DOMSE,EF,ECONCHAR)                  "Technical lifetime of Industry (years)"
$ondelim
$include"./iDataDomTech.csv"
$offdelim
;
table iDataNonEneSec(NENSE,EF,ECONCHAR)                  "Technical data of non energy uses and bunkers (various)"
$ondelim
$include"./iDataNonEneSec.csv"
$offdelim
;
table iIndChar(allCy,INDSE,Indu_Scon_Set)                "Industry sector charactetistics (various)"
$ondelim
$include"./iIndChar.csv"
$offdelim
;
table iInitConsSubAndInitShaNonSubElec(allCy,DOMSE,Indu_Scon_Set) "Initial Consumption per Subsector and Initial Shares of Non Substitutable Electricity in Total Electricity Demand (Mtoe)"
$ondelim
$include"./iInitConsSubAndInitShaNonSubElec.csv"
```

```
$offdelim
;
iShrHeatPumpElecCons(allCy,INDSE) = iIndChar(allCy,INDSE,"SH_HPELC");
iShrHeatPumpElecCons(allCy,DOMSE) = iInitConsSubAndInitShaNonSubElec(allCy,DOMSE,"SH_HPELC");
iExogDemOfBiomass(allCy,DOMSE,YTIME) = 0;
iElastCO2Seq(allCy,CO2SEQELAST) = iCO2SeqData(allCy,CO2SEQELAST,"%fBaseY%");
iRatioImpFinElecDem(runCy,YTIME)$an(YTIME) = iSuppRefCapacity(runCy,"ELC_IMP",YTIME);
iFuelExprts(runCy,EFS,YTIME) = iSuppExports(runCy,EFS,YTIME);
iRatePriProTotPriNeeds(runCy,PPRODEF,YTIME) = iSuppRatePrimProd(runCy,PPRODEF,YTIME);
iResHcNgOilPrProd(runCy,"HCL",YTIME)$an(YTIME)   = iSupResRefCapacity(runCy,"HCL_PPROD",YTIME);
iResHcNgOilPrProd(runCy,"NGS",YTIME)$an(YTIME)   = iSupResRefCapacity(runCy,"NGS_PPROD",YTIME);
iResHcNgOilPrProd(runCy,"CRO",YTIME)$an(YTIME)   = iSupResRefCapacity(runCy,"OIL_PPROD",YTIME);
iResFeedTransfr(runCy,YTIME)$an(YTIME) = iSupResRefCapacity(runCy,"FEED_RES",YTIME);
iFeedTransfr(runCy,EFS,YTIME) = iSuppTransfers(runCy,EFS,YTIME);
iRateEneBranCons(allCy,EFS,YTIME)= iSupRateEneBranCons(allCy,EFS,YTIME)*iEneProdRDscenarios(allCy,"PG",YTIME);
iResTransfOutputRefineries(runCy,EFS,YTIME) = iSupTrnasfOutputRefineries(runCy,EFS,YTIME);
iRefCapacity(runCy,YTIME)= iSuppRefCapacity(runCy,"REF_CAP",YTIME);
iResRefCapacity(runCy,YTIME) = iSupResRefCapacity(runCy,"REF_CAP_RES",YTIME);
iTransfInpGasworks(runCy,EFS,YTIME)= iSuppTransfInputPatFuel(EFS,YTIME);
iShareFueTransfInput(runCy,EFS)$sum(EF$EFS(EF),iTransfInpGasworks(runCy,EF,"%fBaseY%")) =  iTransfInpGasworks(runCy,EFS,"%fBaseY%") / sum(EF$EFS(EF),iTransfInpGasworks(runCy,EF,"%fBaseY%"));
iRateLossesFinCons(runCy,EFS,YTIME)$an(YTIME)  = iRateLossesFinConsSup(runCy,EFS, YTIME)*iEneProdRDscenarios(runCy,"PG",YTIME);
table iPwrLoadFactorDem(allCy,SBS,YTIME)              "Parameters for load factor adjustment (1)"
$ondelim
$include"./iPwrLoadFactorDem.csv"
$offdelim
;
```

```
table iLoadFactorAdjMxm(allCy,VARIOUS_LABELS,YTIME)                    "Parameter for load factor adjustment iMxmLoadFacElecDem (1)"
$ondelim
$include"./iLoadFactorAdjMxm.csv"
$offdelim
;
iBslCorrection(allCy,YTIME)$an(YTIME) = iLoadFactorAdjMxm(allCy,"AMAXBASE",YTIME);
iMxmLoadFacElecDem(allCy,YTIME)$an(YTIME) = iLoadFactorAdjMxm(allCy,"MAXLOADSH",YTIME);
iLoadFacElecDem(allCy,DSBS,YTIME)$(ord(YTIME)>(ordfirst-4)) = iPwrLoadFactorDem(allCy,DSBS,YTIME);
```

*Calculation of consumer size groups and their distribution function

```
iNcon(TRANSE)$(sameas(TRANSE,"PC") or sameas(TRANSE,"GU")) = 10; !! 11 different consumer size groups for cars and trucks
iNcon(TRANSE)$(not (sameas(TRANSE,"PC") or sameas(TRANSE,"GU"))) = 1; !! 2 different consumer size groups for inland navigation, trains, busses and aviation
iNcon(INDSE) = 10; !! 11 different consumer size groups for industrial sectors
iNcon(DOMSE) = 10; !! 11 different consumer size groups for domestic and tertiary sectors
iNcon(NENSE) = 10; !! 11 different consumer size groups for non energy uses
iNcon("BU") = 2;   !! ... except bunkers .
```

- 11 vehicle mileage groups
- 0.952 turned out to be a (constant) ratio between modal and average mileage through iterations in Excel

```
iAnnCons(runCy,'PC','smallest')= 0.5 * 0.952 * iTransChar(runCy,"KM_VEH","%fBaseY%") * 1000 * 1E-6;
iAnnCons(runCy,'PC' ,'modal')=0.952 * iTransChar(runCy,"KM_VEH","%fBaseY%") * 1000 * 1E-6;
iAnnCons(runCy,'PC' ,'largest')= 4 * 0.952 * iTransChar(runCy,"KM_VEH","%fBaseY%") * 1000 * 1E-6;
```

- modal value is assumed to be 2 tonnes/vehicle, min = 1/3*modal and max = 10*modal tkm.
- 0.706 is the constant ratio of modal/average tkm through iterations in Excel

```
iAnnCons(runCy,'GU','smallest')=0.5 * 0.706 * iTransChar(runCy,"KM_VEH_TRUCK","%fBaseY%")* 1000 * 2 / 3  * 1E-6;
iAnnCons(runCy,'GU','modal')=0.706 * iTransChar(runCy,"KM_VEH_TRUCK","%fBaseY%") * 1000 * 2  * 1E-6;
iAnnCons(runCy,'GU','largest')=4 * 0.706 * iTransChar(runCy,"KM_VEH_TRUCK","%fBaseY%") * 1000 * 2 * 10  * 1E-6;
iAnnCons(runCy,'PA','smallest')=40000 * 50 * 1E-6;
iAnnCons(runCy,'PA','modal')=400000 * 100 * 1E-6;
iAnnCons(runCy,'PA','largest')=800000 * 300 * 1E-6;
```

- Size will not play a role in buses, trains, ships and aircraft
- Following values are given only for the sake of uniformity, but iDisFunConSize is not really calculated for non-road transport!

```
iAnnCons(runCy,'PT',"smallest") = 50000 * 50 * 1E-6;
iAnnCons(runCy,'PT',"modal") = 200000 * 150 * 1e-6;
iAnnCons(runCy,'PT',"largest") = 400000 * 500 * 1E-6;
iAnnCons(runCy,'GT',"smallest") = 50000 * 20 * 1E-6;
iAnnCons(runCy,'GT',"modal") = 200000 * 200 * 1e-6;
iAnnCons(runCy,'GT',"largest") = 400000 * 500 * 1E-6;
iAnnCons(runCy,'GN',"smallest") = 10000 * 20 * 1E-6;
iAnnCons(runCy,'GN',"modal") = 50000 * 300 * 1e-6;
iAnnCons(runCy,'GN',"largest") = 100000 * 1000 * 1E-6;
iAnnCons(runCy,INDSE,"smallest") = 0.2  ;
iAnnCons(runCy,INDSE,"largest") = 0.9 ;
```

- assuming an average utilisation rate of 0.6 for iron & steel and 0.5 for other industry (see iterations in Excel):

```
iAnnCons(runCy,"IS","modal") = 0.587;
iAnnCons(runCy,INDSE,"modal")$(not sameas(INDSE,"IS")) = 0.487;
iAnnCons(runCy,DOMSE,"smallest") = iConsSizeDistHeat(runCy,"smallest") ;
iAnnCons(runCy,DOMSE,"largest") = iConsSizeDistHeat(runCy,"largest") ;
iAnnCons(runCy,DOMSE,"modal") = iConsSizeDistHeat(runCy,"modal");
iAnnCons(runCy,NENSE,"smallest") = 0.2  ;
iAnnCons(runCy,NENSE,"largest") = 0.9 ;
```

- assuming an average utilisation rate of 0.5 for non-energy uses:

```
iAnnCons(runCy,NENSE,"modal") = 0.487 ;
iAnnCons(runCy,"BU","smallest") = 0.2 ;
iAnnCons(runCy,"BU","largest") = 1 ;
```

```
iAnnCons(runCy,"BU","modal") = 0.5 ;
iAnnCons(runCy,DSBS,ConSet)=0.9;
iDisFunConSize(runCy,DSBS,rCon) = 1;
```

- Consumer size groups distribution function

```
$ontext
Loop (runCy,DSBS) DO
     Loop rCon$(ord(rCon) le iNcon(DSBS)+1) DO
          iDisFunConSize(runCy,DSBS,rCon) =
                 Prod(nSet$(ord(Nset) le iNcon(DSBS)),ord(nSet))
                 /
                 (
                  (
                   Prod(nSet$(ord(nSet) le iNcon(DSBS)-(ord(rCon)-1)),
                      ord(nSet))$(ord(rCon) lt iNcon(DSBS)+1)
                       +1$(ord(rCon) eq iNcon(DSBS)+1)
                   )
                   *
                   (
                      Prod(nSet$(ord(nSet) le ord(rCon)-1),ord(nSet))$(
                         ord(rCon) ne 1)+1$(ord(rCon) eq 1))
                 )
                 *
                 Power(log(iAnnCons(runCy,DSBS,"modal")/iAnnCons(runCy,
                    DSBS,"smallest")),ord(rCon)-1)
                 *
                 Power(log(iAnnCons(runCy,DSBS,"largest")/iAnnCons(
                    runCy,DSBS,"modal")),iNcon(DSBS)-(ord(rCon)-1))
                 /
                (Power(log(iAnnCons(runCy,DSBS,"largest")/iAnnCons(
                   runCy,DSBS,"smallest")),iNcon(DSBS)) )
                *
                iAnnCons(runCy,DSBS,"smallest")*(iAnnCons(runCy,DSBS,"
                   largest")/iAnnCons(runCy,DSBS,"smallest"))**((ord(
                   rCon)-1)/iNcon(DSBS))
;
     ENDLOOP;
ENDLOOP;
$offtext
iCumDistrFuncConsSize(allCy,DSBS) = sum(rCon, iDisFunConSize(allCy,DSBS
    ,rCon));
iCGI(allCy,YTIME) = 1;
```

```
table iResTotCapMxmLoad(allCy,PGRES,YTIME)              "Residuals for total capacity and maximum load (1)"
$ondelim
$include"./iResTotCapMxmLoad.csv"
$offdelim
;
iResMargTotAvailCap(allCy,PGRES,YTIME)$an(YTIME) = iResTotCapMxmLoad(allCy,PGRES,YTIME);
table iInvCost(PGALL,YTIME)             "Capital gross cost per plant type (kEuro2005/KW)"
$ondelim
$include"./iInvCost.csv"
$offdelim
;
table iVarCost(PGALL,YTIME)             "Variable gross cost other than fuel per Plant Type (Euro2005/KW)"
$ondelim
$include"./iVarCost.csv"
$offdelim
;
table iFixOandMCost(PGALL,YTIME)    "Fixed O&M Gross Cost per Plant Type (Euro2005/KW)"
$ondelim
$include"./iFixOandMCost.csv"
$offdelim
;
table iAvailRate(PGALL,YTIME)       "Plant availability rate (1)"
$ondelim
$include"./iAvailRate.csv"
$offdelim
;
table iDataElecProd(allCy,PGALL,YTIME) "Electricity production past years (GWh)"
$ondelim
$include"./iDataElecProd.csv"
$offdelim
;
table iDataDistrLosses(allCy,EF,YTIME)      "Data for Distribution Losses (Mtoe)"
$ondelim
$include"./iDataDistrLosses.csv"
$offdelim
;
```

```
table iDataOtherTransfOutput(allCy,EF,YTIME)        "Data for Other transformation output  (Mtoe)"
$ondelim
$include"./iDataOtherTransfOutput.csv"
$offdelim
;
iTranfOutGasworks(allCy,EFS,YTIME)$(not An(YTIME)) = iDataOtherTransfOutput(allCy,EFS,YTIME);
iDistrLosses(allCy,EFS,YTIME)$(not An(YTIME))  = iDataDistrLosses(allCy,EFS,YTIME);
table iDataTransfOutputRef(allCy,EF,YTIME)      "Data for Other transformation output  (Mtoe)"
$ondelim
$include"./iDataTransfOutputRef.csv"
$offdelim
;
table iFuelCons(allCy,TRANSE,EF,YTIME)   "Fuel consumption (Mtoe)"
$ondelim
$include"./iFuelCons.csv"
$offdelim
;
iTransfOutputRef(allCy,EFS,YTIME)$(not An(YTIME)) = iDataTransfOutputRef(allCy,EFS,YTIME);
iFuelCons(allCy,TRANSE,EF,YTIME)$(SECTTECH(TRANSE,EF) $(iFuelCons(allCy,TRANSE,EF,YTIME)<=0)) = 1e-6;
iFuelConsPerFueSub(allCy,TRANSE,EF,YTIME)$(not An(YTIME))  = iFuelCons(allCy,TRANSE,EF,YTIME);
table iFuelConsINDSE(allCy,INDSE,EF,YTIME)   "Fuel consumption of industry subsector (Mtoe)"
$ondelim
$include"./iFuelConsINDSE.csv"
$offdelim
;
iFuelConsINDSE(allCy,INDSE,EF,YTIME)$(SECTTECH(INDSE,EF) $(iFuelConsINDSE(allCy,INDSE,EF,YTIME)<=0)) = 1e-6;
table iFuelConsDOMSE(allCy,DOMSE,EF,YTIME)   "Fuel consumption of domestic subsector (Mtoe)"
$ondelim
$include"./iFuelConsDOMSE.csv"
$offdelim
;
iFuelConsDOMSE(allCy,DOMSE,EF,YTIME)$(SECTTECH(DOMSE,EF) $(iFuelConsDOMSE(allCy,DOMSE,EF,YTIME)<=0)) = 1e-6;
```

```
table iFuelConsNENSE(allCy,NENSE,EF,YTIME)   "Fuel consumption of non energy and bunkers (Mtoe)"
$ondelim
$include"./iFuelConsNENSE.csv"
$offdelim
;
iFuelConsNENSE(allCy,NENSE,EF,YTIME)$(SECTTECH(NENSE,EF) $(iFuelConsNENSE(allCy,NENSE,EF,YTIME)<=0)) = 1e-6;
iFuelConsPerFueSub(allCy,INDSE,EF,YTIME)$(not An(YTIME))   = iFuelConsINDSE(allCy,INDSE,EF,YTIME);
iFuelConsPerFueSub(allCy,DOMSE,EF,YTIME)$(not An(YTIME))   = iFuelConsDOMSE(allCy,DOMSE,EF,YTIME);
iFuelConsPerFueSub(allCy,NENSE,EF,YTIME)$(not An(YTIME))   = iFuelConsNENSE(allCy,NENSE,EF,YTIME);
iFinEneCons(runCy,EFS,YTIME) = sum(INDDOM,
                          sum(EF$(EFtoEFS(EF,EFS) $SECTTECH(INDDOM,EF)), iFuelConsPerFueSub(runCy,INDDOM,EF,YTIME))
                        )
                       +
                       sum(TRANSE,
                          sum(EF$(EFtoEFS(EF,EFS) $SECTTECH(TRANSE,EF) $(not plugin(EF)) ), iFuelConsPerFueSub(runCy,TRANSE,EF,YTIME)));
iGrossCapCosSubRen(runCy,PGALL,YTIME)=iInvCost(PGALL,YTIME);
loop(runCy,PGALL,YTIME)$AN(YTIME) DO
         abort $(iGrossCapCosSubRen(runCy,PGALL,YTIME)<0) "CAPITAL COST IS NEGATIVE", iGrossCapCosSubRen
ENDLOOP;
table iDataElecSteamGen(allCy,PGOTH,YTIME)            "Various Data related to electricity and steam generation (1)"
$ondelim
$include"./iDataElecSteamGen.csv"
$offdelim
;
iTotAvailCapBsYr(allCy) = sum(tfirst,iDataElecSteamGen(allCy,"TOTCAP",TFIRST))+sum(tfirst,iDataElecSteamGen(allCy,"CHP_CAP",TFIRST))*0.85;
iElecImp(allCy,YTIME)=0;
iScaleEndogScrap(allCy,PGALL,YTIME) = iPremReplacem(allCy,PGALL);
table iDecomPlants(allCy,PGALL,PG1_set)             "Decomissioning Plants (MW)"
$ondelim
$include"./iDecomPlants.csv"
$offdelim
```

```
;
iPlantDecomSched(allCy,PGALL,YTIME) = iDecomPlants(allCy,PGALL,"DEC_10");
table iInvPlants(allCy,PGALL,PG1_set)                "Investment Plants (MW)"
$ondelim
$include"./iInvPlants.csv"
$offdelim
;
iDecInvPlantSched(allCy,PGALL,"2010") = iInvPlants(allCy,PGALL,"INV_10");
iDecInvPlantSched(allCy,PGALL,"2011") = iInvPlants(allCy,PGALL,"INV_11");
iDecInvPlantSched(allCy,PGALL,"2012") = iInvPlants(allCy,PGALL,"INV_12");
iDecInvPlantSched(allCy,PGALL,"2013") = iInvPlants(allCy,PGALL,"INV_13");
iDecInvPlantSched(allCy,PGALL,"2014") = iInvPlants(allCy,PGALL,"INV_14");
iDecInvPlantSched(allCy,PGALL,"2034") = iInvPlants(allCy,PGALL,"INV_15");
iDecInvPlantSched(allCy,PGALL,"2018") = iInvPlants(allCy,PGALL,"INV_18");
iDecInvPlantSched(allCy,PGALL,"2019") = iInvPlants(allCy,PGALL,"INV_19");
table iCummMxmInstRenCap(allCy,PGRENEF,YTIME)    "Cummulative maximum potential installed Capacity for Renewables (GW)"
$ondelim
$include"./iCummMxmInstRenCap.csv"
$offdelim
;
iCummMxmInstRenCap(allCy,PGRENEF,YTIME)$(not iCummMxmInstRenCap(allCy,PGRENEF,YTIME)) = 1e-4;
iMaxRenPotential(allCy,"LHYD",YTIME) = iCummMxmInstRenCap(allCy,"LHYD",YTIME);
iMaxRenPotential(allCy,"SHYD",YTIME) = iCummMxmInstRenCap(allCy,"SHYD",YTIME);
iMaxRenPotential(allCy,"WND",YTIME)$AN(YTIME) = iCummMxmInstRenCap(allCy,"WND",YTIME);
iMaxRenPotential(allCy,"WNO",YTIME)$AN(YTIME) = iCummMxmInstRenCap(allCy,"WNO",YTIME);
iMaxRenPotential(allCy,"SOL",YTIME)$AN(YTIME) = iCummMxmInstRenCap(allCy,"SOL",YTIME);
```

```
iMaxRenPotential(allCy,"DPV",YTIME)$AN(YTIME) = iCummMxmInstRenCap(allCy,"DPV",YTIME);
iMaxRenPotential(allCy,"BMSWAS",YTIME)$AN(YTIME) = iCummMxmInstRenCap(allCy,"BMSWAS",YTIME);
iMaxRenPotential(allCy,"OTHREN",YTIME)$AN(YTIME) = iCummMxmInstRenCap(allCy,"OTHREN",YTIME);
table iCummMnmInstRenCap(allCy,PGRENEF,YTIME)    "Cummulative minimum potential installed Capacity for Renewables (GW)"
$ondelim
$include"./iCummMnmInstRenCap.csv"
$offdelim
;
iCummMnmInstRenCap(allCy,PGRENEF,YTIME)$(not iCummMnmInstRenCap(allCy,PGRENEF,YTIME)) = 1e-4;
iMinRenPotential(allCy,"LHYD",YTIME) = iCummMnmInstRenCap(allCy,"LHYD",YTIME);
iMinRenPotential(allCy,"SHYD",YTIME) = iCummMnmInstRenCap(allCy,"SHYD",YTIME);
iMinRenPotential(allCy,"WND",YTIME)  = iCummMnmInstRenCap(allCy,"WND",YTIME);
iMinRenPotential(allCy,"WNO",YTIME)  = iCummMnmInstRenCap(allCy,"WNO",YTIME);
iMinRenPotential(allCy,"SOL",YTIME)  = iCummMnmInstRenCap(allCy,"SOL",YTIME);
iMinRenPotential(allCy,"DPV",YTIME)  = iCummMnmInstRenCap(allCy,"DPV",YTIME);
iMinRenPotential(allCy,"BMSWAS",YTIME) = iCummMnmInstRenCap(allCy,"BMSWAS",YTIME);
iMinRenPotential(allCy,"OTHREN",YTIME) = iCummMnmInstRenCap(allCy,"OTHREN",YTIME);
table iMatFacCap(allCy,PGALL,YTIME)  "Maturty factors on Capacity (1)"
$ondelim
$include"./iMatFacCap.csv"
$offdelim
;
iMatFacPlaAvailCap(allCy,PGALL,YTIME)$an(YTIME) = iMatFacCap(allCy,PGALL,YTIME);
iMatFacPlaAvailCap(allCy,CCS,YTIME)$an(YTIME)  =0;
table iMatureFacLoad(allCy,PGALL,YTIME)  "Maturty factors on Load (1)"
$ondelim
$include"./iMatureFacLoad.csv"
$offdelim
;
```

```
iMatureFacPlaDisp(allCy,PGALL,YTIME)$an(YTIME) = iMatureFacLoad(allCy,PGALL,YTIME);
table iMxmShareChpElec(allCy,YTIME)  "Maximum share of CHP electricity in a country (1)"
$ondelim
$include"./iMxmShareChpElec.csv"
$offdelim
;
iEffValueInEuro(allCy,SBS,YTIME)=0;
table iContrElecPrice(allCy,ELCPCHAR,YTIME)  "Parameters controlling electricity price (1)"
$ondelim
$include"./iContrElecPrice.csv"
$offdelim
;
iFacElecPriConsu(allCy,ELCPCHAR,YTIME)$an(YTIME) = iContrElecPrice(allCy,ELCPCHAR,YTIME);
iScenarioPri(WEF,"NOTRADE",YTIME)=0;
table iDataPriceReform(allCy,AGSECT,EF,YTIME)    "Price reform (1)"
$ondelim
$include"./iDataPriceReform.csv"
$offdelim
;
```

- INSERT MECHANSIM FOR PRICE REFORM!!

```
iPriceReform(runCy,INDSE1(SBS),EF,YTIME)=iDataPriceReform(runCy,"INDSE1",EF,YTIME) ;
iPriceReform(runCy,DOMSE1(SBS),EF,YTIME)=iDataPriceReform(runCy,"DOMSE1",EF,YTIME) ;
iPriceReform(runCy,NENSE1(SBS),EF,YTIME)=iDataPriceReform(runCy,"NENSE1",EF,YTIME) ;
iPriceReform(runCy,TRANS1(SBS),EF,YTIME)=iDataPriceReform(runCy,"TRANS1",EF,YTIME) ;
iPriceReform(runCy,PG(SBS),EF,YTIME)=iDataPriceReform(runCy,"INDSE1",EF,YTIME) ;
table iDataPriceTargets(allCy,AGSECT,EF,YTIME)   "Data for the Price targets (1)"
$ondelim
$include"./iDataPriceTargets.csv"
$offdelim
;
iPriceTragets(runCy,INDSE1(SBS),EF,YTIME)=iDataPriceTargets(runCy,"
```

```
INDSE1",EF,"2030") ;
iPriceTragets(runCy,DOMSE1(SBS),EF,YTIME)=iDataPriceTargets(runCy,"DOMSE1",EF,"2030") ;
iPriceTragets(runCy,NENSE1(SBS),EF,YTIME)=iDataPriceTargets(runCy,"NENSE1",EF,"2030") ;
iPriceTragets(runCy,TRANS1(SBS),EF,YTIME)=iDataPriceTargets(runCy,"TRANS1",EF,"2030") ;
iPriceTragets(runCy,PG(SBS),EF,YTIME)=iDataPriceTargets(runCy,"INDSE1",EF,"2030") ;
```

*SPECIFIC CASE FOR NATURAL GAS PRICES!!!

```
iPriceTragets("RAS",INDSE1(SBS),"NGS",YTIME)=iDataPriceTargets("RAS","INDSE1","NGS",YTIME) ;
iPriceTragets("RAS",DOMSE1(SBS),"NGS",YTIME)=iDataPriceTargets("RAS","DOMSE1","NGS",YTIME) ;
iPriceTragets("RAS",NENSE1(SBS),"NGS",YTIME)=iDataPriceTargets("RAS","NENSE1","NGS",YTIME) ;
iPriceTragets("RAS",PG(SBS),"NGS",YTIME)=iDataPriceTargets("RAS","INDSE1","NGS",YTIME) ;
iPriceTragets("MAR",INDSE1(SBS),"NGS",YTIME)=iDataPriceTargets("MAR","INDSE1","NGS",YTIME) ;
iPriceTragets("MAR",DOMSE1(SBS),"NGS",YTIME)=iDataPriceTargets("MAR","DOMSE1","NGS",YTIME) ;
iPriceTragets("MAR",NENSE1(SBS),"NGS",YTIME)=iDataPriceTargets("MAR","NENSE1","NGS",YTIME) ;
iPriceTragets("MAR",PG(SBS),"NGS",YTIME)=iDataPriceTargets("MAR","INDSE1","NGS",YTIME) ;
iHydrogenPri(allCy,SBS,YTIME)=4.3;
table iFuelPrice(allCy,SBS,EF,YTIME)     "Prices of fuels per subsector (k$2015/toe)"
$ondelim
$include"./iFuelPrice.csv"
$offdelim
;
iFuelPrice(allCy,SBS,EF,YTIME) = iFuelPrice(allCy,SBS,EF,YTIME)/1000; !! change units $15 -> k$15
table iDataGrossInlCons(allCy,EF,YTIME)  "Data for Gross Inland Conusmption (Mtoe)"
$ondelim
$include"./iDataGrossInlCons.csv"
$offdelim
;
```

```
table iDataConsEneBranch(allCy,EF,YTIME)     "Data for Consumption of Energy Branch (Mtoe)"
$ondelim
$include"./iDataConsEneBranch.csv"
$offdelim
;
iTotEneBranchCons(allCy,EFS,YTIME)$(not An(YTIME)) = iDataConsEneBranch(allCy,EFS,YTIME);
table iDataImports(allCy,EF,YTIME)             "Data for imports (Mtoe)"
$ondelim
$include"./iDataImports.csv"
$offdelim
;
iFuelImports(allCy,EFS,YTIME)$(not An(YTIME)) = iDataImports(allCy,EFS,YTIME);
iGrosInlCons(allCy,EFS,YTIME)$(not An(YTIME)) = iDataGrossInlCons(allCy,EFS,YTIME);
iGrossInConsNoEneBra(runCy,EFS,YTIME) = iGrosInlCons(runCy,EFS,YTIME) + iTotEneBranchCons(runCy,EFS,YTIME)$EFtoEFA(EFS,"LQD")
                                                - iTotEneBranchCons(runCy,EFS,YTIME)$(not EFtoEFA(EFS,"LQD"));
iPeakBsLoadBy(allCy,PGLOADTYPE) = sum(tfirst, iDataElecSteamGen(allCy,PGLOADTYPE,tfirst));
table iDataElecAndSteamGen(allCy,CHP,YTIME)  "Data releated to electricity and steam generation"
$ondelim
$include"./iDataElecAndSteamGen.csv"
$offdelim
;
iHisChpGrCapData(allCy,CHP,YTIME) = iDataElecAndSteamGen(allCy,CHP,YTIME);
table iDataTotTransfInputRef(allCy,EF,YTIME)     "Total Transformation Input in Refineries (Mtoe)"
$ondelim
$include"./iDataTotTransfInputRef.csv"
$offdelim
;
iTransfInputRef(allCy,EFS,YTIME)$(not An(YTIME)) = iDataTotTransfInputRef(allCy,EFS,YTIME);
```

- Calculation of weights for sector average fuel price

```
iResElecIndex(allCy,YTIME) = 1;
loop SBS do
         iDiffFuelsInSec(SBS) = 0;
         loop EF$(SECTTECH(SBS,EF) $(not plugin(EF)))  do
              iDiffFuelsInSec(SBS) = iDiffFuelsInSec(SBS)+1;
         endloop;
endloop;
iTotFinEneDemSubBaseYr(runCy,TRANSE,YTIME) = sum(EF$(SECTTECH(TRANSE,EF) $(not plugin(EF))), iFuelConsPerFueSub(runCy,TRANSE,EF,YTIME));
iTotFinEneDemSubBaseYr(allCy,INDSE,YTIME)   = SUM(EF,iFuelConsPerFueSub(allCy,INDSE,EF,YTIME));
iTotFinEneDemSubBaseYr(allCy,DOMSE,YTIME)   = SUM(EF,iFuelConsPerFueSub(allCy,DOMSE,EF,YTIME));
iTotFinEneDemSubBaseYr(allCy,NENSE,YTIME)   = SUM(EF,iFuelConsPerFueSub(allCy,NENSE,EF,YTIME));
iWgtSecAvgPriFueCons(runCy,TRANSE,EF)$(SECTTECH(TRANSE,EF) $(not plugin(EF)) ) = (iFuelConsPerFueSub(runCy,TRANSE,EF,"2019") / iTotFinEneDemSubBaseYr(runCy,TRANSE,"2019"))$iTotFinEneDemSubBaseYr(runCy,TRANSE,"2019")
                                                  + (1/iDiffFuelsInSec(TRANSE))$(not iTotFinEneDemSubBaseYr(runCy,TRANSE,"2019"));
iWgtSecAvgPriFueCons(runCy,NENSE,EF)$SECTTECH(NENSE,EF) = ( iFuelConsPerFueSub(runCy,NENSE,EF,"2019") / iTotFinEneDemSubBaseYr(runCy,NENSE,"2019") )$iTotFinEneDemSubBaseYr(runCy,NENSE,"2019")
                                                 + (1/iDiffFuelsInSec(NENSE))$(not iTotFinEneDemSubBaseYr(runCy,NENSE,"2019"));
iWgtSecAvgPriFueCons(runCy,INDDOM,EF)$(SECTTECH(INDDOM,EF)$(not sameas(EF,"ELC"))) = ( iFuelConsPerFueSub(runCy,INDDOM,EF,"2019") / ( iTotFinEneDemSubBaseYr(runCy,INDDOM,"2019") - iFuelConsPerFueSub(runCy,INDDOM,"ELC","2019")) )$( iTotFinEneDemSubBaseYr(runCy,INDDOM,"2019") - iFuelConsPerFueSub(runCy,INDDOM,"ELC","2019") )
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                +
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
(1/(
iDiffFue
(
INDDOM
```

```
)
-1)
)
$
(
not

(
iTotFinE
(
runCy
,
INDDOM
,
"
2019
"
)

-

iFuelCon
(
runCy
,
INDDOM
,
"
ELC
"
,
"
2019
"
)
)
)
;
```

- Rescaling the weights

```
iWgtSecAvgPriFueCons(runCy,SBS,EF)$(SECTTECH(SBS,EF) $sum(ef2$SECTTECH(
```

```
SBS,EF),iWgtSecAvgPriFueCons(runCy,SBS,EF2))) = iWgtSecAvgPriFueCons(runCy,SBS,EF)/sum(ef2$SECTTECH(SBS,EF),iWgtSecAvgPriFueCons(runCy,SBS,EF2));
table iResNonSubElec(allCy,INDSE,YTIME)  "Residuals for non subsitutable electricity (1)"
$ondelim
$include"./iResNonSubElec.csv"
$offdelim
;
table iResNonSubElecCons(allCy,DOMSE,YTIME)  "Residuals in Non Substitutable Electricity Consumption per substector (1)"
$ondelim
$include"./iResNonSubElecCons.csv"
$offdelim
;
iResNonSubsElecDem(allCy,INDSE,YTIME)$an(YTIME) = iResNonSubElec(allCy,INDSE,YTIME);
iResNonSubsElecDem(allCy,DOMSE,YTIME)$an(YTIME) = iResNonSubElecCons(allCy,DOMSE,YTIME);
table iResFuelConsSub(allCy,INDSE,EF,YTIME)  "Residuals for fuel consumption per subsector (1)"
$ondelim
$include"./iResFuelConsSub.csv"
$offdelim
;
table iResFuelConsPerFuelAndSub(allCy,DOMSE,EF,YTIME)    "Residuals in fuel consumption per fuel and subsector (1)"
$ondelim
$include"./iResFuelConsPerFuelAndSub.csv"
$offdelim
;
table iResInFuelConsPerFuelAndSub(allCy,NENSE,EF,YTIME)  "Residuals in fuel consumption per fuel and subsector (1)"
$ondelim
$include"./iResInFuelConsPerFuelAndSub.csv"
$offdelim
;
iResFuelConsPerSubAndFuel(allCy,INDSE,EF,YTIME)$an(YTIME) = iResFuelConsSub(allCy,INDSE,EF,YTIME);
iResFuelConsPerSubAndFuel(allCy,DOMSE,EF,YTIME)$an(YTIME) = iResFuelConsPerFuelAndSub(allCy,DOMSE,EF,YTIME);
iResFuelConsPerSubAndFuel(allCy,NENSE,EF,YTIME)$an(YTIME) = iResInFuelConsPerFuelAndSub(allCy,NENSE,EF,YTIME);
```

```
iTransfOutputGasw(allCy,YTIME)$an(YTIME)  = iSupResRefCapacity(allCy,"TOOTH_RES",YTIME);
table iResTranspFuelConsSubTech(allCy,TRANSE,EF,YTIME)   "Residual Transport on Specific Fuel Consumption per Subsector and Technology (1)"
$ondelim
$include"./iResTranspFuelConsSubTech.csv"
$offdelim
;
iResSpecificFuelConsCost(allCy,TRANSE,TTECH,EF,YTIME)$(sameas(TTECH,EF) $an(YTIME)) = iResTranspFuelConsSubTech(allCy,TRANSE,EF,YTIME)* iEneProdRDscenarios(allCy,TRANSE,YTIME);
iResSpecificFuelConsCost(allCy,TRANSE,"BGDO","BGDO",YTIME)$(an(YTIME) $SECTTECH(TRANSE,"BGDO")) = iResSpecificFuelConsCost(allCy,TRANSE,"GDO","GDO",YTIME);
iResSpecificFuelConsCost(allCy,TRANSE,"PHEVGSL","GSL",YTIME)$(TRANSETTECH(TRANSE,"PHEVGSL") $an(YTIME))= iResSpecificFuelConsCost(allCy,TRANSE,"GSL","GSL",YTIME);
iResSpecificFuelConsCost(allCy,TRANSE,"PHEVGSL","ELC",YTIME)$(TRANSETTECH(TRANSE,"PHEVGSL") $an(YTIME))= iResSpecificFuelConsCost(allCy,TRANSE,"ELC","ELC",YTIME);
iResSpecificFuelConsCost(allCy,TRANSE,"PHEVGDO","GDO",YTIME)$(TRANSETTECH(TRANSE,"PHEVGDO") $an(YTIME))= iResSpecificFuelConsCost(allCy,TRANSE,"GDO","GDO",YTIME);
iResSpecificFuelConsCost(allCy,TRANSE,"PHEVGDO","ELC",YTIME)$(TRANSETTECH(TRANSE,"PHEVGDO") $an(YTIME))= iResSpecificFuelConsCost(allCy,TRANSE,"ELC","ELC",YTIME);
iResSpecificFuelConsCost(allCy,TRANSE,"cHEVGDO","GDO",YTIME)$(TRANSETTECH(TRANSE,"cHEVGDO") $an(YTIME))= iResSpecificFuelConsCost(allCy,TRANSE,"GDO","GDO",YTIME);
iResSpecificFuelConsCost(allCy,TRANSE,"cHEVGsl","gsl",YTIME)$(TRANSETTECH(TRANSE,"cHEVGsl") $an(YTIME))= iResSpecificFuelConsCost(allCy,TRANSE,"Gsl","Gsl",YTIME);
table iPlugHybrFractOfMileage(allCy,ELSH_SET,YTIME)  "Plug in hybrid fraction of mileage covered by electricity, residualls on GDP-Depnd car market ext (1)"
$ondelim
$include"./iPlugHybrFractOfMileage.csv"
$offdelim
;
iShareAnnMilePlugInHybrid(allCy,YTIME)$an(YTIME) = iPlugHybrFractOfMileage(allCy,"ELSH",YTIME);
table iCapDataLoadFacEachTransp(TRANSE,TRANSUSE)     "Capacity data and
```

```
     Load factor for each transportation mode (passenger or tonnes/
    vehicle)"
$ondelim
$include"./iCapDataLoadFacEachTransp.csv"
$offdelim
;
iUtilRateChpPlants(allCy,CHP,YTIME) = 0.5;
$ontext
loop YTIME$((ord(YTIME) gt TF+1) $(ord(YTIME) lt TF+6)) do
         iUtilRateChpPlants(allCy,CHP,YTIME) = (iUtilRateChpPlants(
            allCy,CHP,"2020")-iUtilRateChpPlants(allCy,CHP,"2018"))/5+
            iUtilRateChpPlants(allCy,CHP,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF+6) $(ord(YTIME) lt TF+11)) do
         iUtilRateChpPlants(allCy,CHP,YTIME) = (iUtilRateChpPlants(
            allCy,CHP,"2025")-iUtilRateChpPlants(allCy,CHP,"2020"))/5+
            iUtilRateChpPlants(allCy,CHP,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF+11) $(ord(YTIME) lt TF+16)) do
         iUtilRateChpPlants(allCy,CHP,YTIME) = (iUtilRateChpPlants(
            allCy,CHP,"2030")-iUtilRateChpPlants(allCy,CHP,"2025"))/5+
            iUtilRateChpPlants(allCy,CHP,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF+16) $(ord(YTIME) lt TF+21)) do
         iUtilRateChpPlants(allCy,CHP,YTIME) = (iUtilRateChpPlants(
            allCy,CHP,"2035")-iUtilRateChpPlants(allCy,CHP,"2030"))/5+
            iUtilRateChpPlants(allCy,CHP,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF+21) $(ord(YTIME) lt TF+41)) do
         iUtilRateChpPlants(allCy,CHP,YTIME) = (iUtilRateChpPlants(
            allCy,CHP,"2050")-iUtilRateChpPlants(allCy,CHP,"2030"))/20+
            iUtilRateChpPlants(allCy,CHP,YTIME-1);
endloop;
$offtext
table iDataInstCapElecFuel(allCy,PGALL,PG1_set)      "Installed
    capacity input (GW)"
$ondelim
$include"./iDataInstCapElecFuel.csv"
$offdelim
;
iInstCapPast(allCy,PGALL,"2010") = iDataInstCapElecFuel(allCy,PGALL,"
    2010");
iInstCapPast(allCy,PGALL,"2015") = iDataInstCapElecFuel(allCy,PGALL,"
```

```
2015");
iInstCapPast(allCy,PGALL,"2017") = iDataInstCapElecFuel(allCy,PGALL,"2017");
iInstCapPast(allCy,PGALL,"2011") = iInstCapPast(allCy,PGALL,"2010") +( iInstCapPast(allCy,PGALL,"2015")- iInstCapPast(allCy,PGALL,"2010"))/5 ;
iInstCapPast(allCy,PGALL,"2012") = iInstCapPast(allCy,PGALL,"2011") +( iInstCapPast(allCy,PGALL,"2015")- iInstCapPast(allCy,PGALL,"2010"))/5 ;
iInstCapPast(allCy,PGALL,"2013") = iInstCapPast(allCy,PGALL,"2012") +( iInstCapPast(allCy,PGALL,"2015")- iInstCapPast(allCy,PGALL,"2010"))/5 ;
iInstCapPast(allCy,PGALL,"2014") = iInstCapPast(allCy,PGALL,"2013") +( iInstCapPast(allCy,PGALL,"2015")- iInstCapPast(allCy,PGALL,"2010"))/5 ;
iInstCapPast(allCy,PGALL,"2016") = iInstCapPast(allCy,PGALL,"2015") +( iInstCapPast(allCy,PGALL,"2017")- iInstCapPast(allCy,PGALL,"2015"))/2 ;
table iEnvPolicies(POLICIES_SET,YTIME) "Environmental policies on emissions constraints  and subsidy on renewables (Mtn CO2)"
$ondelim
$include"./iEnvPolicies.csv"
$offdelim
;
table iDataPowGenCost(PGALL,PGECONCHAR) "Data for power generation costs (various)"
$ondelim
$include"./iDataPowGenCost.csv"
$offdelim
;
iCarbValYrExog(YTIME)$an(YTIME) = iEnvPolicies("exogCV",YTIME);
table iMatrFactor(allCy,SBS,EF,YTIME)       "Maturity factor per technology and subsector (1)"
$ondelim
$include"./iMatrFactor.csv"
$offdelim
;
```

** Industry

```
iShrNonSubElecInTotElecDem(allCy,INDSE)  = iIndChar(allCy,INDSE,"SHR_NSE");
iShrNonSubElecInTotElecDem(allCy,INDSE)$(iShrNonSubElecInTotElecDem(
```

```
    allCy,INDSE)>0.98) = 0.98;
```

**Domestic - Tertiary

```
iShrNonSubElecInTotElecDem(allCy,DOMSE) =
    iInitConsSubAndInitShaNonSubElec(allCy,DOMSE,"SHR_NSE");
iShrNonSubElecInTotElecDem(allCy,DOMSE)$(iShrNonSubElecInTotElecDem(
    allCy,DOMSE)>0.98) = 0.98;
```

** Macroeconomic

```
$ontext
INTPRICE(WEF,YTIME)              = IntPricePRN(WEF,YTIME);
INTPRICEB(WEF,YTIME)             = IntPriceBPRN(WEF,YTIME);
$offtext
iCapCostTech(runCy,TRANSE,EF,YTIME)  = iCapCostTechTr(runCy,TRANSE,EF,
    YTIME);
iFixOMCostTech(runCy,TRANSE,EF,YTIME)$(ord(YTIME) eq TF-7)=
    iDataTransTech(TRANSE,EF,"FC_05");
iFixOMCostTech(runCy,TRANSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataTransTech(TRANSE,EF,"FC_25");
iFixOMCostTech(runCy,TRANSE,EF,YTIME)$(ord(YTIME) eq TF+33)=
    iDataTransTech(TRANSE,EF,"FC_50");
iVarCostTech(runCy,TRANSE,EF,YTIME)$(ord(YTIME) eq TF-7)=
    iDataTransTech(TRANSE,EF,"VC_05");
iVarCostTech(runCy,TRANSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataTransTech(TRANSE,EF,"VC_25");
iVarCostTech(runCy,TRANSE,EF,YTIME)$(ord(YTIME) eq TF+33)=
    iDataTransTech(TRANSE,EF,"VC_50");
iTechLft(runCy,TRANSE,EF,YTIME)$(ord(YTIME)>sum(TFIRST,(ord(TFIRST)-3))
    ) = iDataTransTech(TRANSE,EF,"LFT");
iAvgVehCapLoadFac(runCy,TRANSE,TRANSUSE,YTIME)$(ord(YTIME)>sum(TFIRST,(
    ord(TFIRST)-3)))= iCapDataLoadFacEachTransp(TRANSE,TRANSUSE);
```

** Industrial Sector

```
iCapCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF-7) =
    iDataIndTechnology(INDSE,EF,"IC_05");
iCapCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataIndTechnology(INDSE,EF,"IC_25");
iCapCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+33) =
    iDataIndTechnology(INDSE,EF,"IC_50");
```

```
iFixOMCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF-7) =
    iDataIndTechnology(INDSE,EF,"FC_05");
iFixOMCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataIndTechnology(INDSE,EF,"FC_25");
iFixOMCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+33) =
    iDataIndTechnology(INDSE,EF,"FC_50");
iVarCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF-7) =
    iDataIndTechnology(INDSE,EF,"VC_05");
iVarCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataIndTechnology(INDSE,EF,"VC_25");
iVarCostTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+33) =
    iDataIndTechnology(INDSE,EF,"VC_50");
iUsfEneConvSubTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) <= TF-7)    =
    iDataIndTechnology(INDSE,EF,"USC_05");
iUsfEneConvSubTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataIndTechnology(INDSE,EF,"USC_25");
iUsfEneConvSubTech(runCy,INDSE,EF,YTIME)$(ord(YTIME) eq TF+33) =
    iDataIndTechnology(INDSE,EF,"USC_50");
iTechLft(runCy,INDSE,EF,YTIME)$(ord(YTIME)>(ordfirst-8)) =
    iDataIndTechnology(INDSE,EF,"LFT");
```

** Domestic Sector

```
iFixOMCostTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF-7)= iDataDomTech
    (DOMSE,EF,"FC_05");
iFixOMCostTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataDomTech(DOMSE,EF,"FC_25");
iFixOMCostTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF+33)=
    iDataDomTech(DOMSE,EF,"FC_50");
iVarCostTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF-7)= iDataDomTech(
    DOMSE,EF,"VC_05");
iVarCostTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF+8) = iDataDomTech(
    DOMSE,EF,"VC_25");
iVarCostTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF+33)= iDataDomTech(
    DOMSE,EF,"VC_50");
iUsfEneConvSubTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) <= TF-7)=
    iDataDomTech(DOMSE,EF,"USC_05");
iUsfEneConvSubTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataDomTech(DOMSE,EF,"USC_25");
iUsfEneConvSubTech(runCy,DOMSE,EF,YTIME)$(ord(YTIME) eq TF+33)=
    iDataDomTech(DOMSE,EF,"USC_50");
iTechLft(runCy,DOMSE,EF,YTIME)$(ord(YTIME)>(ordfirst-8))  =
    iDataDomTech(DOMSE,EF,"LFT");
```

```
$ontext
NETWORKPARAM(NETWSET,INDDOM,NETEF)$(NETTECH(INDDOM,NETEF))
                                    = NetEffect(NETWSET,NETEF,"OTH");
NETWORKPARAM(NETWSET,INDDOM,"STE1AH2F")$(NETTECH(INDDOM,"STE1AH2F"))
                          = NetEffect(NETWSET,"H2F","OTH");
NETWORKPARAM(NETWSET,TRANSE,NETEF)$(NETTECH(TRANSE,NETEF))
                                    = NetEffect(NETWSET,NETEF,"TRA");
NETWORKPARAM(NETWSET,TRANSE,EF)$(NETTECH(TRANSE,EF)$PLUGIN(EF) )
                              = NetEffect(NETWSET,"ELC","TRA");
$offtext
```

** Non Energy Sector and Bunkers

```
iFixOMCostTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF-7)=
    iDataNonEneSec(NENSE,EF,"FC_05");
iFixOMCostTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataNonEneSec(NENSE,EF,"FC_25");
iFixOMCostTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF+33)=
    iDataNonEneSec(NENSE,EF,"FC_50");
iVarCostTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF-7)= iDataNonEneSec
    (NENSE,EF,"VC_05");
iVarCostTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataNonEneSec(NENSE,EF,"VC_25");
iVarCostTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF+33)=
    iDataNonEneSec(NENSE,EF,"VC_50");
iUsfEneConvSubTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) <= TF-7)=
    iDataNonEneSec(NENSE,EF,"USC_05");
iUsfEneConvSubTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF+8) =
    iDataNonEneSec(NENSE,EF,"USC_25");
iUsfEneConvSubTech(runCy,NENSE,EF,YTIME)$(ord(YTIME) eq TF+33)=
    iDataNonEneSec(NENSE,EF,"USC_50");
iTechLft(runCy,NENSE,EF,YTIME)$(ord(YTIME)>(ordfirst-8)) =
    iDataNonEneSec(NENSE,EF,"LFT");
display TF;
loop YTIME$((ord(YTIME) gt TF-8) $(ord(YTIME) lt TF+8)) do
        iCapCostTech(runCy,SBS,EF,YTIME)= (iCapCostTech(runCy,SBS,EF,"
            2025")-iCapCostTech(runCy,SBS,EF,"2010"))/20+iCapCostTech(
            runCy,SBS,EF,YTIME-1);
        iFixOMCostTech(runCy,SBS,EF,YTIME)= (iFixOMCostTech(runCy,SBS,
            EF,"2025")-iFixOMCostTech(runCy,SBS,EF,"2010"))/20+
            iFixOMCostTech(runCy,SBS,EF,YTIME-1);
        iVarCostTech(runCy,SBS,EF,YTIME)= (iVarCostTech(runCy,SBS,EF,"
            2025")-iVarCostTech(runCy,SBS,EF,"2010"))/20+iVarCostTech(
```

```
                runCy,SBS,EF,YTIME-1);
          iUsfEneConvSubTech(runCy,SBS,EF,YTIME)     = (iUsfEneConvSubTech(runCy,SBS,EF,"2025")-iUsfEneConvSubTech(runCy,SBS,EF,"2010"))/20+iUsfEneConvSubTech(runCy,SBS,EF,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF+8) $(ord(YTIME) lt TF+84)) do
          iCapCostTech(runCy,SBS,EF,YTIME)= (iCapCostTech(runCy,SBS,EF,"2050")-iCapCostTech(runCy,SBS,EF,"2025"))/25+iCapCostTech(runCy,SBS,EF,YTIME-1);
          iFixOMCostTech(runCy,SBS,EF,YTIME)= (iFixOMCostTech(runCy,SBS,EF,"2050")-iFixOMCostTech(runCy,SBS,EF,"2025"))/25+iFixOMCostTech(runCy,SBS,EF,YTIME-1);
          iVarCostTech(runCy,SBS,EF,YTIME)= (iVarCostTech(runCy,SBS,EF,"2050")-iVarCostTech(runCy,SBS,EF,"2025"))/25+iVarCostTech(runCy,SBS,EF,YTIME-1);
          iUsfEneConvSubTech(runCy,SBS,EF,YTIME)     = (iUsfEneConvSubTech(runCy,SBS,EF,"2050")-iUsfEneConvSubTech(runCy,SBS,EF,"2025"))/25+iUsfEneConvSubTech(runCy,SBS,EF,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF-12) $(ord(YTIME) lt TF+3)) do
           iUsfEneConvSubTech(runCy,SBS,EF,YTIME) = (iUsfEneConvSubTech(runCy,SBS,EF,"2025")-iUsfEneConvSubTech(runCy,SBS,EF,"2010"))/
           20+iUsfEneConvSubTech(runCy,SBS,EF,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF+3) $(ord(YTIME) lt TF+84)) do
          iUsfEneConvSubTech(runCy,SBS,EF,YTIME)     = (iUsfEneConvSubTech(runCy,SBS,EF,"2050")-iUsfEneConvSubTech(runCy,SBS,EF,"2025"))/
          25+iUsfEneConvSubTech(runCy,SBS,EF,YTIME-1);
endloop;
```

** Power Generation

```
iTechLftPlaType(runCy,PGALL) = iDataPowGenCost(PGALL, "LFT");
iPlantEffByType(runCy,PGALL,"2010") = iDataPowGenCost(PGALL,"EFF_05");
iPlantEffByType(runCy,PGALL,"2020") = iDataPowGenCost(PGALL,"EFF_20");
iPlantEffByType(runCy,PGALL,"2050") = iDataPowGenCost(PGALL,"EFF_50");
loop YTIME$((ord(YTIME) gt TF-7) $(ord(YTIME) lt TF+12)) do
          iPlantEffByType(runCy,PGALL,YTIME) = (iPlantEffByType(runCy,PGALL,"2020")-iPlantEffByType(runCy,PGALL,"2010"))/15+
```

```
                iPlantEffByType(runCy,PGALL,YTIME-1);
endloop;
loop YTIME$((ord(YTIME) gt TF+11) $(ord(YTIME) lt TF+41)) do
          iPlantEffByType(runCy,PGALL,YTIME) = (iPlantEffByType(runCy,
                PGALL,"2050")-iPlantEffByType(runCy,PGALL,"2020"))/30+
                iPlantEffByType(runCy,PGALL,YTIME-1);
endloop;
iPlantEffByType(runCy,PGALL,YTIME)$(ord(YTIME)>(ordfirst+41)) =
    iPlantEffByType(runCy,PGALL,"2050");
iCO2CaptRate(runCy,PGALL,YTIME)$(ord(YTIME)>(ordfirst-8))  =
    iDataPowGenCost(PGALL,"CR");
iEffDHPlants(runCy,EFS,YTIME)$(ord(YTIME)>(ordfirst-8))  = sum(
    PGEFS$sameas(EFS,PGEFS),iParDHEfficiency(PGEFS,"2010"));
```

** CHP economic and technical data initialisation for electricity production

```
table iDataChpPowGen(EF,YTIME,CHPPGSET)   "Data for power generation
    costs (various)"
$ondelim
$include "./iDataChpPowGen.csv"
$offdelim
;
iInvCostChp(runCy,DSBS,CHP,YTIME) = iDataChpPowGen(CHP,"2010","IC");
iFixOMCostPerChp(runCy,DSBS,CHP,YTIME) = iDataChpPowGen(CHP,"2010","FC"
    );
iVarCostChp(runCy,DSBS,CHP,YTIME) = iDataChpPowGen(CHP,"2010","VOM");
iLifChpPla(runCy,DSBS,CHP) = iDataChpPowGen(CHP,"2010","LFT");
iAvailRateChp(runCy,DSBS,CHP) = iDataChpPowGen(CHP,"2010","AVAIL");
iBoiEffChp(runCy,CHP,YTIME) = iDataChpPowGen(CHP,"2010","BOILEFF");
```

** Policies for climate change and renewables

```
$ontext
EN_PRD_INDX(runCy,SBS,YTIME)$an(YTIME)=EN_PRD_INDX_PRN(runCy,SBS,YTIME)
    ;
CCRES(PGALL,YTIME)$AN(YTIME)=CCRES_PRN(PGALL,YTIME)  ;
FOMRES(PGALL,YTIME)$AN(YTIME)=FOMRES_PRN(PGALL,YTIME) ;
VOMRES(PGALL,YTIME)$AN(YTIME)=VOMRES_PRN(PGALL,YTIME)  ;
EFFRES(PGALL,YTIME)$AN(YTIME)=EFFRES_PRN(PGALL,YTIME);
NUCRES(YTIME)$an(ytime)=NUCRES_PRN("RES",YTIME);
$offtext
```

- Update efficiencies according to energy productivity index

```
iElecIndex(runCy,YTIME) = 0.9;
```

## Preloop (preloop.gms)

### Description

- Calculation of polynomial distribution lags coefficients

```
iNPDL(DSBS) = 6;
loop DSBS do
   loop KPDL$(ord(KPDL) le iNPDL(DSBS)) do
         iFPDL(DSBS,KPDL) = 6 * (iNPDL(DSBS)+1-ord(KPDL)) * ord(KPDL)
                            /
                            (iNPDL(DSBS) * (iNPDL(DSBS)+1) * (iNPDL(DSBS
                               )+2))
   endloop;
endloop;
model openprom /
QDemSub
qDummyObj
/;
option iPop:2:0:6;
display iPop;
display iDisc;
display TF;
display TFIRST;
display runCy;
display iWgtSecAvgPriFueCons;
display iTotFinEneDemSubBaseYr;
VNumVeh.L(allCy,YTIME)=0.1;
VTrnspActiv.l(allCy,TRANSE,YTIME) = 0.1;
VFuelPriceSub.l(allCy,SBS,EF,YTIME) = 0.1;
VElecIndPrices.l(allCy,YTIME)= 0.1;
VTechCostVar.l(allCy,SBS,EF,TEA,YTIME) = 0.1;
VTechCostIntrm.l(allCy,DSBS,rcon,EF,TEA,YTIME) = 0.1;
VLifeTimeTech.l(allCy,DSBS,EF,TEA,YTIME)= 0.1;
VTechSort.l(allCy,DSBS,rCon,YTIME) = 0.1;
VConsFuel.l(allCy,DSBS,EF,YTIME)=0.1;
VRefCapacity.l(allCy,YTIME)=0.1;
VFeCons.l(allCy,EF,YTIME) = 0.1;
VTransfOutputRefineries.l(allCy,EFS,YTIME)=0.1;
VGrsInlConsNotEneBranch.l(allCy,EFS,YTIME)=0.1;
VElecConsAll.l(allCy,DSBS,YTIME)=0.1;
VCapChpPlants.l(allCy,YTIME)=0.1;
VElecPeakLoad.l(allCy,YTIME)=0.1;
```

```
VElecPeakLoad.up(allCy,YTIME)=1e6;
VElecProdPowGenPlants.l(allCy,PGALL,YTIME) = 1;
VSensCcs.l(allCy,YTIME)=1;
VCarVal.fx(allCy,NAP,YTIME)=1;
VFuelPriceSub.l(allCy,"PG",PGEF,YTIME)=1;
VProdCostTechnology.l(allCy,PGALL2,YTIME)=0.1;
VProdCostTechnology.up(allCy,PGALL2,YTIME)=1e6;
VVarCostTech.l(allCy,PGALL,YTIME)=0.1;
VProdCostTechPreReplacAvail.l(allCy,PGALL,PGALL2,YTIME)=0.1;
VTechSortVarCost.l(allCy,TRANSE,Rcon,YTIME)=0.1;
VHourProdCostTechNoCCS.up(allCy,PGALL,HOUR,YTIME)=8000;
VRenPotSupplyCurve.l(allCy,PGRENEF,YTIME)=0.1;
VScrRate.l(allCy,YTIME)=0.1;
VTranspCostPermeanConsSize.l(allCy,TRANSE,RCon,TTECH,TEA,YTIME)=0.1;
VTranspCostPermeanConsSize.lo(allCy,TRANSE,RCon,TTECH,TEA,YTIME)
    =0.0001;
VElecNonSub.l(allCy,DSBS,YTIME)=0.1;
VPowerPlantNewEq.L(allCy,PGALL,TT)=0.1;
VHourProdCostTechAfterCCS.L(allCy,PGALL,HOUR,TT)=0.1;
VPowerPlaShrNewEq.L(allCy,PGALL,TT)=0.1;
VHourProdCostTechNoCCS.L(runCy,PGALL,HOUR,TT) = VPowerPlantNewEq.L(
    runCy,PGALL,TT)*VHourProdCostTechAfterCCS.L(runCy,PGALL,HOUR,TT)+
          sum(CCS$CCS_NOCCS(CCS,PGALL), VPowerPlaShrNewEq.L(runCy,CCS,TT
              )*VHourProdCostTechAfterCCS.L(runCy,CCS,HOUR,TT));
VHourProdCostTechNoCCS.SCALE(runCy,PGALL,HOUR,TT)= max(abs(
    VHourProdCostTechNoCCS.L(runCy,PGALL,HOUR,TT)),1E-20);
VNewInvDecis.l(allCy,YTIME)=0.1;
VVarCostTechnology.l(allCy,PGALL,YTIME)=0.1;
VElecPeakLoads.l(allCy,YTIME)=0.1;
VNewCapYearly.l(allCy,PGALL,YTIME)=0.1;
VAvgCapFacRes.l(allCy,PGALL,YTIME)=0.1;
VOverallCap.scale(allCy,PGALL,YTIME)=1;
VPowPlantSorting.l(runCy,PGALL,YTIME)=0.01;
VReqElecProd.l(runCy,YTIME)=0.1;
VPowPlantSorting.up(runCy,PGALL,YTIME)=0.001;
VPowPlantSorting.scale(runCy,PGALL,YTIME)=1;
VElecDem.l(allCy,YTIME)=0.1;
VRenTechMatMult.l(allCy,PGALL,YTIME)=0.1;
VGoodsTranspActiv.l(allCy,TRANSE,YTIME)=0.1;
VRenShareElecProdSub.FX(runCy,YTIME)$(NOT AN(YTIME))=0;
VRenValue.l(YTIME)=1;
VCO2CO2SeqCsts.l(allCy,YTIME)=1;
VScalWeibullSum.l(allCy,PGALL,YTIME)=2;
```

```
VHourProdCostTech.l(runCy,PGALL,HOUR,TT) = 0.0001;
```

```
                    VARIABLE INITIALISATION
                                                        *
```

```
VFuelPriceSub.FX(runCy,SBS,EF,YTIME)$(SECTTECH(SBS,EF) $(not HEATPUMP(EF))  $(not An(YTIME))) = iFuelPrice(runCy,SBS,EF,YTIME);
VFuelPriceSub.FX(runCy,SBS,ALTEF,YTIME)$(SECTTECH(SBS,ALTEF) $(not An(YTIME))) = sum(EF$ALTMAP(SBS,ALTEF,EF),iFuelPrice(runCy,SBS,EF,YTIME));
VFuelPriceSub.FX(runCy,"PG","NUC",YTIME) = 0.025; !! fixed price for nuclear fuel to 25Euro/toe
VFuelPriceSub.FX(runCy,"H2P","NUC",YTIME) = 0.025; !! fixed price for nuclear fuel to 25Euro/toe
VFuelPriceSub.fx(runCy,INDDOM,"HEATPUMP",YTIME)$(SECTTECH(INDDOM,"HEATPUMP") $(not An(YTIME))) = iFuelPrice(runCy,INDDOM,"ELC",YTIME);
VFuelPriceSub.fx(runCy,"H2P",EF,YTIME)$(SECTTECH("H2P",EF) $(not An(YTIME))) = VFuelPriceSub.l(runCy,"PG",EF,YTIME);
VFuelPriceSub.fx(runCy,"H2P","ELC",YTIME)$(not An(YTIME)) = VFuelPriceSub.l(runCy,"OI","ELC",YTIME);
VElecPriInduResConsu.FX(runCy,"i",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*0.086;
VElecPriInduResConsu.FX(runCy,"r",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"HOU","ELC",YTIME)*0.086;
VElecPriIndResNoCliPol.FX(runCy,"i",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*0.086;
VElecPriIndResNoCliPol.FX(runCy,"r",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"HOU","ELC",YTIME)*0.086;
VFuelPriSubNoCarb.FX(runCy,SBS,EF,YTIME)$(SECTTECH(SBS,EF) $(not HEATPUMP(EF))  $(not An(YTIME))) = iFuelPrice(runCy,SBS,EF,YTIME);
VFuelPriSubNoCarb.FX(runCy,SBS,ALTEF,YTIME)$(SECTTECH(SBS,ALTEF) $(not An(YTIME))) = sum(EF$ALTMAP(SBS,ALTEF,EF),iFuelPrice(runCy,SBS,EF,YTIME));
VFuelPriSubNoCarb.FX(runCy,"PG","NUC",YTIME) = 0.025; !! fixed price for nuclear fuel to 25Euro/toe
VFuelPriSubNoCarb.fx(runCy,INDDOM,"HEATPUMP",YTIME)$(SECTTECH(INDDOM,"HEATPUMP") $(not An(YTIME))) = iFuelPrice(runCy,INDDOM,"ELC",YTIME);
VFuelPriceAvg.FX(runCy,DSBS,YTIME) = sum(EF$SECTTECH(DSBS,EF),iWgtSecAvgPriFueCons(runCy,DSBS,EF) * iFuelPrice(runCy,DSBS,EF,YTIME
```

```
    ));
VNumVeh.UP(runCy,YTIME) = 10000; !! upper bound of VNumVeh is 10000 million vehicles
VNumVeh.FX(runCy,YTIME)$(not An(YTIME)) = iActv(YTIME,runCy,"PC");
VLamda.UP(runCy,YTIME) = 1;
VLamda.FX(runCy,YTIME)$((not An(YTIME)) $(ord(YTIME) gt 1) ) = (VNumVeh.l(runCy,YTIME-1) / (iPop(YTIME-1,runCy)*1000) /
            iPassCarsMarkSat(runCy))$(iPop(YTIME-1,runCy))+VLamda.l(runCy,YTIME-1)$(not iPop(YTIME-1,runCy));
VMExtF.l(runCy,YTIME)$((not An(YTIME)) $(ord(YTIME) gt 1)  ) = (iTransChar(runCy,"RES_MEXTF",YTIME) * iSigma(runCy,"S1") * EXP(iSigma(runCy,"S2") *
            EXP(iSigma(runCy,"S3") * VLamda.l(runCy,YTIME)))
                * VNumVeh.l(runCy,YTIME-1) /(iPop(YTIME-1,runCy) * 1000)
                    )$(iPop(YTIME-1,runCy));
VMExtF.FX(runCy,YTIME)$((not An(YTIME)) $(ord(YTIME) gt 1)  ) = (iTransChar(runCy,"RES_MEXTF",YTIME) * iSigma(runCy,"S1") * EXP(iSigma(runCy,"S2") * EXP(iSigma(runCy,"S3") *
                            VLamda.l(runCy,YTIME)))*
                            VNumVeh.l(runCy,YTIME-1) /(iPop(YTIME-1,runCy) * 1000) )$(iPop(YTIME-1,runCy))+VMExtF.l(runCy,YTIME-1)$(not iPop(YTIME-1,runCy));
VMExtV.FX(runCy,YTIME)$(not An(YTIME)) = iDataPassCars(runCy,"PC","MEXTV");
VScrRate.UP(runCy,YTIME) = 1;
VScrRate.FX(runCy,YTIME)$(not An(YTIME)) = iDataPassCars(runCy,"PC","scr");
VGapTranspFillNewTech.FX(runCy,TRANSE,YTIME)$(not AN(YTIME))=0;
VTrnspActiv.FX(runCy,"PC",YTIME)$(not An(YTIME)) = iTransChar(runCy,"KM_VEH",YTIME);
VTrnspActiv.FX(runCy,TRANP,YTIME)$((not An(YTIME)) $(not sameas(TRANP,"PC"))) = iActv(YTIME,runCy,TRANP);
VTrnspActiv.FX(runCy,TRANSE,YTIME)$(not TRANP(TRANSE)) = 0;
VGoodsTranspActiv.FX(runCy,TRANG,YTIME)$(not An(YTIME)) = iActv(YTIME,runCy,TRANG);
VGoodsTranspActiv.FX(runCy,TRANSE,YTIME)$(not TRANG(TRANSE)) = 0;
VLifeTimeTech.FX(runCy,DSBS,EF,TEA,YTIME)$(SECTTECH(DSBS,EF)  $(not TRANSE(DSBS)) $(not sameas(DSBS,"PC"))) = iTechLft(runCy,DSBS,EF,YTIME);
VLifeTimeTech.FX(runCy,TRANSE,TTECH,TEA,YTIME)$(SECTTECH(TRANSE,TTECH) $(not sameas(TRANSE,"PC"))) = iTechLft(runCy,TRANSE,TTECH,YTIME);
VLifeTimeTech.FX(runCy,DSBS,EF,TEA,YTIME)$(not SECTTECH(DSBS,EF))=0;
VLifeTimeTech.FX(runCy,"PC",TTECH,TEA,YTIME)$( (not AN(YTIME))
```

```
$SECTTECH("PC",TTECH))=10;
VSpecificFuelCons.FX(runCy,TRANSE,TTECH,TEA,EF,"2017")$(SECTTECH(TRANSE,EF) ) = iSpeFuelConsCostBy(runCy,TRANSE,TTECH,TEA,EF);
VTechSortVarCostNewEquip.FX(runCy,TRANSE,TTECH,TEA,YTIME)$( SECTTECH(TRANSE,TTECH) $(not AN(YTIME))) = 0;
VConsEachTechTransp.FX(runCy,TRANSE,TTECH,EF,TEA,YTIME)$(SECTTECH(TRANSE,TTECH)  $(not PLUGIN(TTECH)) $TTECHtoEF(TTECH,EF)  $(not An(YTIME))) = iFuelConsPerFueSub(runCy,TRANSE,EF,YTIME);
VConsEachTechTransp.FX(runCy,TRANSE,TTECH,EF,TEA,YTIME)$(SECTTECH(TRANSE,TTECH)  $PLUGIN(TTECH) $(not An(YTIME))) = 0;
VConsEachTechTransp.FX(runCy,TRANSE,TTECH,EF,TEA,YTIME)$(SECTTECH(TRANSE,TTECH)  $CHYBV(TTECH) $(not An(YTIME))) =0;
VDemTr.FX(runCy,TRANSE,EF,YTIME) $(SECTTECH(TRANSE,EF) $(not An(YTIME)) ) = iFuelConsPerFueSub(runCy,TRANSE,EF,YTIME);
VElecNonSub.FX(runCy,INDDOM,YTIME)$(not An(YTIME)) = iFuelConsPerFueSub(runCy,INDDOM,"ELC",YTIME) * iShrNonSubElecInTotElecDem(runCy,INDDOM);
VElecConsInd.FX(runCy,YTIME)$(not An(YTIME))= SUM(INDSE,VElecNonSub.l(runCy,INDSE,YTIME));
VFuePriSubChp.FX(runCy,DSBS,EF,TEA,YTIME)$((not An(YTIME)) $(not TRANSE(DSBS))  $SECTTECH(DSBS,EF)) =
(((VFuelPriceSub.l(runCy,DSBS,EF,YTIME)+iVarCostTech(runCy,DSBS,EF,YTIME)/1000)/iUsfEneConvSubTech(runCy,DSBS,EF,YTIME)-
(0$(not CHP(EF)) + (VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*iFracElecPriChp(runCy,YTIME)*iElecIndex(runCy,"2010"))$CHP(EF))) + (0.003) +
SQRT( SQR(((VFuelPriceSub.l(runCy,DSBS,EF,YTIME)+iVarCostTech(runCy,DSBS,EF,YTIME)/1000)/iUsfEneConvSubTech(runCy,DSBS,EF,YTIME)- (0$(not CHP(EF)) +
(VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*iFracElecPriChp(runCy,YTIME)*iElecIndex(runCy,"2010"))$CHP(EF)))-(0.003)) + SQR(1e-7) ) )/2;
VDemSub.FX(runCy,INDDOM,YTIME)$(not An(YTIME)) = max(iTotFinEneDemSubBaseYr(runCy,INDDOM,YTIME) - VElecNonSub.L(runCy,INDDOM,YTIME),1e-5);
VDemSub.FX(runCy,NENSE,YTIME)$(not An(YTIME)) = max(iTotFinEneDemSubBaseYr(runCy,NENSE,YTIME),1e-5);
VDemSub.FX(runCy,"HOU",YTIME)$(not An(YTIME)) = max(iTotFinEneDemSubBaseYr(runCy,"HOU",YTIME) - VElecNonSub.L(runCy,"HOU",YTIME)-iExogDemOfBiomass(runCy,"HOU",YTIME),1e-5);
VDemInd.FX(runCy,YTIME)$(not An(YTIME))= SUM(INDSE,VDemSub.L(runCy,INDSE,YTIME));
VElecIndPrices.FX(runCy,YTIME)$TFIRST(YTIME) = iElecIndex(runCy,YTIME);
VElecConsHeatPla.FX(runCy,INDDOM,YTIME)$(not An(YTIME)) =
```

```
iFuelConsPerFueSub(runCy,INDDOM,"ELC",YTIME)*(1-iShrNonSubElecInTotElecDem(runCy,INDDOM))*iShrHeatPumpElecCons(runCy,INDDOM);
VConsFuel.FX(runCy,DSBS,EF,YTIME)$(SECTTECH(DSBS,EF) $(not HEATPUMP(EF)) $(not TRANSE(DSBS)) $(not An(YTIME))) = iFuelConsPerFueSub(runCy,DSBS,EF,YTIME);
VConsFuelSub.FX(runCy,DSBS,EF,YTIME)$(SECTTECH(DSBS,EF) $(not TRANSE(DSBS)) $(not An(YTIME))) =
(iFuelConsPerFueSub(runCy,DSBS,EF,YTIME))$((not ELCEF(EF)) $(not HEATPUMP(EF)))
+(VElecConsHeatPla.l(runCy,DSBS,YTIME)*iUsfEneConvSubTech(runCy,DSBS,"HEATPUMP",YTIME))$HEATPUMP(EF)+
(iFuelConsPerFueSub(runCy,DSBS,EF,YTIME)-VElecConsHeatPla.l(runCy,DSBS,YTIME))$ELCEF(EF)
+1e-7$(H2EF(EF) or sameas("STE1AH2F",EF))
;
VTechShareNewEquip.FX(runCy,DSBS,EF,TEA,YTIME)$(SECTTECH(DSBS,EF)$(not An(YTIME))) = 0;
VConsRemSubEquip.FX(runCy,DSBS,EF,YTIME)$(SECTTECH(DSBS,EF) $(not An(ytime))) =
(VConsFuelSub.l(runCy ,DSBS,EF,YTIME) - (VElecNonSub.l(runCy,DSBS,YTIME)$(ELCEF(EF) $INDDOM(DSBS)) + 0$(not (ELCEF(EF) $INDDOM(DSBS)) )))
;
VFeCons.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iFinEneCons(runCy,EFS,YTIME);
iFinEneConsPrevYear(runCy,EFS,YTIME)$(not An(YTIME)) = iFinEneCons(runCy,EFS,YTIME);
VLosses.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iDistrLosses(runCy,EFS,YTIME);
VTransfOutputPatFuel.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iTranfOutGasworks(runCy,EFS,YTIME);
VTransfOutThermPowSta.FX(runCy,EFS,YTIME)$(not TOCTEF(EFS)) = 0;
VRefCapacity.FX(runCy,YTIME)$(not An(YTIME)) = iRefCapacity(runCy,YTIME);
VTransfOutputRefineries.FX(runCy,EFS,YTIME)$(EFtoEFA(EFS,"LQD") $(not An(YTIME))) = iTransfOutputRef(runCy,EFS,YTIME);
VTransfInputRefineries.FX(runCy,"CRO",YTIME)$(not An(YTIME)) = iTransfInputRef(runCy,"CRO",YTIME);
VGrsInlConsNotEneBranch.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iGrossInConsNoEneBra(runCy,EFS,YTIME);
VGrssInCons.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iGrosInlCons(runCy,EFS,YTIME);
VTransfers.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iFeedTransfr(runCy,EFS
```

```
    ,YTIME);
VPrimProd.FX(runCy,PPRODEF,YTIME)$(not An(YTIME)) = iFuelPriPro(runCy,
    PPRODEF,YTIME);
VEnCons.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iTotEneBranchCons(runCy,
    EFS,YTIME);
VExportsFake.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iFuelExprts(runCy,
    EFS,YTIME);
VFkImpAllFuelsNotNatGas.FX(runCy,"NGS",YTIME)$(not An(YTIME)) =
    iFuelImports(runCy,"NGS",YTIME);
VExportsFake.FX(runCy,"NGS",YTIME)$(not An(YTIME)) = iFuelExprts(runCy,
    "NGS",YTIME);
VElecDem.FX(runCy,YTIME)$(not An(YTIME)) =  1/0.086 * ( iFinEneCons(
    runCy,"ELC",YTIME) + sum(NENSE, iFuelConsPerFueSub(runCy,NENSE,"ELC"
    ,YTIME)) + iDistrLosses(runCy,"ELC",YTIME)
                                                + iTotEneBranchCons(runCy,
                                                    "ELC",YTIME) - (
                                                    iFuelImports(runCy,"ELC
                                                    ",YTIME)-iFuelExprts(
                                                    runCy,"ELC",YTIME)));
VCorrBaseLoad.FX(runCy,YTIME)$(not An(YTIME)) = iPeakBsLoadBy(runCy,"
    BASELOAD");
$ontext
VCapChpPlants.FX(runCy,YTIME)$(datay(YTIME)) =
         (sum(INDDOM,VConsFuel.l(runCy,INDDOM,"ELC",YTIME)) + sum(
            TRANSE, VDemTr.l(runCy,TRANSE,"ELC",YTIME)))/
         (sum(INDDOM,VConsFuel.l(runCy,INDDOM,"ELC",YTIME)/
            iLoadFacElecDem(runCy,INDDOM,"2015")) + sum(TRANSE, VDemTr.
            l(runCy,TRANSE,"ELC",YTIME)/
         iLoadFacElecDem(runCy,TRANSE,"2015")));
$offtext
VElecPeakLoad.FX(runCy,YTIME)$(datay(YTIME)) = VElecDem.l(runCy,YTIME)
    /(VCapChpPlants.l(runCy,YTIME)*8.76);
VTotElecGenCap.FX(runCy,YTIME)$(not An(YTIME)) = iTotAvailCapBsYr(runCy
    );
VElecGenNoChp.FX(runCy,YTIME)$(not An(YTIME)) = iTotAvailCapBsYr(runCy)
    ;
VElecCapChpPla.FX(runCy,CHP,YTIME)$(not An(YTIME)) = iHisChpGrCapData(
    runCy,CHP,YTIME);
VPowPlaShaNewEquip.FX(runCy,PGALL,YTIME)$((NOT AN(YTIME)) )=0;
VHourProdCostTech.FX(runCy,PGALL,HOUR,YTIME)$((NOT AN(YTIME)))=0;
VPowerPlaShrNewEq.FX(runCy,PGALL,YTIME)$((NOT AN(YTIME)) )=0;
VPowerPlantNewEq.FX(runCy,PGALL,YTIME)$((NOT AN(YTIME)) )=0;
VPowerPlaShrNewEq.FX(runCy,PGALL,YTIME)$(AN(YTIME) $(NOT CCS(PGALL)))
```

```
=0;
VPowerPlantNewEq.FX(runCy,PGALL,YTIME)$(AN(YTIME) $(NOT NOCCS(PGALL)) )=0;
VNewInvDecis.FX(runCy,YTIME)$(NOT AN(YTIME))=1;
VElecGenPlanCap.FX(runCy,PGALL,YTIME)$DATAY(YTIME) =  iInstCapPast(runCy,PGALL,YTIME);
VElecGenPlantsCapac.FX(runCy,PGALL,YTIME)$DATAY(YTIME) =  iInstCapPast(runCy,PGALL,YTIME);
VOverallCap.FX(runCy,PGALL,YTIME)$TFIRST(YTIME) =  iInstCapPast(runCy,PGALL,YTIME)$TFIRST(YTIME);
VNewCapYearly.FX(runCy,PGALL,"2011")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2011")- iInstCapPast(runCy,PGALL,"2010") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2012")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2012")- iInstCapPast(runCy,PGALL,"2011") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2013")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2013")- iInstCapPast(runCy,PGALL,"2012") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2014")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2014")- iInstCapPast(runCy,PGALL,"2013") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2015")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2015")- iInstCapPast(runCy,PGALL,"2014") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2016")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2016")- iInstCapPast(runCy,PGALL,"2015") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2017")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2017")- iInstCapPast(runCy,PGALL,"2016") +1E-10;
VNewCapYearly.FX(runCy,"PGLHYD",YTIME)$TFIRST(YTIME) = +1E-10;
VAvgCapFacRes.FX(runCy,PGALL,YTIME)$DATAY(YTIME) =iAvailRate(PGALL,YTIME);
VElecProdPowGenPlants.FX(runCy,pgall,YTIME)=iDataElecProd(runCy,pgall,YTIME)/1000;
VEndogScrapIndex.FX(runCy,PGALL,YTIME)$(not an(YTIME) ) = 1;
VEndogScrapIndex.FX(runCy,PGSCRN,YTIME) = 1;            !! premature replacement it is not allowed for all new plants
VCO2ElcHrgProd.FX(runCy,YTIME)$(not An(YTIME)) = 0;
VRenShareElecProdSub.FX(runCy,YTIME)$(NOT AN(YTIME))=0;
VRenPotSupplyCurve.FX(runCy,PGRENEF, YTIME) $(NOT AN(YTIME)) = iMinRenPotential(runCy,PGRENEF,YTIME);
VAvgPowerGenLongTrm.L(runCy,ESET,"2010") = 0;
VLonPowGenCostNoClimPol.L(runCy,ESET,"2010") = 0;
VAvgPowerGenCostShoTrm.L(runCy,ESET,"2010") = 0;
VLongPowGenCost.L(runCy,PGALL,ESET,"2010") = 0;
VLonAvgPowGenCostNoClimPol.L(runCy,PGALL,ESET,"2010") = 0;
VAvgPowerGenLongTrm.L(runCy,ESET,"2017") = 0;
VLonPowGenCostNoClimPol.L(runCy,ESET,"2017") = 0;
```

```
VAvgPowerGenCostShoTrm.L(runCy,ESET,"2017") = 0;
VLongPowGenCost.L(runCy,PGALL,ESET,"2017") = 0;
VLonAvgPowGenCostNoClimPol.L(runCy,PGALL,ESET,"2017") = 0;
VCarVal.fx(runCy,NAP,YTIME)$(not An(YTIME))=0;
VCarVal.FX(runCy,"TRADE",YTIME)$an(YTIME) = sExogCarbValue*
    iCarbValYrExog(YTIME);
VCarVal.FX(runCy,"NOTRADE",YTIME)$an(YTIME) =sExogCarbValue*
    iCarbValYrExog(YTIME);
VCumCO2Capt.FX(runCy,YTIME)$(not an(YTIME)) = 0 ;
VCO2CO2SeqCsts.FX(runCy,YTIME)$(not an(YTIME)) =
        (iElastCO2Seq(runCy,"mc_a") *iElastCO2Seq(runCy,"mc_b"));
VDemTr.FX(runCy,TRANSE,EF,YTIME)$(not SECTTECH(TRANSE,EF)) = 0;
VTransfOutputDHPlants.FX(runCy,EFS,YTIME)$(not STEAM(EFS)) = 0;
VTransfOutputRefineries.FX(runCy,EFS,YTIME)$(not EFtoEFA(EFS,"LQD")) =
    0;
VTransfInputRefineries.FX(runCy,EFS,YTIME)$(not sameas("CRO",EFS)) = 0;
VTransfOutputNuclear.FX(runCy,EFS,YTIME)$(not sameas("ELC",EFS)) = 0;
VTransfInNuclear.FX(runCy,EFS,YTIME)$(not sameas("NUC",EFS)) = 0;
VTransfInThermPowPls.FX(runCy,EFS,YTIME)$(not PGEF(EFS)) = 0;
VExportsFake.FX(runCy,EFS,YTIME)$(not IMPEF(EFS)) = 0;
VFkImpAllFuelsNotNatGas.FX(runCy,EFS,YTIME)$(not IMPEF(EFS)) = 0;
VScalFacPlaDisp.LO(runCy, HOUR, YTIME)=-1;
VLoadCurveConstr.LO(runCy,YTIME)=0;
VLoadCurveConstr.L(runCy,YTIME)=0.01;
VRenValue.FX(YTIME) = 0 ;
VTotReqElecProd.fx(runCy,YTIME)$TFIRST(YTIME)=sum(pgall,
    VElecProdPowGenPlants.L(runCy,pgall,YTIME)$TFIRST(YTIME));
display VCarVal.l;
execute_unload "blabla.gdx";
loop an do
   s = s + 1;
   TIME(YTIME) = NO;
   TIME(AN)$(ord(an)=s) = YES;
   display TIME;
```

## sets (sets.gms)

### Description

```
sets
```

*** Geographic Coverage *

```
allCy    Countries for which the model is applied
/
RAS       REST OF ASIA
MAR       MOROCCO
RWO
REU
REP
RAF
RLA
ARG
OCE
AUT
BEL
BRA
BGR
CAN
CHA
HRV
CYP
CZE
DNK
EGY
EST
FIN
FRA
DEU
GRC
HUN
NSI
IND
IDN
IRN
IRL
ISR
ITA
```

```
JPN
KOR
LVA
LTU
LUX
MLT
MEX
NLD
NGA
POL
PRT
ROU
RUS
SAU
SVK
SVN
ZAF
ESP
SWE
TUN
TUR
GBR
USA
/
runCy(allCy) Countries for which the model is running
/
%fCountries%
/
```

*** Model Time Horizon *

```
ytime           Model time horizon                                  /%fStartHorizon%*%fEndHorizon%/
an(ytime)       Years for which the model is running                /%fStartY%*%fEndY%/
twenties(ytime) THE DECADE FROM 2020-2030 /2021*2030/
thirties(ytime) THE DECADE FROM 2030-2040 /2031*2040/
after2020(ytime) The period 2021 to 2050 /2021*2050/
after2030(ytime) The period 2031 to 2050 /2031*2050/
```

```
after2035(ytime) The period 2036 to 2050 /2036*2050/
an1(ytime) /2010*2015/
an2(ytime) /2016*2030/
an3(ytime) /2016*2020/
an4(ytime) /2021*2025/
an5(ytime) /2026*2030/
carbon(ytime)   Years for which carbon tax is applied
period(ytime)   Model can also run for periods of years
tFirst(ytime)   Base year                                        /%fBaseY%/
tFirstAn(ytime) First year for which the model is running        /%fStartY%/
time(ytime)     Model time horizon used in equation definitions  /%fStartY%*%fEndY%/
datay(ytime)    Historical year before the base year of the model /%fStartHorizon%*%fBaseY%/
timeRep(ytime)  Model time horizon used in report
hour            "Segments of hours in a year (250,1250,...,8250)" /h0*h8/
```

*** Consumer Sizes *

```
conSet       Consumer size groups related to space heating
/
smallest
modal
largest
/
conSizeSet /size/
eSet         Electricity consumers used for average electricity price calculations /i,r/
iSet(eSet)   Industrial consumer /i/
rSet(eSet)   Residential consumer /r/
```

*** Auxiliary Counters *

```
rCon         counter for the number of consumers              /0,1*19/
nSet         auxiliary counter for the definition of Vr       /b1*b20/
kpdl         counter for Polynomial Distribution Lag          /a1*a6/
rc                                                            /1*3/
rcc                                                           /1*4/
rres                                                          /r1*r4/
```

*** Sectoral Structure *

```
sct          Model Sectors
/
INDU   Industry
TRAN   Transport
TERT   Tertiary
PSG    Power and Steam Generation
BUN    Bunkers
PCH    Petrochemical Industry
OTHNEN Other non energy uses
/
ALLSBS   All sectors
/
IS    "Iron and Steel"
NF    "Non Ferrous Metals"
CH    "Chemicals"
BM    "Non Metallic Minerals"
PP    "Paper and Pulp"
FD    "Food Drink and Tobacco"
EN    "Engineering"
TX    "Textiles"
OE    "Ore Extraction"
OI    "Other Industrial sectors"
SE    "Services and Trade"
AG    "Agriculture, Fishing, Forestry etc."
HOU   "Households"
PC    "Passenger Transport - Cars"
PB    "Passenger Transport - Busses"
PT    "Passenger Transport - Rail"
PN    "Passenger Transport - Inland Navigation"
PA    "Passenger Transport - Aviation"
GU    "Goods Transport - Trucks"
GT    "Goods Transport - Rail"
GN    "Goods Transport - Inland Navigation"
BU    "Bunkers"
PCH   "Petrochemicals Industry"
NEN   "Other Non Energy Uses"
PG    "Power and Steam Generation"
H2P   "Hydrogen Production"
H2INFR "Hydrogen storage and delivery"
LGN_PRD_CH4    "Lignite Primary Production, related with CH4"
HCL_PRD_CH4    "Hard coal primary production, related with CH4"
```

```
GAS_PRD_CH4    "Natural gas primary production, related with CH4"
TERT_CH4       "Natural gas distribution in tertiary, related with CH4"
TRAN_CH4       "Gasoline in transport, related with CH4"
AG_CH4         "Agriculture activities, related with CH4"
SE_CH4         "Waste used in housing and water charges in services,
    related with CH4"
TRAN_N2O       "Oil activities in transport, related with N2O"
TX_N2O         "Textile activities, related with N2O"
AG_N2O         "Agriculture activities, related with N2O"
OI_HFC         "Other Industry activities, related with HFC"
OI_PFC         "Semiconductor manufacturing, related with PFC"
NF_PFC         "Aluminium production, related with PFC"
BM_CO2         "Non metallic minerals production, related with CO2"
PG_SF6         "Electricity production, related with SF6"
OI_SF6         "Other Industries activities, related with SF6"
/
SBS(ALLSBS)          Model Subsectors
/
IS    "Iron and Steel"
NF    "Non Ferrous Metals"
CH    "Chemicals"
BM    "Non Metallic Minerals"
PP    "Paper and Pulp"
FD    "Food Drink and Tobacco"
EN    "Engineering"
TX    "Textiles"
OE    "Ore Extraction"
OI    "Other Industrial sectors"
SE    "Services and Trade"
AG    "Agriculture, Fishing, Forestry etc."
HOU   "Households"
PC    "Passenger Transport - Cars"
PT    "Passenger Transport - Rail"
PA    "Passenger Transport - Aviation"
GU    "Goods Transport - Trucks"
GT    "Goods Transport - Rail"
GN    "Goods Transport - Inland Navigation"
BU    "Bunkers"
PCH   "Petrochemicals Industry"
NEN   "Other Non Energy Uses"
PG    "Power and Steam Generation"
H2P   "Hydrogen production"
H2INFR "Hydrogen storage and delivery"
```

```
/
SCT_GHG(ALLSBS)      Aggregate Sectors used in non-energy related GHG emissions
/
LGN_PRD_CH4    "Lignite Primary Production, related with CH4"
HCL_PRD_CH4    "Hard coal primary production, related with CH4"
GAS_PRD_CH4    "Natural gas primary production, related with CH4"
TERT_CH4       "Natural gas distribution in tertiary, related with CH4"
TRAN_CH4       "Gasoline in transport, related with CH4"
AG_CH4         "Agriculture activities, related with CH4"
SE_CH4         "Waste used in housing and water charges in services, related with CH4"
TRAN_N2O       "Oil activities in transport, related with N2O"
TX_N2O         "Textile activities, related with N2O"
AG_N2O         "Agriculture activities, related with N2O"
OI_HFC         "Other Industry activities, related with HFC"
OI_PFC         "Semiconductor manufacturing, related with PFC"
NF_PFC         "Aluminium production, related with PFC"
BM_CO2         "Non metallic minerals production, related with CO2"
PG_SF6         "Electricity production, related with SF6"
OI_SF6         "Other Industries activities, related with SF6"
/
SCTtoSBS(SCT, SBS)  Sectors to Subsectors
/
INDU.(IS,NF,CH,BM,PP,FD,EN,TX,OE,OI)
TRAN.(PC,PT,PA,GU,GT,GN)
TERT.(SE,AG,HOU)
PSG.PG
BUN.BU
PCH.PCH
OTHNEN.NEN
/
POLICIES_set
/
TRADE
NoTrade
OPT
REN
EFF
NONE
RENE_SUBS
BIO_SUBS
BIOM_SUBS
```

```
SOLRES_SUBS
exogCV
/
RegulaPolicies(POLICIES_set) Set of policies entering in the regula falsi loops
/ OPT           Carbon Value for Optimal CO2 permits allocation
  TRADE         Carbon Value for Trade sectors
  REN           Renewable value for renewable target in final demand
  EFF           Efficiency value for energy savings in final demand
  NONE          No policy
/
APolicies(RegulaPolicies) Regula falsi policies that are active
PolicyYears(RegulaPolicies, ytime) Years for which a policy is effective
NAP(Policies_set) National Allocation Plan sector categories
/
Trade    Carbon Value for trading sectors
NoTrade  Carbon Value for non-trading sectors
/
NAPtoALLSBS(NAP,ALLSBS) Energy sectors corresponding to NAP sectors
/
Trade.(FD,EN,TX,OE,OI,NF,CH,IS,BM,PP,PG,BM_CO2,H2P)
NoTrade.(SE,AG,HOU,PC,PB,PT,PN,PA,GU,GT,GN,BU,PCH,NEN,LGN_PRD_CH4,HCL_PRD_CH4,GAS_PRD_CH4,TERT_CH4,TRAN_CH4,AG_CH4,SE_CH4,TRAN_N2O,TX_N2O,AG_N2O,OI_HFC,OI_PFC,NF_PFC,PG_SF6,OI_SF6)
/
DSBS(SBS)         All Demand Subsectors         /PC,PT,PA,GU,GT,GN, IS,NF,CH,BM,PP,FD,EN,TX,OE,OI,SE,AG,HOU,PCH,NEN,BU/
TRANSE(DSBS)      All Transport Subsectors      /PC,PT,PA,GU,GT,GN/
TRANS1(SBS)       All Transport Subsectors      /PC,PT,PA,GU,GT,GN/
TRANP(TRANSE)     Passenger Transport           /PC,PT,PA/
TRANP1(SBS)       Passenger Transport           /PC,PT,PA/
TRANG(TRANSE)     Goods Transport               /GU,GT,GN/
TRANG1(SBS)       Goods Transport               /GU,GT,GN/
TRANR(TRANSE)     Road Transport                /PC,GU/
TRANR1(SBS)       Road Transport                /PC,GU/
TRANT(TRANSE)     Train Transport               /PT,GT/
TRANA(TRANSE)     Aviation                      /PA/
INDSE(DSBS)       Industrial SubSectors         /IS,NF,CH,BM,PP,FD,EN,TX,OE,OI/
DOMSE(DSBS)       Tertiary SubSectors           /SE,AG,HOU/
INDSE1(SBS)       Industrial SubSectors         /IS,NF,CH,BM,PP,FD,EN,TX,OE,OI/
```

```
DOMSE1(SBS)        Tertiary SubSectors           /SE,AG,HOU/
HOU(DSBS)          Households                     /HOU/
TERSE(DSBS)        Services and Agriculture      /SE,AG/
NENSE(DSBS)        Non Energy and Bunkers        /PCH,NEN,BU/
NENSE1(SBS)        Non Energy and Bunkers        /PCH,NEN,BU/
BUN(DSBS)          Bunkers                       /BU/
INDDOM(DSBS)       Industry and Tertiary         /IS,NF,CH,BM,PP,FD,EN,
    TX,OE,OI,SE,AG,HOU/
INDDOM1(SBS)       Industry and Tertiary         /IS,NF,CH,BM,PP,FD,EN,
    TX,OE,OI,SE,AG,HOU/
```

- the following sets are used in price equation for electricity

```
INDTRANS(SBS)      Industry and Transport        /IS,NF,CH,BM,PP,FD,EN,
    TX,OE,OI ,PC,PT,PA,GU,GT, GN /
RESIDENT(SBS)      Residential                   /SE,AG,HOU/
AGSECT             aggregate sectors             /INDSE1,DOMSE1,NENSE1,
    TRANS1,PG/
```

- 

```
     Energy Forms            *
```

```
EF           Energy Forms
/
HCL     "Hard Coal, Coke and Other Solids"
LGN     "Lignite"
CRO     "Crude Oil and Feedstocks"
LPG     "Liquefied Petroleum Gas"
GSL     "Gasoline"
KRS     "Kerosene"
GDO     "Diesel Oil"
RFO     "Residual Fuel Oil"
OLQ     "Other Liquids"
NGS     "Natural Gas"
OGS     "Other Gases"
NUC     "Nuclear"
STE     "Steam"
HYD     "Hydro"
WND     "Wind"
SOL     "Solar"
BMSWAS  "Biomass and Waste"
GEO     "Geothermal and other renewable sources eg. Tidal, etc."
MET     "Methanol"
```

```
ETH     "Ethanol"
BGDO    "Biodiesel"
H2F     "Hydrogen"
ELC     "Electricity"
STE1AL  "Steam produced from CHP advanced lgn"
STE1AH  "Steam produced from CHP advanced hcl"
STE1AD  "Steam produced from CHP advanced gdo"
STE1AR  "Steam produced from CHP advanced rfo"
STE1AG  "Steam produced from CHP advanced ngs"
STE1AB  "Steam produced from CHP advanced bmswas"
STE1AH2F "Steam produced from Hydrogen powered CHP fuel cells"
STE2LGN "Steam coming from district heating plants burning lgn"
STE2OSL "Steam produced from district heating plants burning hcl"
STE2GDO "Steam produced from district heating plants burning gdo"
STE2RFO "Steam produced from district heating plants burning rfo"
STE2OLQ "Steam produced from district heating plants burning olq"
STE2NGS "Steam produced from district heating plants burning ngs"
STE2OGS "Steam produced from district heating plants burning ogs"
STE2BMS "Steam produced from district heating plants burning bmswas"
PHEVGSL  "Plug in Hybrid engine - gasoline"
PHEVGDO  "Plug in Hybrid engine - diesel"
```

*hybrid cars with gasoline and diesel

```
CHEVGSL  "conventional Hybrid engine - gasoline"
CHEVGDO  "conventional Hybrid engine - diesel"
```

- Aggregate Fuels

```
SLD     "Solid Fuels"
GAS     "Gases"
LQD     "All Liquids"
OLQT    "All liquids but GDO, RFO, GSL"
REN     "Renewables except Hydro"
NFF     "Non Fossil Fuels"
NEF     "New energy forms"
HEATPUMP "Low enthalpy heat produced by heatpumps reducing the total
    final consumption of the sector"
/
HEATPUMP(EF) Heatpumps are reducing the heat requirements of the sector
     but increasing electricity consumption
/HEATPUMP/
EFA(EF)          Aggregate Energy Forms
```

```
/
SLD   "Solid Fuels"
LQD   "Liquids"
OLQT  "All liquids but GDO, RFO, GSL"
GAS   "Gases"
REN   "Renewables except Hydro"
STE   "Steam"
NFF   "Non Fossil Fuels"
NEF   "New energy forms"
/
REFORM(EF) FUELS CONSIDERED IN PRICE REFORM
/
CRO     "Crude Oil and Feedstocks"
LPG     "Liquefied Petroleum Gas"
GSL     "Gasoline"
KRS     "Kerosene"
GDO     "Diesel Oil"
RFO     "Residual Fuel Oil"
OLQ     "Other Liquids"
NGS     "Natural Gas"
OGS     "Other Gases"
/
REFORM1(EF) FUELS CONSIDERED IN PRICE REFORM
/HCL
LGN/
RENEF(EF)        Renewable technologies in demand side
/
HYD     "Hydro"
WND     "Wind"
SOL     "Solar"
BMSWAS  "Biomass and Waste"
GEO     "Geothermal and other renewable sources eg. Tidal, etc."
BGDO    "Biodiesel"
STE1AB  "Steam produced from CHP advanced bmswas"
STE2BMS "Steam produced from district heating plants burning bmswas"
/
OIL(EF) Liquid fuels in private road transport
/
GSL     "Gasoline"
GDO     "Diesel Oil"
/
EFtoEFA(EF,EFA)  Energy Forms Aggregations (for summary balance report)
/
```

```
(HCL,LGN).SLD
(GSL,GDO,RFO,LPG,KRS,OLQ).LQD
(LPG,KRS,OLQ).OLQT
(NGS,OGS).GAS
(STE1AL,STE1AH,STE1AD,STE1AR,STE1AG,STE1AB,STE1AH2F,STE2LGN,STE2OSL,
    STE2GDO,STE2RFO,STE2OLQ,
 STE2NGS,STE2OGS,STE2BMS).STE
(WND,SOL,GEO).REN
(HYD,WND,SOL,GEO,NUC,BMSWAS).NFF
(H2F,MET,ETH).NEF
/
WEF              Imported Energy Forms (affecting fuel prices)
/
WHCL    Imported Hard Coal
WCOKE   Imported Coke
WCRO    Imported Crude Oil
WNGS    Imported Natural Gas
WLGN    Lignite for Industry
WGSL    Spot price of gasoline Rotterdam
WGDO    Spot price of diesel Rotterdam
WRFO    Spot price of heavy fuel oil Rotterdam
/
WEFMAP(EF,WEF)          Link between Imported Energy Forms and Energy
    Forms used in Model Subsectors
/(LGN,STE1AL,STE2LGN).WLGN
(HCL,STE1AH,STE2OSL).WHCL
(NGS,OGS,STE1AG,STE2NGS,STE2OGS).WNGS
HCL.WCOKE
(GSL,KRS).WGSL
(GDO,STE1AD,STE2GDO).WGDO
(RFO,STE1AR,STE2RFO).WRFO
(OLQ,LPG,STE2OLQ).WCRO
/
EFtoWEF(SBS,EF,WEF) Link between Imported Energy Forms and Energy Forms
     used in Model Subsectors
EFS(EF)          Energy Forms used in Supply Side
/
LGN
HCL
CRO
GSL
GDO
RFO
```

```
LPG
KRS
OLQ
NGS
OGS
NUC
STE
HYD
BMSWAS
SOL
WND
GEO
ELC
H2F
MET
ETH
/
EFtoEFS(EF,EFS)    Fuel Aggregation for Supply Side
/
LGN.LGN
HCL.HCL
CRO.CRO
GSL.GSL
GDO.GDO
RFO.RFO
LPG.LPG
KRS.KRS
OLQ.OLQ
NGS.NGS
OGS.OGS
NUC.NUC
(STE1AL,STE1AH,STE1AD,STE1AR,STE1AG,STE1AB,STE1AH2F,STE2LGN,STE2OSL,
   STE2GDO,STE2RFO,STE2OLQ,STE2NGS,STE2OGS,STE2BMS).STE
HYD.HYD
(BGDO,ETH,BMSWAS).BMSWAS
SOL.SOL
GEO.GEO
WND.WND
ELC.ELC
MET.MET
H2F.H2F
/
ELCEF(EF)        Electricity                                        /
```

```
    ELC/
H2EF(EF)          Hydrogen                                          /
    H2F/
STEAM(EFS)        Steam                                             /
    STE/
TOCTEF(EFS)       Energy forms produced by power plants and boilers   /
    ELC,STE/
ALTEF(EF)         Alternative Fuels used in transport               /
    BGDO,MET,ETH/
ALTMAP(SBS,EF,EF)    Fuels whose prices affect the prices of
    alternative fuels
/
(PC,GU,PT,GT,GN).MET.GDO
(PC,GU,PT,GT,GN).ETH.GDO
(PC,GU,GN).BGDO.GDO
/
PGEF(EFS)         Energy forms used for power generation
/
LGN
HCL
GDO
RFO
OLQ
NGS
NUC
HYD
BMSWAS
SOL
GEO
WND
H2F
/
h2f1(pgef)
/h2f/
PPRODEF(EFS)      Fuels considered in primary production
/
LGN
HCL
CRO
HYD
BMSWAS
NUC
SOL
```

```
GEO
WND
NGS
/
IMPEF(EFS)       Fuels considered in Imports and Exports
/
LGN
HCL
CRO
GSL
GDO
RFO
LPG
KRS
OLQ
NGS
OGS
ELC
/
EFIS(EF)        Fuels in Iron and Steel sector in which useful energy
    conversion factor is expressed in tn per toe
/HCL, ELC/
```

```
     Technologies            *
```

```
TEAALL              Technology progress (Demand Side)
/
ORD   Ordinary
IMP   Improved
/
TEA(TEAALL)         Technologies currently used inside the model
/ORD/
TTECH(EF)        Transport Technologies
/
GSL      "Internal Combustion Engine fueled by Gasoline"
LPG      "Internal Combustion Engine fueled by Liquified Petroleum Gas"
GDO      "Internal Combustion Engine fueled by Diesel Oil"
NGS      "Internal Combustion Engine fueled by Natural Gas"
ELC      "Pure Electirc Engine"
KRS      "Gas Turbine fueled by Kerosene"
ETH      "Internal Combustion Engine fueled by Ethanol"
MET      "Methanol (85% gasoline 15% methanol) coming either from ngs
```

```
    or bms"
BGDO     "Biodiesel internal combustion engine"
PHEVGSL  "Plug in Hybrid engine - gasoline"
PHEVGDO  "Plug in Hybrid engine - diesel"
H2F      "Fuel Cells: Hydrogen"
CHEVGSL  "conventional Hybrid engine - gasoline"
CHEVGDO  "conventional Hybrid engine - diesel"
/
TTECHtoEF(TTECH,EF) Fuels consumed by transport technologies
/
GSL.GSL
LPG.LPG
GDO.GDO
NGS.NGS
ELC.ELC
KRS.KRS
ETH.ETH
MET.MET
BGDO.BGDO
PHEVGSL.(GSL,ELC)
PHEVGDO.(GDO,ELC)
H2F.H2F
CHEVGSL.GSL
CHEVGDO.GDO
/
PLUGIN(EF) Plug-in hybrids
/
PHEVGSL
PHEVGDO
/
CHYBV(EF) CONVENTIONAL hybrids
/
CHEVGSL
CHEVGDO
/
ITECH(EF)        Industrial Technologies
/
LGN,HCL,GDO,RFO,OLQ,NGS,OGS,STE1AL,
STE1AH,STE1AD,STE1AR,STE1AG,STE1AB,STE1AH2F, HEATPUMP, BMSWAS,ELC
/
DTECH(EF)        Tertiary Technologies
/
LGN,HCL,GSL,GDO,RFO,OLQ,NGS,OGS,STE1AL,
```

```
STE1AH,STE1AD,STE1AR,STE1AG,STE1AB,STE1AH2F, HEATPUMP,STE2LGN,STE2OSL,
    STE2GDO,STE2RFO,STE2OLQ,STE2NGS,
STE2OGS,STE2BMS,SOL,BMSWAS,ELC
/
BTECH(EF)        Bunkers Technologies                    /GDO,RFO,OLQ/
PNTECH(EF)       Petrochemical Technologies               /LGN,HCL,GDO,
    RFO,OLQ,NGS,OGS/
SECTTECH(SBS,EF) Link between Model Subsectors and Fuels
/
PC.(GSL,LPG,GDO,NGS,ELC,ETH,MET,BGDO,PHEVGSL,PHEVGDO,CHEVGSL,CHEVGDO)
GU.(GSL,LPG,GDO,NGS,ELC,ETH,MET,BGDO,PHEVGSL,PHEVGDO,CHEVGDO)
(PT,GT).(GDO,ELC,MET)
PA.(KRS)
GN.(GSL,GDO)
(IS,NF,CH,BM,PP,FD,EN,TX,OE,OI).(LGN,HCL,GDO,RFO,LPG,KRS,OLQ,NGS,OGS,
    ELC,STE1AL,
                                 STE1AH,STE1AD,STE1AR,STE1AG,STE1AB)
(SE,HOU,AG).                    (LGN,HCL,GSL,GDO,RFO,LPG,KRS,OLQ,NGS,
    OGS,ELC,STE1AL,
                                 STE1AH,STE1AD,STE1AR,STE1AG,STE1AB,
                                     STE2LGN,STE2OSL,STE2GDO,STE2RFO,
                                     STE2OLQ,STE2NGS,
                                 STE2OGS,STE2BMS, BMSWAS)
BU.(GDO,RFO)
(PCH,NEN).(LGN,HCL,GDO,RFO,LPG,OLQ,NGS,OGS)
PG.(LGN,HCL,GDO,RFO,NGS,NUC,HYD,BMSWAS,SOL,GEO,WND)
H2P.(HCL,RFO,NGS,NUC,BMSWAS,SOL,WND,ELC)
/
NETTECH(SBS,EF) Link between model subsectors and technologies
    participating in network effect
/
(PC,GU).(ELC,H2F,PHEVGSL,PHEVGDO)
(PT,GT).(H2F)
PA.(H2F)
(IS,NF,CH,BM,PP,FD,EN,TX,OE,OI).(STE1AH2F,HEATPUMP,ELC)
(SE,HOU,AG).                    (STE1AH2F,HEATPUMP,ELC)
/
NETTECHTRA(SBS,EF) Link between transport subsectors and technologies
    participating in network effect
/
(PC).(ELC,H2F,PHEVGSL,PHEVGDO)
PT.H2F
PA.H2F
```

```
GU.(ELC,H2F,PHEVGSL,PHEVGDO)
GT.H2F
GN.H2F
/
NETEF(EF)
/
ELC,H2F,PHEVGSL,PHEVGDO,STE1AH2F,HEATPUMP
/
NETEF2(EF)
/
ELC,H2F,HEATPUMP
/
NETEF3(EF)
/
ELC,H2F,PHEVGSL,PHEVGDO
/
NETEF4
/
OTH,TRA
/
TRANSFINAL(TRANSE)
/PC,PA,PT,GU,GT,GN/
TRANSPFINAL(TRANSFINAL)/PC,PA,PT/
TRANSGFINAL(TRANSFINAL)/GU,GT,GN/
TRANSNONPC(TRANSE)
/PA,PT/
TRANSGOOD(TRANSE)
/GU,GT,GN/
TRANSPC(TRANSE)
/PC/
TRANSPASS(TRANSE)
/PA,PC,PT/
NETEFtoNETEF2(EF,EF)
/
ELC.(ELC,PHEVGSL,PHEVGDO)
HEATPUMP.HEATPUMP
H2F.(H2F,STE1AH2F)
/
NETEFtoNETEF3(EF,EF)
/
ELC.(ELC,PHEVGSL,PHEVGDO)
H2F.H2F
/
```

```
TRANSETTECH(TRANSE,TTECH) Technologies used in transport sectors
/
PC.(GSL,LPG,GDO,NGS,ELC,ETH,MET,H2F,BGDO,PHEVGSL,PHEVGDO,CHEVGSL,
    CHEVGDO)
GU.(GSL,LPG,GDO,NGS,ELC,ETH,MET,H2F,BGDO,PHEVGSL,PHEVGDO,CHEVGDO)
(PT,GT).(GDO,ELC,MET,H2F)
PA.(KRS,H2F)
GN.(GSL,GDO,H2F)
/
PGALL            Power Generation Plant Types
/
CTHBMSWAS Conventional thermal monovalent biomass and waste
ATHLGN  Advanced thermal monovalent lignite
ATHHCL Advanced thermal monovalent hard coal
ATHRFO Advanced thermal monovalent fuel oil
ATHNGS Advanced thermal monovalent natural gas
ATHBMSWAS Advanced thermal monovalent biomass and waste
SUPCRL  Supercritical lignite
SUPCR  Supercritical coal
FBCLGN Fluidised bed lignite
FBCHCL Fluidised bed hard coal
IGCCLGN Integrated gasification combine cycle lignite
IGCCHCL Integrtaed gasification combine cycle hard coal
IGCCBMS Integrated gasification combine cycle biomass
CCCGT  Conventional combined cycle gas turbine
ACCGT  Advanced combined cycle gas turbine
AGTGDO Advanced gas turbines (peak devices) diesel oil
AGTNGS Advanced gas turbines (peak devices) natural gas
PGLHYD Large Hydro Plants
PGSHYD Small Hydro Plants
PGWND Wind Plants
PGSOL Solar Photovoltaic Plants
PGASHYD Advanced Small Hydro Plants
PGAWND Advanced Wind Plants
PGASOL Advanced Solar Thermal Plants
PGADPV Advanced Building Integrated PV Plants
PGAOTHREN Advanced geothermal Plants
PGANUC New Nuclear Designs
PGAPSS Supercritical coal with CCS
PGAPSSL Supercritical lignite with CCS
PGACGSL Integrated lignite gasification with CCS
PGACGS Integrated coal gasification with CCS
PGAGGS Gas turbine combined cycle with CCS
```

```
PGAWNO Wind offshore
/
PGASOL(PGALL)
/PGASOL/
CONVPGALL(PGALL)            Conventional power generation plant types
/
CTHBMSWAS Conventional thermal monovalent biomass and waste
CCCGT  Conventional combined cycle gas turbine
PGLHYD Large Hydro Plants
PGSHYD Small Hydro Plants
PGWND Wind Plants
PGSOL Solar Photovoltaic Plants
/
pg(sbs)
/pg/
BMSPGALL(PGALL)
/
ATHBMSWAS
IGCCBMS
/
CCS(PGALL) Plants which can be equipped with CCS
/
PGAPSS
PGAPSSL
PGACGSL
PGACGS
PGAGGS
/
NOCCS(PGALL) Plants which can be equipped with CCS but they are not
/
SUPCRL
SUPCR
IGCCHCL
IGCCLGN
ACCGT
/
CCS_NOCCS(PGALL,PGALL) mapping
/
PGAPSSL.SUPCRL
PGAPSS.SUPCR
PGACGSL.IGCCLGN
PGACGS.IGCCHCL
PGAGGS.ACCGT
```

```
/
CHP(EF)       CHP Plants
/
STE1AL,STE1AH,STE1AD,STE1AR,STE1AG,STE1AB
/
DH(EF)       District Heating
/
STE2LGN
STE2OSL
STE2GDO
STE2RFO
STE2OLQ
STE2NGS
STE2OGS
STE2BMS
/
PGNUCL(PGALL)    Nuclear plants                          / PGANUC/
PGEXO(allCy,PGALL,Ytime)     Plant types for exogenous investment
PGREN(PGALL)     Renewable Plants                        /PGLHYD,
    PGSHYD,PGWND,PGSOL,PGASHYD,PGAWND,PGASOL,PGAOTHREN, PGAWNO, PGADPV/
PGNREN(PGALL)    Advanced Renewable Plants potential      /PGASHYD,
    PGAWND,PGASOL,PGAOTHREN,PGAWNO,PGADPV,ATHBMSWAS,IGCCBMS/
PGGEO(PGALL)     Geothermal Plants                       /PGAOTHREN/
PGRENEF          Renewable energy forms in power generation  /LHYD,SHYD
    ,WND,WNO,SOL,DPV,BMSWAS,OTHREN/
PGHYDRO(PGALL) Hydro power technologies  /PGLHYD,PGSHYD,PGASHYD/
PGRENSUBS(PGALL) RES technologies to be subsidized in a scenario /
    PGAWND,PGASOL,PGAWNO,PGADPV,ATHBMSWAS,IGCCBMS,PGASHYD/
PGALLtoPGRENEF(PGALL,PGRENEF)     Correspondence between renewable
    plants and renewable energy forms
/
PGLHYD.LHYD
(PGSHYD,PGASHYD).SHYD
(PGWND,PGAWND).WND
PGAWNO.WNO
(PGSOL,PGASOL).SOL
PGADPV.DPV
(PGAOTHREN).OTHREN
(CTHBMSWAS,ATHBMSWAS,IGCCBMS).BMSWAS
/
PGALLtoEF(PGALL,PGEF)     Correspondence between plants and energy
    forms
/
```

```
(ATHLGN,FBCLGN,IGCCLGN,SUPCRL,PGAPSSL,PGACGSL).LGN
(ATHHCL,SUPCR,FBCHCL,IGCCHCL, PGAPSS, PGACGS).HCL
(ATHRFO).RFO
(AGTGDO).GDO
(ATHNGS,CCCGT,ACCGT,AGTNGS,PGAGGS).NGS
(CTHBMSWAS,ATHBMSWAS,IGCCBMS).BMSWAS
(PGANUC).NUC
(PGLHYD,PGSHYD,PGASHYD).HYD
(PGWND,PGAWND,PGAWNO).WND
(PGSOL,PGASOL,PGADPV).SOL
(PGAOTHREN).GEO
/
CHPtoEF(EF,EF)           correspondence of CHP plant types to fuels
/
(STE1AL).LGN
(STE1AH).HCL
(STE1AD).GDO
(STE1AR).RFO
(STE1AG).NGS
(STE1AB).BMSWAS
STE1AH2F.H2F
/
DHtoEF(EF,EF)          correspondence of district heating plant types
    to fuels
/
STE2LGN.LGN,
STE2OSL.HCL,
STE2GDO.GDO,
STE2RFO.RFO,
STE2OLQ.OLQ,
STE2NGS.NGS,
STE2OGS.OGS,
STE2BMS.BMSWAS
/
CHPtoEON(EF,PGALL)       Mapping of chp technologies to elec-only
    technologies
/
STE1AL.SUPCRL,
STE1AH.SUPCR,
STE1AD.AGTGDO,
STE1AR.ATHRFO,
STE1AG.ACCGT,
STE1AB.ATHBMSWAS
```

```
/
PGALLAG        Aggregations of power plants used in the aggregated
    report
/
NUC      "Nuclear Power Plants"
REN      "Renewables"
HYD      "Hydro Plants"
WND      "Wind Plants"
BMS      "Biomass and waste"
SOL      "Other renewables (mainly solar)"
THE      "Thermal plants"
HCL      "Solids"
ICG      "Clean coal"
SUP      "Supercritical"
OIL      "oil"
GAS      "Gas plants"
GGC      "Combined cycles"
OTH      "Hydrogen"
/
PGALLtoPGALLAG(PGALL,PGALLAG)
/
CTHBMSWAS.(BMS,REN)
ATHLGN.(HCL,THE)
ATHHCL.(HCL,THE)
ATHRFO.(OIL,THE)
ATHNGS.(GAS,THE)
ATHBMSWAS.(BMS,REN)
SUPCRL.(HCL,SUP,THE)
SUPCR.(HCL,SUP,THE)
FBCLGN.(HCL,SUP,THE)
FBCHCL.(HCL,SUP,THE)
IGCCLGN.(HCL,ICG,THE)
IGCCHCL.(HCL,ICG,THE)
IGCCBMS.(BMS,REN)
CCCGT.(GAS,GGC,THE)
ACCGT.(GAS,GGC,THE)
AGTGDO.(OIL,THE)
AGTNGS.(GAS,THE)
PGLHYD.(HYD,REN)
PGSHYD.(HYD,REN)
PGWND.(WND,REN)
PGSOL.(SOL,REN)
PGASHYD.(HYD,REN)
```

```
PGAWND.(WND,REN)
PGASOL.(SOL,REN)
PGADPV.(SOL,REN)
PGAOTHREN.(SOL,REN)
PGANUC.NUC
PGAPSS.(HCL,SUP,THE)
PGAPSSL.(HCL,SUP,THE)
PGACGSL.(HCL,ICG,THE)
PGACGS.(HCL,ICG,THE)
PGAGGS.(GAS,GGC,THE)
PGAWNO.(WND,REN)
/
geme3tech
/
coalconv
gasconv
oilconv
gasadv
nuc
hyd
bms
wind
spv
csp
coalccs
gasccs
/
pgmap(geme3tech,pgall)
/
coalconv.(ATHLGN,ATHHCL,SUPCRL,SUPCR,FBCLGN,FBCHCL,IGCCLGN,IGCCHCL)
gasconv.(ATHNGS,AGTNGS)
oilconv.(ATHRFO,AGTGDO)
gasadv.(CCCGT,ACCGT)
nuc.(PGANUC)
hyd.(PGLHYD,PGSHYD,PGASHYD)
bms.(CTHBMSWAS,ATHBMSWAS,IGCCBMS)
wind.(PGWND,PGAWND,PGAWNO)
spv.(PGSOL,PGADPV)
csp.(PGASOL)
coalccs.(PGAPSS,PGAPSSL,PGACGSL,PGACGS)
gasccs.(PGAGGS )
/
PGSCRN(PGALL)     New plants involved in endogenous scrapping (these
```

```
      plants are not scrapped)
/
ATHBMSWAS
SUPCRL
SUPCR
FBCLGN
FBCHCL
IGCCLGN
IGCCHCL
IGCCBMS
PGLHYD
PGSHYD
PGWND
PGSOL
PGASHYD
PGAWND
PGASOL
PGANUC
PGAPSS
PGAPSSL
PGACGSL
PGACGS
PGAGGS
PGAWNO
PGADPV
PGAOTHREN
/
NEWTECHS(PGALL) new technologies
/
ATHLGN  Advanced thermal monovalent lignite
ATHHCL Advanced thermal monovalent hard coal
ATHRFO Advanced thermal monovalent fuel oil
ATHNGS Advanced thermal monovalent natural gas
ATHBMSWAS Advanced thermal monovalent biomass and waste
SUPCRL  Supercritical lignite
SUPCR  Supercritical coal
FBCLGN Fluidised bed lignite
FBCHCL Fluidised bed hard coal
IGCCLGN Integrated gasification combine cycle lignite
IGCCHCL Integrtaed gasification combine cycle hard coal
IGCCBMS Integrated gasification combine cycle biomass
ACCGT  Advanced combined cycle gas turbine
AGTGDO Advanced gas turbines (peak devices) diesel oil
```

```
AGTNGS Advanced gas turbines (peak devices) natural gas
PGASHYD Advanced Small Hydro Plants
PGAWND Advanced Wind Plants
PGASOL Advanced Solar Thermal Plants
PGADPV Advanced Building Integrated PV Plants
PGAOTHREN Advanced geothermal Plants
PGANUC New Nuclear Designs
PGAPSSL Supercritical lignite with CCS
PGAPSS Supercritical coal with CCS
PGACGSL Integrated lignite gasification with CCS
PGACGS Integrated coal gasification with CCS
PGAGGS Gas turbine combined cycle with CCS
PGAWNO Wind offshore
/
```

```
        Emissions             *
```

```
EMISS            "Pollutant Emissions"
/
CO2      "Carbon dioxide"
CH4      "Methanio"
N2O      "Nitrogen dioxide"
HFC      "Hydrofluorocarbons"
PFC      "Perfluorinated Compounds "
SF6      "Sulfur Hexafluoride"
/
SCT_GHGtoEMISS(SCT_GHG,EMISS)  Mapping between sectors and emissions
/
(LGN_PRD_CH4,HCL_PRD_CH4,GAS_PRD_CH4,TERT_CH4,TRAN_CH4,AG_CH4,SE_CH4).
    CH4
(TRAN_N2O,TX_N2O,AG_N2O).N2O
(OI_PFC,NF_PFC).PFC
OI_HFC.HFC
BM_CO2.CO2
(PG_SF6,OI_SF6).SF6
/
CO2SEQELAST Elasticities for CO2 sequestration cost curve
/
POT      MAXIMUM POTENTIAL
mc_a     linear slope
mc_b     initial cost at x=0
mc_c
```

```
mc_d
mc_s     speed to transition to exponential
mc_m     value for the ratio of x to potential after which exponential
    is taking over
/
```

- DUMMY SETS USED FOR DATA INPUT *

```
Macro            "Elements used for reading Pop, GDP, Consumption
    Expenditure and Value Added"
/GDP,CONS_EXP,IS,NF,CH,BM,PP,FD,TX,EN,OE,OI,SE,AG,PCH,NEN,BU,HOU,POP,
    HOU_SIZE,HOUSEHOLDS/
HOUCHAR(Macro)   "Elements used for reading inhabitants per Households
    and number of Households"
/CONS_EXP,HOU_SIZE,HOUSEHOLDS/
ETYPES           "Elasticities types"
/a,b1,b2,b3,b4,b5,c,c1,c2,c3,c4,c5,aend,b3end,b4end/
ELCPCHAR         "Factors controlling electricity price"
/
INDU     "Elecprice-eleccost in Industry"
TERT     "Elecprice-eleccost in Tertiary"
W_INDU   "Weight for production cost in industry"
W_TERT   "Weight for production cost in tertiary"
IND_RES  "Additive residual to electricity price for industry"
TERT_RES  "Additive residual to electricity price for tertiary"
VAT      "VAT"
/
ECONCHAR         "Technical - Economic characteristics for demand
    technologies"
/
IC_00    "Capital Cost in 2000"
IC_05    "Capital Cost in 2005"
IC_10    "Capital Cost in 2010"
IC_15    "Capital Cost in 2015"
IC_20    "Capital Cost in 2020"
IC_25    "Capital Cost in 2025"
IC_30    "Capital Cost in 2030"
IC_50    "Capital cost in 2050"
FC_00    "Fixed O&M Cost in 2000"
FC_05    "Fixed O&M Cost in 2005"
FC_10    "Fixed O&M Cost in 2010"
FC_15    "Fixed O&M Cost in 2015"
FC_20    "Fixed O&M Cost in 2020"
```

```
FC_25    "Fixed O&M Cost in 2025"
FC_30    "Fixed O&M Cost in 2030"
FC_50    "Fixed O&M Cost in 2050"
VC_00    "Variable Cost in 2000"
VC_05    "Variable Cost in 2005"
VC_10    "Variable Cost in 2010"
VC_15    "Variable Cost in 2015"
VC_20    "Variable Cost in 2020"
VC_25    "Variable Cost in 2025"
VC_30    "Variable Cost in 2030"
VC_50    "Variable Cost in 2050"
LFT      "Technical Lifetime"
USC_00   "Useful Energy Conversion Factor in 2000"
USC_05   "Useful Energy Conversion Factor in 2005"
USC_10   "Useful Energy Conversion Factor in 2010"
USC_15   "Useful Energy Conversion Factor in 2015"
USC_20   "Useful Energy Conversion Factor in 2020"
USC_25   "Useful Energy Conversion Factor in 2025"
USC_30   "Useful Energy Conversion Factor in 2030"
USC_50   "Useful Energy Conversion Factor in 2050"
/
PGECONCHAR       "Technical - economic characteristics for power
    generation plants"
/
IC_05
IC_20
IC_50
FC_05
FC_20
FC_50
VC_05
VC_20
VC_50
EFF_05
EFF_20
EFF_50
AVAIL_05
AVAIL_20
AVAIL_50
LFT
CR
/
SG               "S parameters in Gompertz function for passenger cars
```

```
vehicle km"
/S1,S2,S3,SAT/
TRANSPCHAR        "Various transport data used in equations and post-solution calculations"
/
KM_VEH            "Thousands km per passenger car vehicle"
KM_VEH_TRUCK      "Thousands km per truck vehicle"
OCCUP_CAR         "Passengers per car (car occupancy rate)"
RES_MEXTV         "Residual for controlling the GDP-Dependent market extension of passenger cars"
RES_MEXTF         "Residual for controlling the GDP-Independent market extension of passenger cars"
/
TRANSUSE          "Usage (average) data concerning capacity and load factor"
/
CAP               "Capacity in tn/vehicle or passengers/vehicle"
LF                "Load factor"
/
TOCTSET           "Total Transformation Output and Total District Heating Output"
/
TOCT_ELC          "Electricity Total Transformation Output in Mtoe"
TOCT_STE          "Steam Total Transformation Output in Mtoe"
TONUC_ELC         "Electricity production from nuclear plants in Mtoe"
TODH_STE          "Steam produced from District Heating Plants in Mtoe"
/
SUPOTH            "Indicators related to supply side"
/
REF_CAP           "Refineries Capacity in Mbl/day"
REF_CAP_RES       "Residuals of Refineries Capacity"
HCL_PPROD         "Residual of Hard Coal Primary Production"
NGS_PPROD         "Residual of Natural Gas Primary Production"
OIL_PPROD         "Residual of Crude Oil Primary Production"
NGS_EXPORT        "Residual of Natural Gas Exports"
ELC_IMP           "Ratio of electricity imports to total final demand"
TIOTH_RES         "Residual of Total Transformation Input in Gasworks, Blast Furnances, etc."
TOOTH_RES         "Residual of Total Transformation Output from Gasworks, etc."
FEED_RES          "Residual for Feedstocks in Transfers"
/
PGTECH           "Indicators about Power Generation Plants"
```

```
/
CAPP_95          "1995 Installed Capacity in GW"
CAPP_00          "2000 Installed Capacity in GW"
CAPP_05          "2005 Installed Capacity in GW"
CAPP_09          "2009 Installed Capacity in GW"
CAPP_10          "2010 Installed Capacity in GW"
CAPP_15          "2015 Installed Capacity in GW"
CAPP_17          "2017 Installed Capacity in GW"
CAPI_00          "2000 Installed AVAILABLE Capacity in GW"
LFT              "Technical Lifetime in years"
AVAIL_00         "Plant availability in 2000"
AVAIL_10         "Plant availability in 2010"
AVAIL_30         "Plant availability in 2030"
EFF_00           "Plant Efficiency in 2000"
EFF_10           "Plant Efficiency in 2010"
EFF_30           "Plant Efficiency in 2030"
PROD_95          "Electricity Production in 1995"
PROD_00          "Electricity Production in 2000"
PROD_09          "Electricity Production in 2005"
PROD_10          "Electricity Production in 2009"
PROD_05          "Electricity Production in 2010"
PROD_15          "Electricity Production in 2015"
PROD_17          "Electricity Production in 2017"
H                "Scale parameter for endogenous scrapping (applied to
    the sum of full costs)"
S                "Weight parameter for endogenous scrapping"
/
PGCAPAST(PGTECH)         /CAPP_00,CAPP_05,CAPP_09,CAPP_10,CAPP_15,
    CAPP_17/
PGPRODPAST(PGTECH)       /PROD_00,PROD_05,PROD_09, PROD_10,PROD_15,
    PROD_17/
PGCAP            "Capacity decommisioning and expansion decided
    schedule"
/
DEC_05           "GW decommisioned until 2005"
DEC_10           "GW decommisioned until 2010"
DEC_15           "GW decommisioned until 2015"
DEC_18           "GW decommisioned until 2018"
DEC_20           "GW decommisioned until 2020"
DEC_25           "GW decommisioned until 2025"
DEC_30           "GW decommisioned until 2030"
INV_05           "GW decided to be installed until 2005"
INV_10           "GW decided to be installed until 2010"
```

```
INV_15            "GW decided to be installed until 2015"
INV_18            "GW decided to be installed until 2018"
INV_19            "GW decided to be installed until 2019"
INV_20            "GW decided to be installed until 2020"
INV_25            "GW decided to be installed until 2025"
INV_30            "GW decided to be installed until 2030"
/
PGOTH             "Various data related to power generation plants"
/
TOTCAP            "Total Available capacity in Base Year (GW)"
PEAKLOAD          "Peak Load in Base Year (GW)"
BASELOAD          "Base Load in Base Year (GW)"
NON_CHP_PER       "Non-CHP capacity (percentage of total (gross)
    capacity)"
CHP_CAP           "CHP Capacity (gross) in GW"
CHP_ELC           "CHP electricity"
STE1AL            "Utilisation rate of Lignite powered advanced CHP"
STE1AH            "Utilisation rate of Hard Coal powered advanced CHP"
STE1AD            "Utilisation rate of Diesel Oil powered advanced CHP"
STE1AR            "Utilisation rate of Fuel Oil powered advanced CHP"
STE1AG            "Utilisation rate of Natural Gas powered advanced CHP"
STE1AB            "Utilisation rate of Biomass-Waste powered advanced
    CHP"
STE1AH2F          "Utilisation rate of HYDROGEN powered FUEL CELL CHP"
```

- The following fuels are related with district heating efficiency

```
LGN
HCL
GDO
RFO
OLQ
NGS
OGS
BMSWAS
/
PGLOADTYPE(PGOTH)   "Peak and Base load of total electricity demand in
    GW"
/PEAKLOAD, BASELOAD/
CHPSET(PGOTH)       "Indicators related to CHP Production"
                          /NON_CHP_PER,CHP_CAP/
PGEFS(PGOTH)        "Fuels used as Input to District Heating"
                        /LGN,HCL,GDO,RFO,OLQ,NGS,OGS,BMSWAS/
```

```
PG_CHP(PGOTH)/
STE1AL,STE1AH,STE1AD,STE1AR,STE1AG,STE1AB,STE1AH2F/
VARIOUS_LABELS /AMAXBASE, MAXLOADSH/
PGRES
/
TOT_CAP_RES         "Residual on Total Capacity (Reserve margin on capacity)"
BASE_LOAD_RES       "Residual on Base Load "
MAX_LOAD_RES        "Residual on Peak Load (Peak Load margin)"
/
CHPRES_SET
/
CHP_CC           Capital cost RD residual adjustment
CHP_FOM          FOM cost RD residual adjustment
CHP_VOM          VOM cost RD residual adjustment
/
nucres_set
/res/
cgiset /Baseline, Scenario/
chpparset /CHPPAR, ELCINDEX/
Gompset1 /PC/
Gompset2 /s1,s2,s3,sat,mextv,scr /
Indu_SCon_Set /Base, SHR_NSE, SH_HPELC/
PG1_set / 2000,2005,2009,2010,DEC_05,DEC_10,DEC_15, DEC_18,DEC_20, DEC_25,DEC_30,INV_05,INV_10,INV_15,INV_18, INV_19, INV_20,INV_25, INV_30,INV_11,INV_12,INV_13,INV_14,PROD_00,PROD_05,PROD_09,PROD_10, 2015,2017, PROD_15, PROD_17/
CHPPGSET /IC,FC,VOM,LFT,AVAIL,BOILEFF,MAXCHPSHARE/
ELSH_SET /elsh/
PMR_SET/H,S/
falsi_iter Maximum iterations for requla-falsi /rf1*rf7/
TAXSET
/
MaxTax
Mid
Speed
/
NETWSET
/
MULT
Mid
Speed
/
```

```
;
```

```
     SET ASSIGNMENTS           *
```

```
alias(TT, ytime);
```

- TT set is used as an index to main time loop

```
alias (YYTIME,ytime);
alias(EF, EF2);
```

- this alias is used in alternative transport fuels price calculations

```
alias(PGALL2,PGALL);
```

*this alias is used in plant dispatching equation

```
scalar TF order of base year in set ytime;
TF=sum((TFIRST,ytime), ord(ytime)$TFIRST(ytime));
```

- Allocate imported fuels to fuels used in demand subsectors

```
EFtoWEF(SBS,EF,WEF)=NO;
loop WEF do
  loop EF$WEFMAP(EF,WEF) do
    loop SBS$SECTTECH(SBS,EF) do
         if not sameas(WEF,"WCOKE") then
             EFtoWEF(SBS,EF,WEF)=yes;
         endif;
    endloop;
  endloop;
endloop;
EFtoWEF("IS","HCL","WCOKE")=yes; !! special case for Iron and Steel ->
   hard coal is in fact coke
EFtoWEF("IS","HCL","WHCL")=no; !! special case for Iron and Steel ->
   hard coal is in fact coke
Parameter TECHS(DSBS) Number of technologies in transport subsectors;
loop DSBS$TRANSE(DSBS) do
     TECHS(DSBS)=0;
     loop EF$SECTTECH(DSBS,EF) do
          loop TEA do
              TECHS(DSBS) = TECHS(DSBS)+1;
          endloop
```

```
     endloop;
endloop;
alias(NAP2,NAP);
ALIAS (EFS2,EFS);
ALIAS (CHP2,CHP);
ALIAS (INDDOM2,INDDOM);
ALIAS (YYTIME2,ytime);
scalar ordfirst /0/;
ordfirst=sum((ytime,YYTIME2)$((ord(ytime)<=ord(YYTIME2)) $TFIRST(
    YYTIME2)),1);
sets
BALEF fuels in balance report
/
"Total"
"Solids"
"Hard coal"
"Lignite"
"Crude oil and Feedstocks"
"Liquids"
"Liquified petroleum gas"
"Gasoline"
"Kerosene"
"Diesel oil"
"Fuel oil"
"Other liquids"
"Gas fuels"
"Natural gas"
"Derived gases"
"Nuclear heat"
"Steam"
"Hydro"
"Wind"
"Solar energy"
"Biomass"
"Geothermal heat"
"Methanol"
"Hydrogen"
"Electricity"
/
biomass(balef)
/"biomass"/
TOTAL(BALEF)
/"TOTAL"/
```

```
BALEF2EFS(BALEF, EFS) Mapping from balance fuels to model fuels
/
"Total".(HCL,LGN,CRO,LPG,GSL,KRS,GDO,RFO,OLQ,NGS,OGS,NUC,STE,HYD,WND,
    SOL,BMSWAS,GEO,MET,ETH,H2F,ELC)
"Solids".(HCL,LGN)
"Hard coal".HCL
"Lignite".LGN
"Crude oil and Feedstocks".CRO
"Liquids".(LPG,GSL,KRS,GDO,RFO,OLQ)
"Liquified petroleum gas".LPG
"Gasoline".GSL
"Kerosene".KRS
"Diesel oil".GDO
"Fuel oil".RFO
"Other liquids".OLQ
"Gas fuels".(NGS,OGS)
"Natural gas".NGS
"Derived gases".OGS
"Nuclear heat".NUC
"Steam".STE
"Hydro".HYD
"Wind".WND
"Solar energy".SOL
"Biomass".BMSWAS
"Geothermal heat".GEO
"Methanol".MET
"Hydrogen".H2F
"Electricity".ELC
/
REPORT_LINES Aggregate Report lines
/
r_dem
r_pop
r_hou
r_hsize
r_blank1
r_econ
r_gdp
r_cons
r_va
r_va_i
r_va_s
r_va_a
```

```
r_blank2
r_prd
r_prd_s
r_prd_o
r_prd_g
r_prd_n
r_prd_r
r_prd_h
r_prd_b
r_prd_w
r_prd_e
r_blank3
r_imp
r_imp_s
r_imp_o
r_imp_g
r_imp_e
r_imp_h
r_blank4
r_gic
r_gic_s
r_gic_o
r_gic_g
r_gic_n
r_gic_e
r_gic_r
r_gic_h
r_blank5
r_fcon
r_fcon_f
r_fcon_c
r_fcon_o
r_fcon_g
r_fcon_h
r_fcon_b
r_fcon_w
r_fcon_e
r_fcon_m
r_fcon_sf
r_fcon_i
r_fcon_i_s
r_fcon_i_o
r_fcon_i_g
```

```
r_fcon_i_h
r_fcon_i_b
r_fcon_i_e
r_fcon_r
r_fcon_r_s
r_fcon_r_o
r_fcon_r_g
r_fcon_r_h
r_fcon_r_b
r_fcon_r_r
r_fcon_r_e
r_fcon_r_m
r_fcon_s
r_fcon_s_s
r_fcon_s_o
r_fcon_s_g
r_fcon_s_h
r_fcon_s_b
r_fcon_s_r
r_fcon_s_e
r_fcon_s_m
r_fcon_a
r_fcon_a_s
r_fcon_a_o
r_fcon_a_g
r_fcon_a_h
r_fcon_a_b
r_fcon_a_r
r_fcon_a_e
r_fcon_a_m
r_fcon_t
r_fcon_t_s
r_fcon_t_o
r_fcon_t_g
r_fcon_t_e
r_fcon_t_m
r_blank6
r_eprod
r_eprod_n
r_eprod_r
r_eprod_h
r_eprod_w
r_eprod_t_g_b
```

```
r_eprod_o
r_eprod_t
r_eprod_t_s
r_eprod_t_cl
r_eprod_t_sp
r_eprod_t_o
r_eprod_t_g
r_eprod_t_g_cc
r_eprod_t_g_h
r_blank7
r_cap
r_cap_n
r_cap_r
r_cap_h
r_cap_w
r_cap_t_g_b
r_cap_csp
r_cap_dpv
r_cap_t
r_cap_t_s
r_cap_t_cl
r_cap_t_sp
r_cap_t_o
r_cap_t_g
r_cap_t_g_cc
r_cap_t_g_h
r_blank8
r_pginp
r_pginp_s
r_pginp_o
r_pginp_g
r_pginp_b
r_pginp_h
r_blank9
TIMEREP
r_prind
r_prres
r_blank11
r_co2
r_co2_p
r_co2_e
r_co2_i
r_co2_r
```

```
r_co2_s
r_co2_t
r_co2_90
r_blank12
r_blank13
r_gic_gdp
r_gic_cap
r_elc_cap
r_co2_gic
r_co2_cap
r_co2_gdp
r_imp_dep
r_blank14
r_blank15
r_int_i
r_int_r
r_int_s
r_int_t
r_blank16
r_blank17
r_blank18
r_blank19
r_blank20
r_actv_p
r_actv_pb
r_actv_pc
r_actv_pt
r_actv_pa
r_actv_pn
r_actv_cap
r_actv_cars
r_actv_km
r_actv_f
r_actv_gu
r_actv_gt
r_actv_gi
r_actv_ga
r_ccs
r_h2inp
r_h2inp_s
r_h2inp_o
r_h2inp_g
r_h2inp_b
```

```
r_co2_h
/
;
Set stringcy(allCy)
Set smallpg(pgall) /AGTGDO,AGTNGS/
Alias (smallpg,smallpgg),(geme3tech,geme3tech1),(EF1,EF),(allCy2,allCy)
;
```

## solve (solve.gms)

### Description

```
    openprom.optfile=1;
    sModelStat = 100;
    loop rcc$(rcc.val <= sSolverTryMax) do
        if sModelStat ne 2 then
            solve openprom using nlp maximizing vDummyObj;
            sModelStat = openprom.modelstat;
        endif;
    endloop;    !! close inner iteration loop
endloop;  !! close outer iteration loop
```

## References